

Example Problem 3-11

Compute the maximum shearing stress in the beam described in Example Problem 3-10 using the special shearing stress formula for a rectangular section.

Solution

Objective Compute the maximum shearing stress τ in the beam in Figure 3-14.

Given The data are the same as stated in Example Problem 3-10 and as shown in Figure 3-14.

Analysis Use Equation (3-18) to compute $\tau = 3V/2A$. For the rectangle, $A = th$.

Results

$$\tau_{\max} = \frac{3V}{2A} = \frac{3(1000 \text{ lb})}{2[(2.0 \text{ in})(8.0 \text{ in})]} = 93.8 \text{ psi}$$

Comment This result is the same as that obtained for Example Problem 3-10, as expected.

3-15 NORMAL STRESS DUE TO BENDING

A *beam* is a member that carries loads transverse to its axis. Such loads produce bending moments in the beam, which result in the development of bending stresses. Bending stresses are *normal stresses*, that is, either tensile or compressive. The maximum bending stress in a beam cross section will occur in the part farthest from the neutral axis of the section. At that point, the *flexure formula* gives the stress:

⇒ **Flexure Formula for Maximum Bending Stress**

$$\sigma = Mc/I \quad (3-22)$$

where M = magnitude of the bending moment at the section

I = moment of inertia of the cross section with respect to its neutral axis

c = distance from the neutral axis to the outermost fiber of the beam cross section

The magnitude of the bending stress varies linearly within the cross section from a value of zero at the neutral axis, to the maximum tensile stress on one side of the neutral axis, and to the maximum compressive stress on the other side. Figure 3-16 shows a typical stress distribution in a beam cross section. Note that the stress distribution is independent of the shape of the cross section.

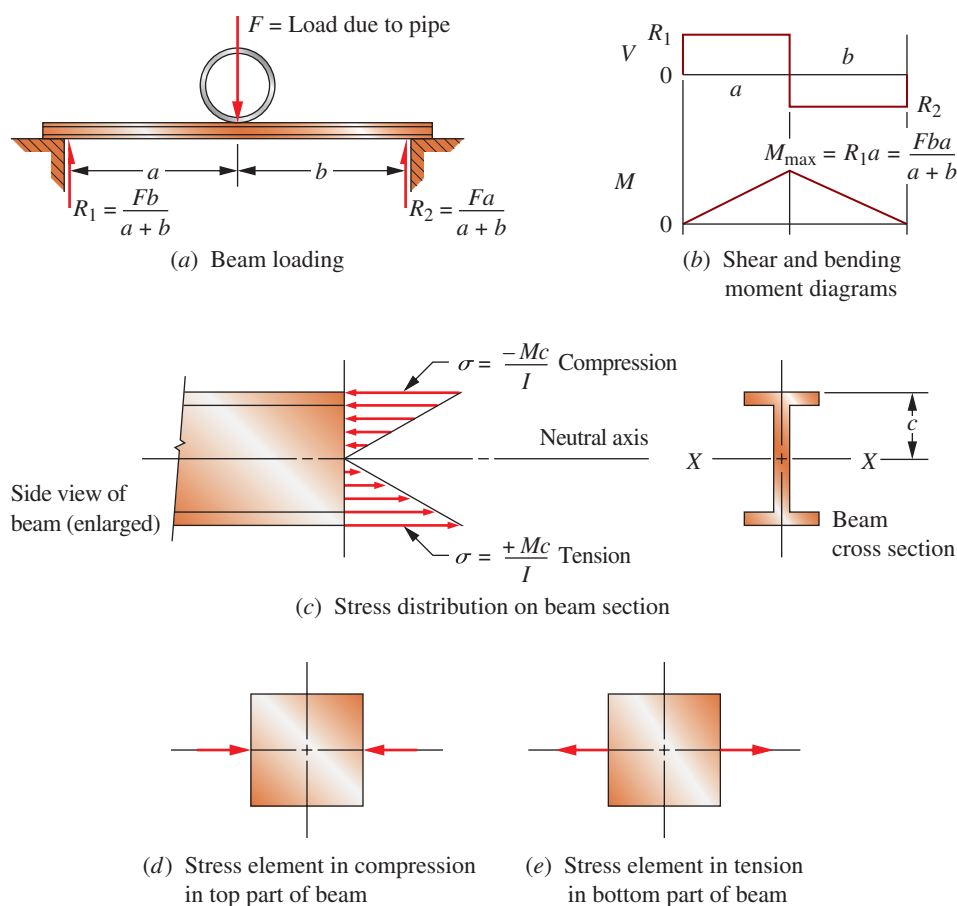


FIGURE 3-16 Typical bending stress distribution in a beam cross section

Note that *positive bending* occurs when the deflected shape of the beam is concave upward, resulting in compression on the upper part of the cross section and tension on the lower part. Conversely, *negative bending* causes the beam to be concave downward.

The flexure formula was developed subject to the following conditions:

1. The beam must be in pure bending. Shearing stresses must be zero or negligible. No axial loads are present.
2. The beam must not twist or be subjected to a torsional load.
3. The material of the beam must obey Hooke's law.
4. The modulus of elasticity of the material must be the same in both tension and compression.
5. The beam is initially straight and has a constant cross section.
6. Any plane cross section of the beam remains plane during bending.
7. No part of the beam shape fails because of local buckling or wrinkling.

If condition 1 is not strictly met, you can continue the analysis by using the method of combined stresses presented in Chapter 4. In most practical beams, which are long relative to their height, shear stresses are

sufficiently small as to be negligible. Furthermore, the maximum bending stress occurs at the outermost fibers of the beam section, where the shear stress is in fact zero. A beam with varying cross section, which would violate condition 5, can be analyzed by the use of stress concentration factors discussed later in this chapter.

For design, it is convenient to define the term *section modulus*, S , as

$$S = I/c \quad (3-23)$$

The flexure formula then becomes

⇒ Flexure Formula

$$\sigma = M/S \quad (3-24)$$

Since I and c are geometrical properties of the cross section of the beam, S is also. Then, in design, it is usual to define a design stress, σ_d , and, with the bending moment known, solve for S :

⇒ Required Section Modulus

$$S = M/\sigma_d \quad (3-25)$$

This results in the required value of the section modulus. From this, the required dimensions of the beam cross section can be determined.

Example Problem 3-12

For the beam shown in Figure 3-16, the load F due to the pipe is 12 000 lb. The distances are $a = 4$ ft and $b = 6$ ft. Determine the required section modulus for the beam to limit the stress due to bending to 30 000 psi, the recommended design stress for a typical structural steel in static bending. Then specify the lightest suitable steel beam.

Solution

Objective Compute the required section modulus S for the beam in Figure 3-16.

Given The layout and the loading pattern are shown in Figure 3-16.

Lengths: Overall length = $L = 10$ ft; $a = 4$ ft; $b = 6$ ft.

Load = $F = 12\,000$ lb.

Design stress = $\sigma_d = 30\,000$ psi.

Analysis Use Equation (3-25) to compute the required section modulus S . Compute the maximum bending moment that occurs at the point of application of the load using the formula shown in Part (b) of Figure 3-16.

$$\begin{aligned} \text{Results} \quad M_{\max} &= R_1 a = \frac{Fba}{a+b} = \frac{(12\,000 \text{ lb})(6 \text{ ft})(4 \text{ ft})}{(6 \text{ ft} + 4 \text{ ft})} = 28\,800 \text{ lb} \cdot \text{ft} \\ S &= \frac{M}{\sigma_d} = \frac{28\,800 \text{ lb} \cdot \text{ft}}{30\,000 \text{ lb/in}^2} \frac{12 \text{ in}}{\text{ft}} = 11.5 \text{ in}^3 \end{aligned}$$

Comments A steel beam section can now be selected from Tables A15-9 and A15-10 that has at least this value for the section modulus. The lightest section, typically preferred, is the W8×15 wide-flange shape with $S = 11.8 \text{ in}^3$.

3-16 BEAMS WITH CONCENTRATED BENDING MOMENTS

Figures 3-16 and 3-17 show beams loaded only with concentrated forces or distributed loads. For such loading in any combination, the moment diagram is

continuous. That is, there are no points of abrupt change in the value of the bending moment. Many machine elements such as cranks, levers, helical gears, and brackets carry loads whose line of action is offset from the centroidal axis of the beam in such a way that a concentrated moment is exerted on the beam.

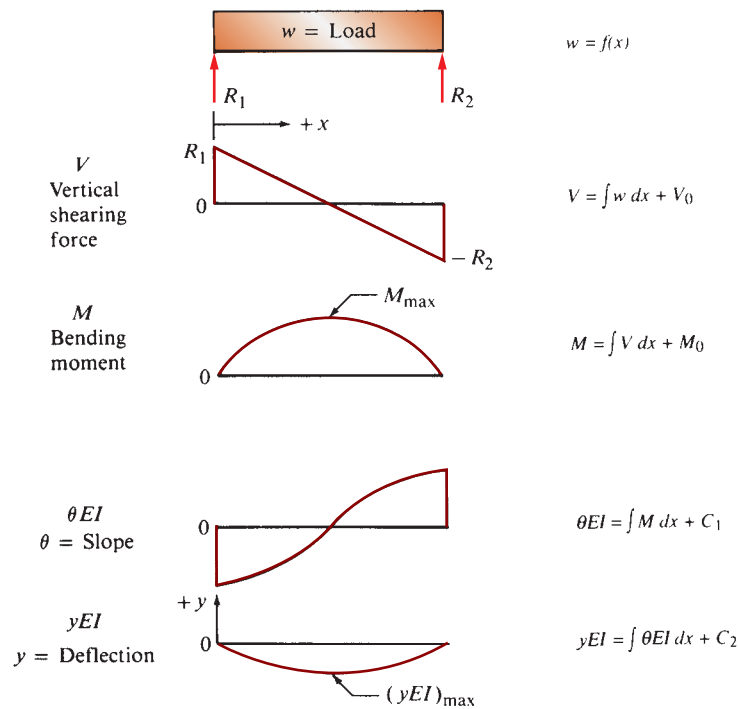


FIGURE 3-17 Relationships of load, vertical shearing force, bending moment, slope of deflected beam shape, and actual deflection curve of a beam

Figures 3-18, 3-19, and 3-20 show three different examples where concentrated moments are created on machine elements. The bell crank in Figure 3-18 pivots

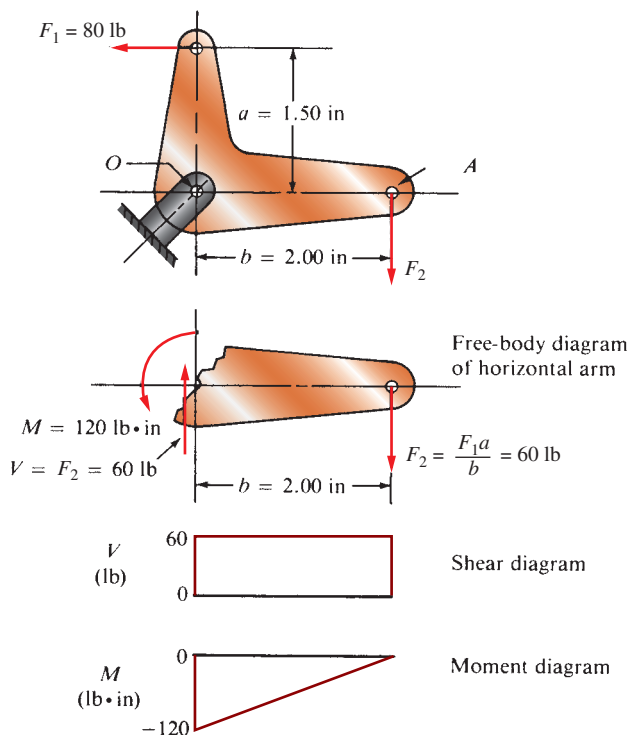


FIGURE 3-18 Bending moment in a bell crank

around point O and is used to transfer an applied force to a different line of action. Each arm behaves similar to a cantilever beam, bending with respect to an axis through the pivot. For analysis, we can isolate an arm by making an imaginary cut through the pivot and showing the reaction force at the pivot pin and the internal moment in the arm. The shearing force and bending moment diagrams included in Figure 3-18 show the results, and Example Problem 3-13 gives the details of the analysis. Note the similarity to a cantilever beam with the internal concentrated moment at the pivot reacting to the force, F_2 , acting at the end of the arm.

Figure 3-19 shows a print head for an impact-type printer in which the applied force, F , is offset from the neutral axis of the print head itself. Thus the force creates a concentrated bending moment at the right end where the vertical lever arm attaches to the horizontal part. The free-body diagram shows the vertical arm cut off and an internal axial force and moment replacing the effect of the extended arm. The concentrated moment causes the abrupt change in the value of the bending moment at the right end of the arm as shown in the bending moment diagram. Example Problem 3-14 gives the details of the analysis.

Figure 3-20 shows an isometric view of a crankshaft that is actuated by the vertical force acting at the end of the crank. One result is an applied torque that tends to rotate the shaft ABC clockwise about its x -axis. The reaction torque is shown acting at the forward end of the crank. A second result is that the vertical force acting

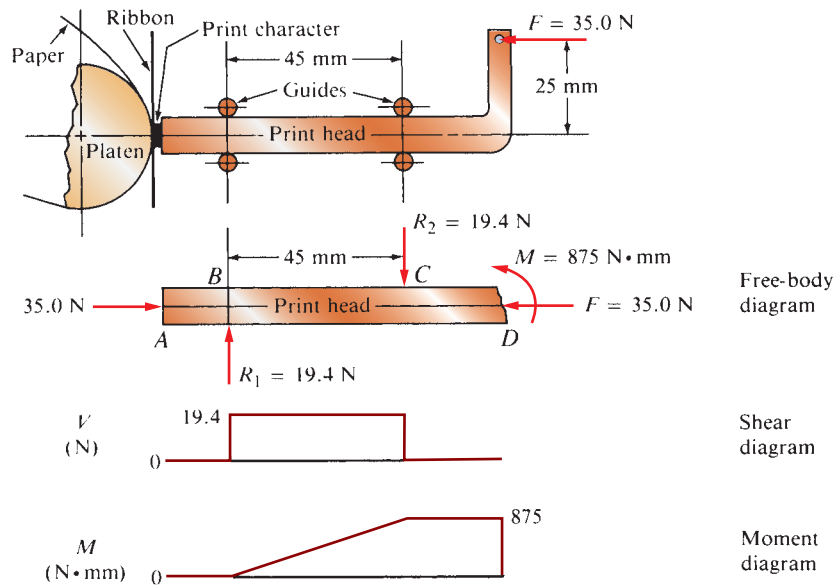


FIGURE 3-19 Bending moment on a print head

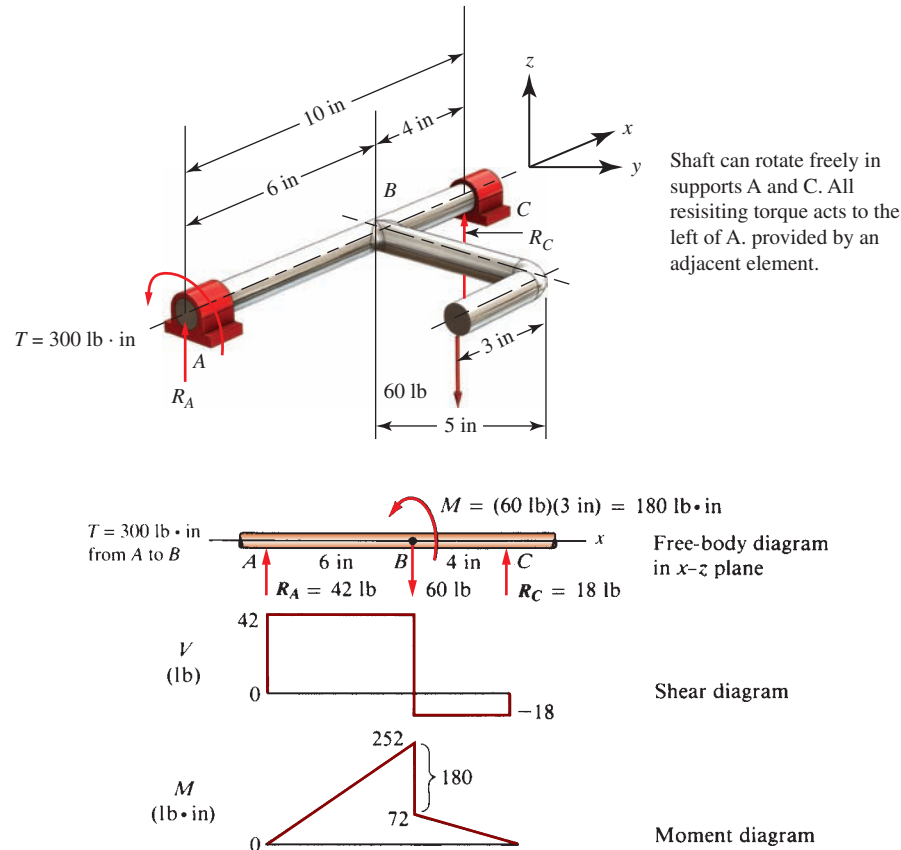


FIGURE 3-20 Bending moment on a shaft carrying a crank

at the end of the crank creates a twisting moment in the rod attached at B and thus tends to bend the shaft ABC in the x - z plane. The twisting moment is treated as a concentrated moment acting at B with the resulting abrupt change in the bending moment at that location as can be seen in the bending moment diagram. Example Problem 3-15 gives the details of the analysis.

When drawing the bending moment diagram for a member to which a concentrated moment is applied, the following sign convention will be used.

When a concentrated bending moment acts on a beam in a counterclockwise direction, the moment diagram drops; when a clockwise concentrated moment acts, the moment diagram rises.

Example Problem 3–13

The bell crank shown in Figure 3–18 is part of a linkage in which the 80-lb horizontal force is transferred to F_2 acting vertically. The crank can pivot about the pin at O . Draw a free-body diagram of the horizontal part of the crank from O to A . Then draw the shearing force and bending moment diagrams that are necessary to complete the design of the horizontal arm of the crank.

Solution

Objective Draw the free-body diagram of the horizontal part of the crank in Figure 3–18. Draw the shearing force and bending moment diagrams for that part.

Given The layout from Figure 3–18.

Analysis Use the entire crank first as a free body to determine the downward force F_2 that reacts to the applied horizontal force F_1 of 80 lb by summing moments about the pin at O .

Then create the free-body diagram for the horizontal part by breaking it through the pin and replacing the removed part with the internal force and moment acting at the break.

Results We can first find the value of F_2 by summing moments about the pin at O using the entire crank:

$$\begin{aligned} F_1 \cdot a &= F_2 \cdot b \\ F_2 &= F_1(a/b) = 80 \text{ lb}(1.50/2.00) = 60 \text{ lb} \end{aligned}$$

Below the drawing of the complete crank, we have drawn a sketch of the horizontal part, isolating it from the vertical part. The internal force and moment at the cut section are shown. The externally applied downward force F_2 is reacted by the upward reaction at the pin. Also, because F_2 causes a moment with respect to the section at the pin, an internal reaction moment exists, where

$$M = F_2 \cdot b = (60 \text{ lb})(2.00 \text{ in}) = 120 \text{ lb} \cdot \text{in}$$

The shear and moment diagrams can then be shown in the conventional manner. The result looks much like a cantilever that is built into a rigid support. The difference here is that the reaction moment at the section through the pin is developed in the vertical arm of the crank.

Comments Note that the shape of the moment diagram for the horizontal part shows that the maximum moment occurs at the section through the pin and that the moment decreases linearly as we move out toward point A . As a result, the shape of the crank is optimized, having its largest cross section (and section modulus) at the section of highest bending moment. You could complete the design of the crank using the techniques reviewed in Section 3–15.

Example Problem 3–14

Figure 3–19 represents a print head for a computer printer. The force F moves the print head toward the left against the ribbon, imprinting the character on the paper that is backed up by the platen. Draw the free-body diagram for the horizontal portion of the print head, along with the shearing force and bending moment diagrams.

Solution

Objective Draw the free-body diagram of the horizontal part of the print head in Figure 3–19. Draw the shearing force and bending moment diagrams for that part.

Given The layout from Figure 3–19.

Analysis The horizontal force of 35 N acting to the left is reacted by an equal 35 N horizontal force produced by the platen pushing back to the right on the print head. The guides provide simple supports in the vertical direction. The applied force also produces a moment at the base of the vertical arm where it joins the horizontal part of the print head.

We create the free-body diagram for the horizontal part by breaking it at its right end and replacing the removed part with the internal force and moment acting at the break. The shearing force and bending moment diagrams can then be drawn.

Results The free-body diagram for the horizontal portion is shown below the complete sketch. Note that at the right end (section D) of the print head, the vertical arm has been removed and replaced with the internal horizontal force of 35.0 N and a moment of 875 N·mm caused by the 35.0 N force acting 25 mm above it. Also note that the 25 mm-moment arm for the force is taken from the line of action of the force *to the neutral axis of the horizontal part*. The 35.0 N reaction of the platen on the print head tends to place the head in compression over the entire length. The rotational tendency of the moment is reacted by the couple created by R_1 and R_2 acting 45 mm apart at B and C .

Below the free-body diagram is the vertical shearing force diagram in which a constant shear of 19.4 N occurs only between the two supports.

The bending moment diagram can be derived from either the left end or the right end. If we choose to start at the left end at *A*, there is no shearing force from *A* to *B*, and therefore there is no change in bending moment. From *B* to *C*, the positive shear causes an increase in bending moment from 0 to 875 N·mm. Because there is no shear from *C* to *D*, there is no change in bending moment, and the value remains at 875 N·mm. The counterclockwise-directed concentrated moment at *D* causes the moment diagram to drop abruptly, closing the diagram.

Example Problem 3–15

Figure 3–20 shows a crank in which it is necessary to visualize the three-dimensional arrangement. The 60-lb downward force tends to rotate the shaft *ABC* around the *x*-axis. The reaction torque acts only at the end of the shaft outboard of the bearing support at *A*. Bearings *A* and *C* provide simple supports. Draw the complete free-body diagram for the shaft *ABC*, along with the shearing force and bending moment diagrams.

Solution

Objective Draw the free-body diagram of the shaft *ABC* in Figure 3–20. Draw the shearing force and bending moment diagrams for that part.

Given The layout from Figure 3–20.

Analysis The analysis will take the following steps:

1. Determine the magnitude of the torque in the shaft between the left end and point *B* where the crank arm is attached.
2. Analyze the connection of the crank at point *B* to determine the force and moment transferred to the shaft *ABC* by the crank.
3. Compute the vertical reactions at supports *A* and *C*.
4. Draw the shearing force and bending moment diagrams considering the concentrated moment applied at point *B*, along with the familiar relationships between shearing force and bending moments.

Results The free-body diagram is shown as viewed looking at the *x*–*z* plane. Note that the free body must be in equilibrium in all force and moment directions. Considering first the torque (rotating moment) about the *x*-axis, note that the crank force of 60 lb acts 5.0 in from the axis. The torque, then, is

$$T = (60 \text{ lb})(5.0 \text{ in}) = 300 \text{ lb} \cdot \text{in}$$

This level of torque acts from the left end of the shaft to section *B*, where the crank is attached to the shaft.

Now the loading at *B* should be described. One way to do so is to visualize that the crank itself is separated from the shaft and is replaced with a force and moment caused by the crank. First, the downward force of 60 lb pulls down at *B*. Also, because the 60-lb applied force acts 3.0 in to the left of *B*, it causes a concentrated moment in the *x*–*z* plane of 180 lb·in to be applied at *B*.

Both the downward force and the moment at *B* affect the magnitude and direction of the reaction forces at *A* and *C*. First, summing moments about *A*,

$$\begin{aligned} (60 \text{ lb})(6.0 \text{ in}) - 180 \text{ lb} \cdot \text{in} - R_C(10.0 \text{ in}) &= 0 \\ R_C &= [(360 - 180) \text{ lb} \cdot \text{in}]/(10.0 \text{ in}) = 18.0 \text{ lb upward} \end{aligned}$$

Now, summing moments about *C*,

$$\begin{aligned} (60 \text{ lb})(4.0 \text{ in}) + 180 \text{ lb} \cdot \text{in} - R_A(10.0 \text{ in}) &= 0 \\ R_A &= [(240 + 180) \text{ lb} \cdot \text{in}]/(10.0 \text{ in}) = 42.0 \text{ lb upward} \end{aligned}$$

Now the shear and bending moment diagrams can be completed. The moment starts at zero at the simple support at *A*, rises to 252 lb·in at *B* under the influence of the 42-lb shear force, then drops by 180 lb·in due to the counterclockwise concentrated moment at *B*, and finally returns to zero at the simple support at *C*.

Comments In summary, shaft *ABC* carries a torque of 300 lb·in from point *B* to its left end. The maximum bending moment of 252 lb·in occurs at point *B* where the crank is attached. The bending moment then suddenly drops to 72 lb·in under the influence of the concentrated moment of 180 lb·in applied by the crank.

3-17 FLEXURAL CENTER FOR BEAM BENDING

A beam section must be loaded in a way that ensures symmetrical bending; that is, there must be no tendency for the section to twist under the load. Figure 3-21 shows several shapes that are typically used for beams having a vertical axis of symmetry. If the line of action of the loads on such sections passes through the axis of symmetry, then there is no tendency for the section to twist, and the flexure formula applies.

When there is no vertical axis of symmetry, as with the sections shown in Figure 3-22, care must be exercised in placement of the loads. If the line of action of the loads were shown as F_1 in the figure, the beam would twist and bend, so the flexure formula would not give accurate results for the stress in the section. For such sections, the load must be placed in line with the *flexural center*, sometimes called the *shear center*. Figure 3-22 shows the approximate location of the flexural center for these shapes (indicated by the symbol Q). Applying the load in line with Q , as shown with the forces labeled F_2 , would result in pure bending. A table of formulas for the location of the flexural center is available (see Reference 7).

3-18 BEAM DEFLECTIONS

The bending loads applied to a beam cause it to deflect in a direction perpendicular to its axis. A beam that was originally straight will deform to a slightly curved shape. In most cases, the critical factor is either the maximum deflection of the beam or its deflection at specific locations.

Consider the double-reduction speed reducer shown in Figure 3-23. The four gears (A , B , C , and D) are mounted on three shafts, each of which is supported by two bearings. The action of the gears in transmitting power creates a set of forces that in turn act on the shafts to cause bending. One component of the total force on the gear teeth acts in a direction that tends to separate the two gears. Thus, gear A is forced upward, while gear B is forced downward. For good gear performance, the net deflection of one gear relative to the other should not exceed 0.005 in (0.13 mm) for medium-sized industrial gearing.

To evaluate the design, there are many methods of computing shaft deflections. We will review briefly those methods using deflection formulas, superposition, and a general analytical approach.

A set of formulas for computing the deflection of beams at any point or at selected points is useful in many practical problems. Appendix 14 includes several cases.

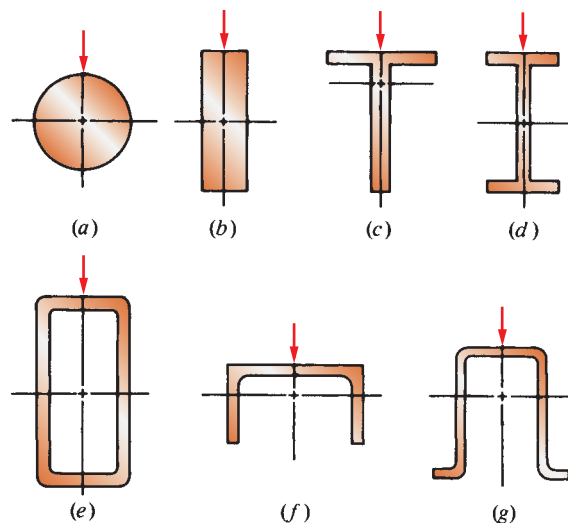


FIGURE 3-21 Symmetrical sections. A load applied through the axis of symmetry results in pure bending in the beam.

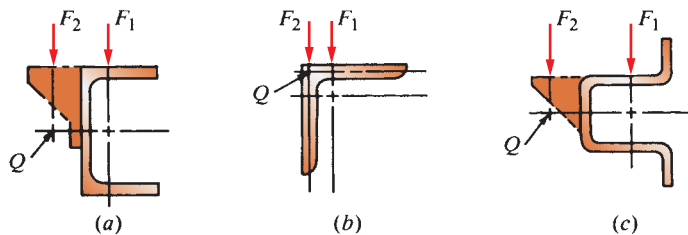


FIGURE 3-22 Nonsymmetrical sections. A load applied as at F_1 would cause twisting; loads applied as at F_2 through the flexural center Q would cause pure bending.

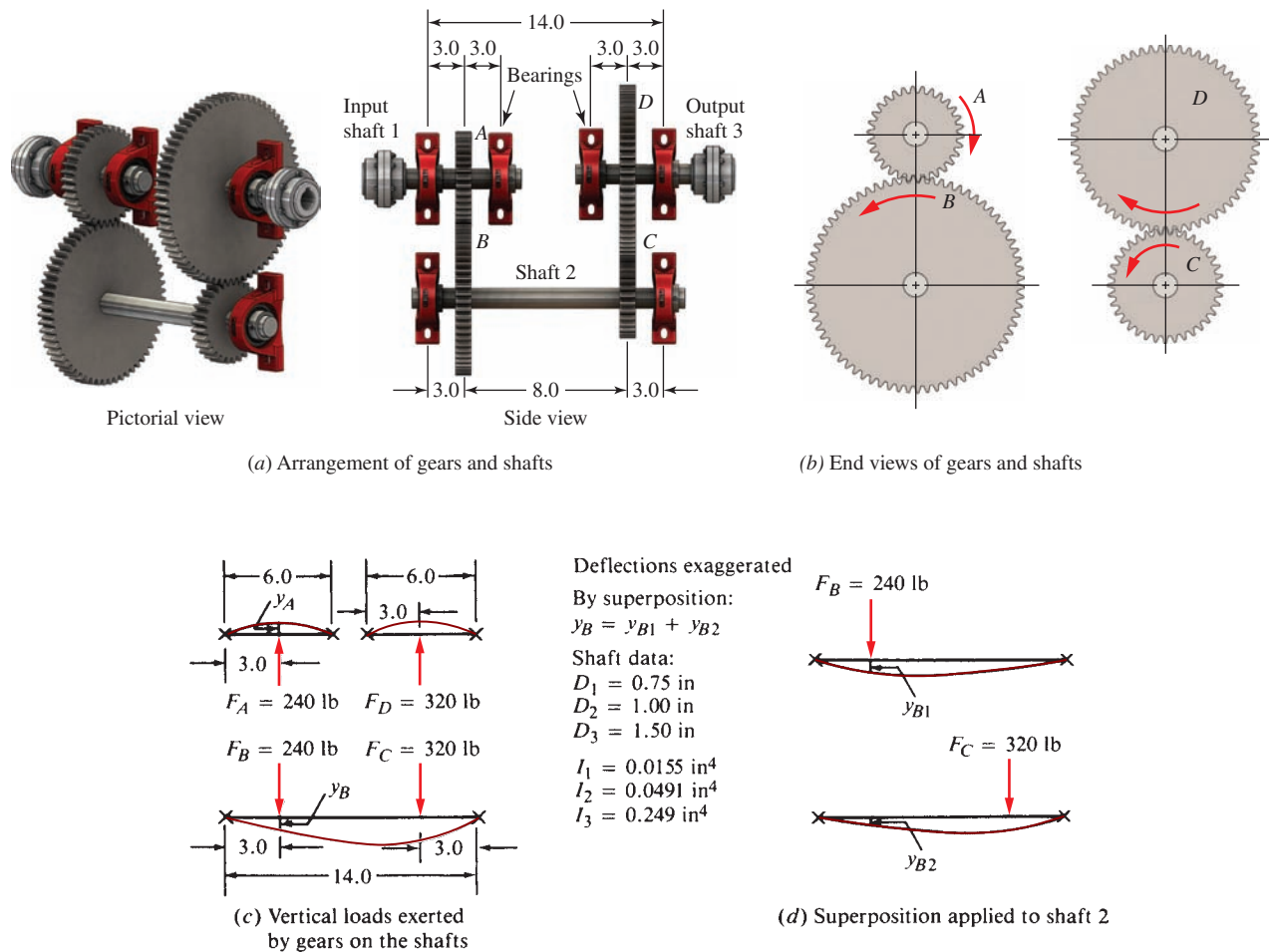


FIGURE 3-23 Shaft deflection analysis for a double-reduction speed reducer

For many additional cases, superposition is useful if the actual loading can be divided into parts that can be computed by available formulas. The deflection for each loading is computed separately, and then the individual deflections are summed at the points of interest.

Many commercially available computer software programs allow the modeling of beams having rather

complex loading patterns and varying geometry. The results include reaction forces, shearing force and bending moment diagrams, and deflections at any point. It is important that you understand the principles of beam deflection, studied in strength of materials and reviewed here, so that you can apply such programs accurately and interpret the results carefully.

Example Problem 3-16

For the two gears, *A* and *B*, in Figure 3-23, compute the relative deflection between them in the plane of the paper that is due to the forces shown in Figure 3-23 (c). These *separating forces*, or *normal forces*, are discussed in Chapters 9 and 10. It is customary to consider the loads at the gears and the reactions at the bearings to be concentrated. The shafts carrying the gears are steel and have uniform diameters as listed in the figure.

Solution

Objective Compute the relative deflection between gears *A* and *B* in Figure 3-23.

Given The layout and loading pattern are shown in Figure 3-23. The separating force between gears *A* and *B* is 240 lb. Gear *A* pushes downward on gear *B*, and the reaction force of gear *B* pushes upward on gear *A*. Shaft 1 has a diameter of 0.75 in and a moment of inertia of 0.0155 in⁴. Shaft 2 has a diameter of 1.00 in and a moment of inertia of 0.0491 in⁴. Both shafts are steel. Use $E = 30 \times 10^6$ psi.

Analysis Use the deflection formulas from Appendix 14 to compute the upward deflection of shaft 1 at gear *A* and the downward deflection of shaft 2 at gear *B*. The sum of the two deflections is the total deflection of gear *A* with respect to gear *B*.

Case (a) from Table A14–1 applies to Shaft 1 because there is a single concentrated force acting at the midpoint of the shaft between the supporting bearings. We will call that deflection y_A .

Shaft 2 is a simply supported beam carrying two nonsymmetrical loads. No single formula from Appendix 14 matches that loading pattern. But we can use superposition to compute the deflection of the shaft at gear B by considering the two forces separately as shown in Part (d) of Figure 3–23. Case (b) from Table A14–1 is used for each load.

We first compute the deflection at B due only to the 240-lb force, calling it y_{B1} . Then we compute the deflection at B due to the 320-lb force, calling it y_{B2} . The total deflection at B is $y_B = y_{B1} + y_{B2}$.

Results The deflection of shaft 1 at gear A is

$$y_A = \frac{F_A L_1^3}{48 EI} = \frac{(240)(6.0)^3}{48(30 \times 10^6)(0.0155)} = 0.0023 \text{ in}$$

The deflection of shaft 2 at B due only to the 240-lb force is

$$y_{B1} = -\frac{F_B a^2 b^2}{3 E I_2 L_2} = -\frac{(240)(3.0)^2(11.0)^2}{3(30 \times 10^6)(0.0491)(14)} = -0.0042 \text{ in}$$

The deflection of shaft 2 at B due only to the 320-lb force at C is

$$y_{B2} = -\frac{F_C b x}{6 E I_2 L_2} (L_2^2 - b^2 - x^2)$$

$$y_{B2} = -\frac{(320)(3.0)(3.0)}{6(30 \times 10^6)(0.0491)(14)} [(14)^2 - (3.0)^2 - (3.0)^2]$$

$$y_{B2} = -0.0041 \text{ in}$$

Then the total deflection at gear B is

$$y_B = y_{B1} + y_{B2} = -0.0042 - 0.0041 = -0.0083 \text{ in}$$

Because shaft 1 deflects upward and shaft 2 deflects downward, the total relative deflection is the sum of y_A and y_B :

$$y_{\text{total}} = y_A + y_B = 0.0023 + 0.0083 = 0.0106 \text{ in}$$

Comment This deflection is very large for this application. How could the deflection be reduced?

3-19 EQUATIONS FOR DEFLECTED BEAM SHAPE

The general principles relating the deflection of a beam to the loading on the beam and its manner of support are presented here. The result will be a set of relationships among the load, the vertical shearing force, the bending moment, the slope of the deflected beam shape, and the actual deflection curve for the beam. Figure 3–17 shows diagrams for these five factors, with θ as the slope and y indicating deflection of the beam from its initial straight position. The product of modulus of elasticity and the moment of inertia, EI , for the beam is a measure of its stiffness or resistance to bending deflection. It is convenient to combine EI with the slope and deflection values to maintain a proper relationship, as discussed next.

One fundamental concept for beams in bending is

$$\frac{M}{EI} = \frac{d^2 y}{dx^2}$$

where M = bending moment

x = position on the beam measured along its length

y = deflection

Thus, if it is desired to create an equation of the form $y = f(x)$ (i.e., y as a function of x), it would be related to the other factors as follows:

$$y = f(x)$$

$$\theta = \frac{dy}{dx}$$

$$\frac{M}{EI} = \frac{d^2 y}{dx^2}$$

$$\frac{V}{EI} = \frac{d^3 y}{dx^3}$$

$$\frac{w}{EI} = \frac{d^4 y}{dx^4}$$

where w = general term for the load distribution on the beam

The last two equations follow from the observation that there is a derivative (slope) relationship between shear and bending moment and between load and shear.

In practice, the fundamental equations just given are used in reverse. That is, the load distribution as a function of x is known, and the equations for the other

factors are derived by successive integrations. The results are

$$\begin{aligned}w &= f(x) \\V &= \int w \, dx + V_0 \\M &= \int V \, dx + M_0\end{aligned}$$

where V_0 and M_0 = constants of integration evaluated from the boundary conditions.

In many cases, the load, shear, and bending moment diagrams can be drawn in the conventional manner, and the equations for shear or bending moment can be created directly by the principles of analytic geometry. With M as a function of x , the slope and deflection relations can be found:

$$\begin{aligned}\theta EI &= \int M \, dx + C_1 \\y EI &= \int \theta EI \, dx + C_2\end{aligned}$$

The constants of integration must be evaluated from boundary conditions. Texts on strength of materials show the details. (See Reference 4.)

3-20 CURVED BEAMS

The analysis of stress due to bending in beams summarized earlier in this chapter is limited to beams that are straight or very nearly so. It was noted that when a beam, that is originally straight, is bent:

1. The radius of curvature of the neutral axis of the beam is very large.
2. Plane sections perpendicular to the axis of the beam remain plane as the beam bends and they rotate in opposite directions under the influence of the bending moment placing part of the section in compression and part in tension.
3. The location of the neutral axis of the cross section of the beam is:
 - a. Where the bending stress is zero.
 - b. Coincident with the centroidal axis of the cross section of the beam.
4. The maximum tensile and compressive stresses occur at the outermost fibers of the beam.
5. There is a linear variation of the bending stress from the place of maximum compression to the place of maximum tension.
6. Under these conditions, the maximum bending stress is computed from the familiar flexure formula:
 - a. $\sigma = Mc/I$.
 - b. where M is the bending moment at the section; c is the distance from the neutral axis to the outermost fiber; and I is the moment of inertia of the cross section with respect to its centroidal axis.

The flexure formula applies reasonably well when the ratio of the radius of curvature, R , to the depth of the cross section is larger than 10. That is $R/h > 10$ (Reference 6).

When a component has a smaller radius of curvature, the maximum stress due to bending is significantly higher than that which would be predicted from the flexure formula and different analysis procedures apply. There are many practical cases where smaller radii of curvature for beams occur. Examples include:

- Crane hooks used for lifting loads.
- Clamps such as the C-clamp.
- Open, C-section frames for machinery such as punch presses.
- Building beams that are curved for architectural and aesthetic reasons.
- Parts of hand tools such as saws, pliers, and wire cutters.
- Structural or functional members of machinery where the curvature of the member allows clearance of obstructions.
- Parts of some types of furniture such as tables and chairs.
- Automotive frames, suspension elements, and actuation arms for window lifts and seats.

Major features and the general nature of the results of the analysis for curved beams include:

1. When the applied bending moment tends to *straighten the beam* by increasing the radius of curvature, it is considered to be *positive*.
2. When the applied bending moment tends to decrease the radius of curvature, it is considered to be *negative*.
3. The stress distribution within the cross section of the member is *not* linear; rather it is hyperbolic. Figure 3-24 shows the general pattern of the stress distribution for the case of a curved beam with a rectangular cross section when a positive bending moment is applied. Note the following:
 - a. The inside surface is placed in tension and the maximum tensile stress occurs on that surface.
 - b. The outside surface is placed in compression and the maximum compressive stress occurs at that surface.
 - c. The place within the cross section where the stress becomes zero is called the *neutral axis* as it is for straight beams. However, the neutral axis is *not* coincident with the centroidal axis of the shape of the cross section. For the case shown in Figure 3-24, it is displaced toward the center of curvature by an amount called e .

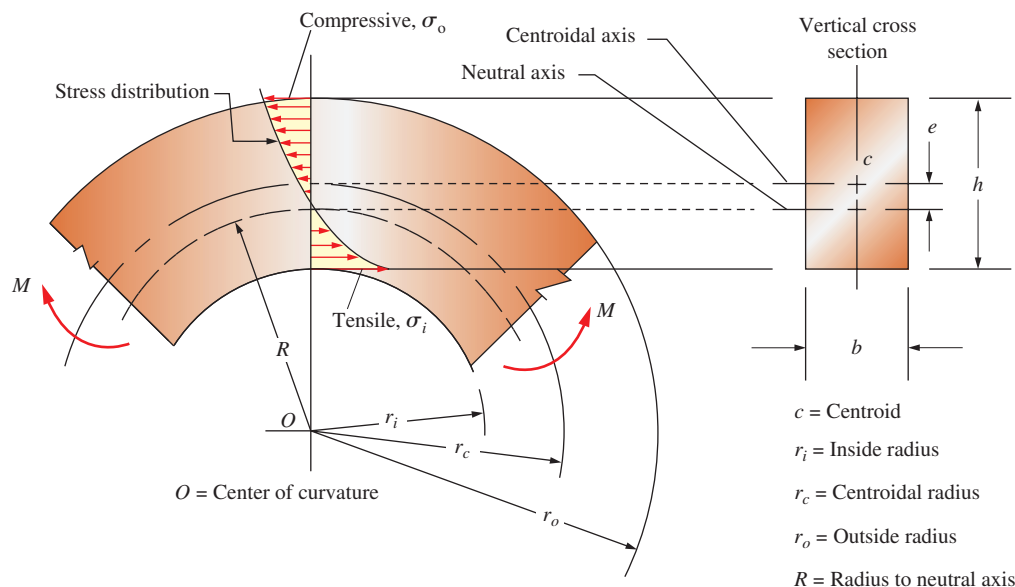


FIGURE 3-24 Segment of a curved beam carrying a positive bending moment

4. The *curved beam formula* given below for the stress distribution has been derived in References 1, 2, and 6.

$$\sigma = \frac{M(R - r)}{Ar(r_c - R)} \quad (3-26)$$

where:

A = cross-sectional area of the shape

M = bending moment applied

The three radii are measured from the center of curvature of the beam:

r to the place where the stress is to be computed

r_c to the centroid of the area

R to the *neutral axis* of the cross section

R depends on the shape of the cross section using the *area shape factor*, ASF , found from the relations shown in Figure 3-25. Then compute:

$$R = A/ASF \quad (3-27)$$

The quantity $(r_c - R)$ in the denominator of Equation (3-26) is the radial distance from the centroidal axis of the cross section to the neutral axis. This value, called e , is typically a very small number, requiring that it be computed with high accuracy, at least three significant figures.

General Procedure for Analyzing Curved Beams Carrying a Pure Bending Moment

This procedure is used to compute the maximum tensile and compressive stresses for a curved beam at the inside (at r_i) and outside (at r_o) surfaces. The results are then compared to determine which value is the true maximum stress.

1. Determine the value of the applied bending moment, M , including its sign.

2. For the cross-sectional area of the beam:

a. Compute the total area, A .

b. Compute the location of the centroid of the area.

c. Determine the value of four radii from the center of curvature of the beam; r_i , r_o , r_c , and R , using Equation (3-27).

d. Compute the area shape factor, ASF . See Figure 3-25.

3. Compute the stress at the outside surface from:

$$\sigma_o = \frac{M(R - r_o)}{Ar_o(r_c - R)} \quad (3-28)$$

4. Compute the stress at the inside surface from:

$$\sigma_i = \frac{M(R - r_i)}{Ar_i(r_c - R)} \quad (3-29)$$

5. Compare σ_o and σ_i to determine the maximum value.

Cross Sections Comprised Two or More Shapes.

Designers often work toward optimizing the shape of the cross section of curved beams to more nearly balance the maximum tensile and compressive stresses when the material is isotropic. Similarly, for materials like cast irons having different strengths in compression and tension, the goal is to achieve a nearly equal design factor for both the tensile and compressive parts of the section. The cross-sectional shape of the curved beam may also be modified to optimize the casting or machining processes or to make the shape more pleasing. Adjusting the shape and dimensions of the cross section of the curved beam can accomplish these goals. Reaching these goals typically results in a cross section that is a composite of two or more of the standard shapes shown in Figure 3-25. Examples of the composite shapes are shown in Figure 3-27.

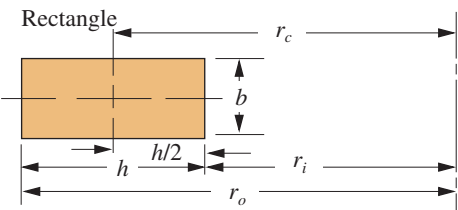
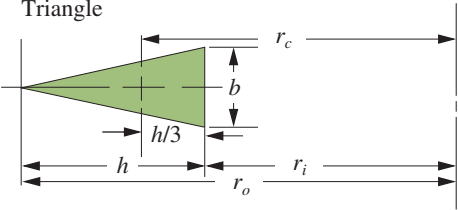
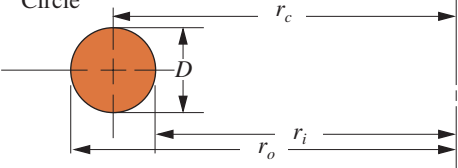
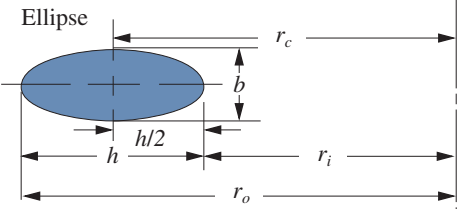
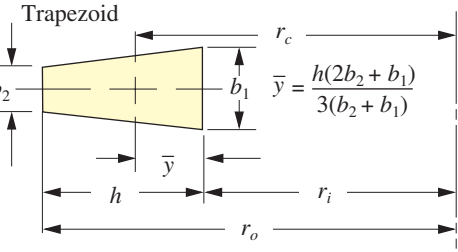
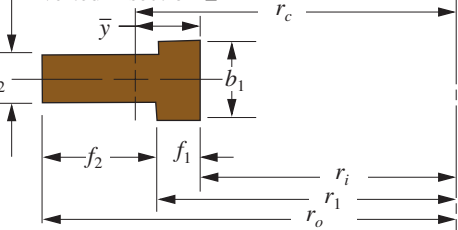
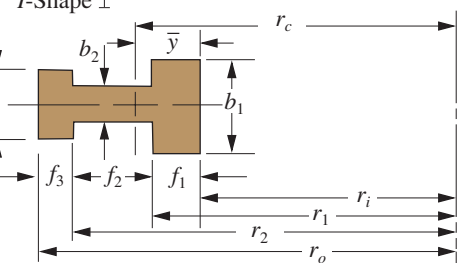
Shape of cross section	Area, A	Area shape factor, ASF
 Rectangle	$bh = b(r_o - r_i)$	$b \ln(r_o / r_i)$
 Triangle	$\frac{1}{2} b(r_o - r_i)$	$\frac{b r_o}{(r_o - r_i)} [\ln(r_o / r_i)] - b$
 Circle	$\pi D^2 / 4$	$2 \pi [r_c - \sqrt{r_c^2 - D^2 / 4}]$
 Ellipse	$\pi h b / 4$	$(2 \pi b / h) [r_c - \sqrt{r_c^2 - h^2 / 4}]$
 Trapezoid	$\frac{1}{2} (r_o - r_i)(b_1 + b_2)$	$\frac{b_1 r_o - b_2 r_i}{(r_o - r_i)} \ln(r_o / r_i) - (b_1 - b_2)$
 Inverted T-section $\perp$	$b_1 f_1 + b_2 f_2$ $\bar{y} = \frac{1}{A} [(b_1 f_1)(f_1 / 2) + (b_2 f_2)(f_1 + f_2 / 2)]$	$b_1 \ln(r_i / r_i) + b_2 \ln(r_o / r_i)$
 I-Shape $\perp$	$b_1 f_1 + b_2 f_2 + b_3 f_3$ $\bar{y} = \frac{1}{A} [(b_1 f_1)(f_1 / 2) + (b_2 f_2)(f_1 + f_2 / 2) + (b_3 f_3)(f_1 + f_2 + f_3 / 2)]$	$b_1 \ln(r_i / r_i) + b_2 \ln(r_2 / r_1) + b_3 \ln(r_o / r_2)$

FIGURE 3-25 Area shape factors for selected cross sections of curved bars

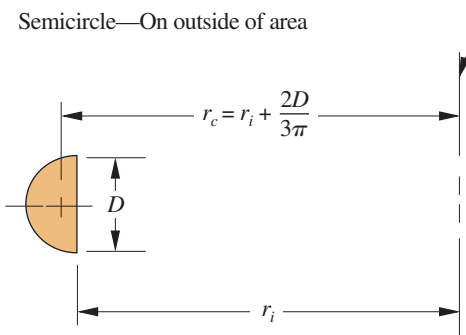
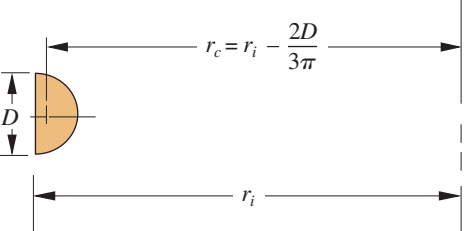
Shape of cross section	Area, A	Area shape factor, ASF
Semicircle—On outside of area 	$\pi D^2/8$	For $D/2 > r_i$ $r_i \pi - D + 2 \sqrt{D^2/4 - r_i^2} \ln \left[\frac{D/2 + \sqrt{D^2/4 - r_i^2}}{r_i} \right]$ For $r_i > D/2$ $r_i \pi - D - \pi \sqrt{r_i^2 - D^2/4} + 2 \sqrt{r_i^2 - D^2/4} \sin^{-1}(D/2r_i)$ [Note: Argument of inverse sine is in radians]
Semicircle—On inside of area 	$\pi D^2/8$	$r_i \pi + D - \pi \sqrt{r_i^2 - D^2/4} - 2 \sqrt{r_i^2 - D^2/4} \sin^{-1}(D/r_i)$ [Note: Argument of inverse sine is in radians]

FIGURE 3-25 Area shape factors for selected cross sections of curved bars (*Continued*)**Example Problem 3-17**

A curved bar has a rectangular cross section 15.0 mm thick by 25.0 mm deep as shown in Figure 3-26. The bar is bent into a circular arc producing an inside radius of 25.0 mm. For an applied bending moment of +400 N·m, compute the maximum tensile and compressive stresses in the bar.

Solution

Objective Compute the maximum tensile and compressive stresses.

Given $M = +400 \text{ N} \cdot \text{m}$ that tends to straighten the bar. $r_i = 25.0 \text{ mm}$.

Analysis Apply Equations (3-28) and (3-29).

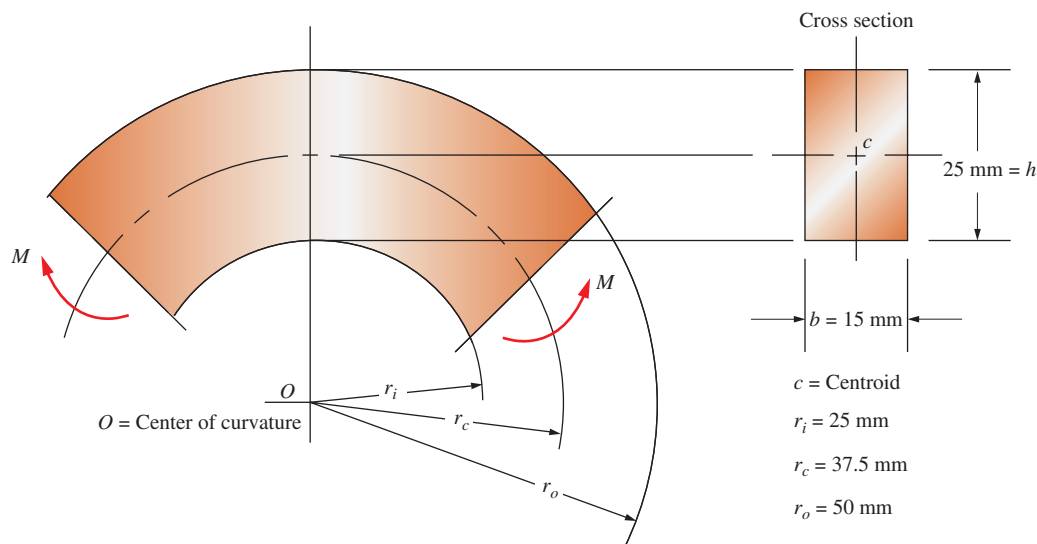


FIGURE 3-26 Curved bar with a rectangular cross section for Example Problem 3-17

Results First compute the cross-sectional area:

$$A = bh = (15.0 \text{ mm})(25.0 \text{ mm}) = 375.0 \text{ mm}^2$$

Now compute the quantities involving radii. See Figure 3–26 for related dimensions:

$$r_o = r_i + h = 25.0 \text{ mm} + 25.0 \text{ mm} = 50.0 \text{ mm}$$

$$r_c = r_i + h/2 = 25.0 \text{ mm} + (25.0/2) \text{ mm} = 37.5 \text{ mm}$$

$$R = A/ASF$$

From Figure 3–25, for a rectangular cross section:

$$ASF = b \cdot \ln(r_o/r_i) = (15 \text{ mm})[\ln(50.0/25.0)] = 10.3972 \text{ mm}$$

Then, $R = A/ASF = (375 \text{ mm}^2)/10.3972 \text{ mm} = 36.0674 \text{ mm}$

Quantities needed in the stress equations include:

$$r_c - R = 37.5 \text{ mm} - 36.0674 \text{ mm} = 1.4326 \text{ mm}$$

This is the distance e as shown in Figure 3–24.

$$R - r_o = 36.0674 \text{ mm} - 50.0 \text{ mm} = -13.9326 \text{ mm}$$

$$R - r_i = 36.0674 \text{ mm} - 25.0 \text{ mm} = 11.0674 \text{ mm}$$

This is the distance from the inside surface to the neutral axis.

Stress at outer surface, using Equation (3–28):

$$\sigma_o = \frac{M(R - r_o)}{Ar_o(r_c - R)} = \frac{(400 \text{ N} \cdot \text{m})(-13.9326 \text{ mm})[1000 \text{ mm/m}]}{(375 \text{ mm}^2)(50.0 \text{ mm})(1.4326 \text{ mm})}$$

$$\sigma_o = -207.5 \text{ N/mm}^2 = -207.5 \text{ MPa}$$

This is the maximum compressive stress in the bar.

Stress at inner surface, using Equation (3–29):

$$\sigma_i = \frac{M(R - r_i)}{Ar_i(r_c - R)} = \frac{(400 \text{ N} \cdot \text{m})(11.0674 \text{ mm})[1000 \text{ mm/m}]}{(375 \text{ mm}^2)(25.0 \text{ mm})(1.4326 \text{ mm})}$$

$$\sigma_i = 329.6 \text{ N/mm}^2 = 329.6 \text{ MPa}$$

This is the maximum tensile stress in the bar.

Comment The stress distribution between the outside and the inside is similar to that shown in Figure 3–24.

The analysis process introduced earlier can be modified to consider a composite shape for the cross section of the curved beam.

Procedure for Analyzing Curved Beams with Composite Cross-sectional Shapes Carrying a Pure Bending Moment

This procedure is used to compute the maximum tensile and compressive stresses for a curved beam at the inside (at r_i) and outside (at r_o) surfaces. The results are then compared to determine which value is the true maximum stress.

1. Determine the value of the applied bending moment, M , including its sign.
2. For the composite cross-sectional area of the beam:
 - a. Determine the inside radius, r_i , and the outside radius, r_o .
 - b. Separate the composite area into two or more parts that are shapes from Figure 3–25.
 - c. Compute the area of each component part, A_i , and the total area, A .

- d. Locate of the centroid of each component area.
- e. Compute the radius of the centroid of the composite area, r_c .
- f. Compute the value of the area shape factor, ASF_i , for each component area using the equations in Figure 3–25.
- g. Compute the radius, R , from the center of curvature to the neutral axis from:

$$R = A/\Sigma(ASF_i)$$

3. Compute the stress at the outside surface from:

$$\sigma_o = \frac{M(R - r_o)}{Ar_o(r_c - R)} \quad (3-30)$$

4. Compute the stress at the inside surface from:

$$\sigma_i = \frac{M(R - r_i)}{Ar_i(r_c - R)} \quad (3-31)$$

5. Compare σ_o and σ_i to determine the maximum value.

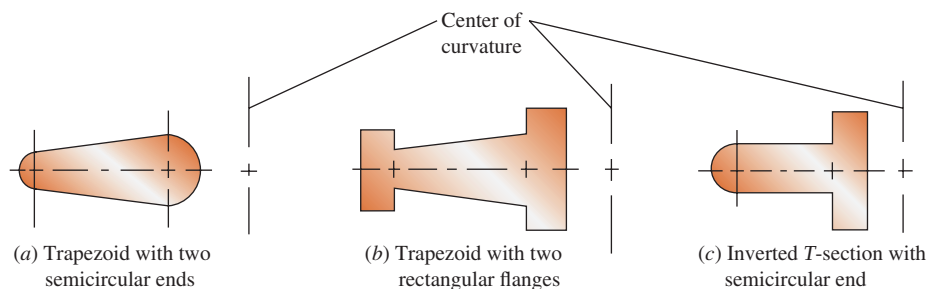


FIGURE 3-27 Examples of composite shapes for cross sections of curved beams

Example Problem 3-18

A curved beam has the shape shown in Figure 3-28 and is subjected to a pure bending moment of $-640 \text{ lb}\cdot\text{in}$. The inside radius is 2.90 in . Compute the maximum tensile and compressive stresses in the beam.

Solution

Objective Compute the maximum tensile and compressive stresses.

Given $M = -640 \text{ lb}\cdot\text{in}$ that tends to decrease the radius of curvature.
 $r_i = 2.90 \text{ in}$ (see Figure 3-28 for related dimensions).

Analysis Apply Equations (3-28) and (3-29)

Results Separate the shape into a composite of an inverted T-section 1 and a semicircular area 2. Compute cross-sectional area for each part:

$$A_1 = b_1 f_1 + b_2 f_2 = (0.80 \text{ in})(0.20 \text{ in}) + (1.0 \text{ in})(0.40 \text{ in})$$

$$A_1 = 0.560 \text{ in}^2$$

$$A_2 = \pi D^2/8 = \pi(0.40 \text{ in})^2/8 = 0.06283 \text{ in}^2$$

$$\text{Total area} = A = A_1 + A_2 = 0.560 + 0.06283 = 0.62283 \text{ in}^2$$

Now locate the centroid of each area with respect to the inside surface; from Figure 3-25:

$$\bar{y}_1 = [1/A_1][(b_1 f_1)(f_1/2) + (b_2 f_2)(f_1 + f_2/2)]$$

$$\bar{y}_1 = [1/0.56 \text{ in}^2][(0.80)(0.20)(0.10) + (1.0)(0.40)(0.70)] \text{ in}^3$$

$$\bar{y}_1 = 0.5286 \text{ in}$$

$$\bar{y}_2 = 1.20 \text{ in} + 2D/3\pi = 1.20 \text{ in} + (2)(0.40 \text{ in})/(3)(\pi)$$

$$\bar{y}_2 = 1.2849 \text{ in}$$

Now locate the centroid of the composite area:

$$\bar{y}_c = [1/A][A_1 \bar{y}_1 + A_2 \bar{y}_2]$$

$$\bar{y}_c = [1/0.62283 \text{ in}^2][(0.560)(0.5286) + (0.06283)(1.2849)] \text{ in}^3$$

$$\bar{y}_c = 0.60489 \text{ in}$$

Now define the pertinent radii from the center of curvature:

$$r_o = r_i + 1.40 \text{ in} = 2.90 \text{ in} + 1.40 \text{ in} = 4.30 \text{ in}$$

$$r_c = r_i + \bar{y}_c = 2.90 \text{ in} + 0.60489 \text{ in} = 3.5049 \text{ in}$$

Compute the value of the area shape factor, ASF_i , for each part:

For the T-shape:

$$ASF_1 = b_1 \ln(r_i/r_i) + b_2 \ln(r_o/r_i)$$

$$ASF_1 = (0.80 \text{ in}) \ln(3.1/2.9) + (0.40 \text{ in}) \ln(4.1/3.1)$$

$$ASF_1 = 0.16519 \text{ in}$$

For the semicircular area, we must determine the relationship between r_{i2} and $D/2$:

$$r_{i2} = 4.10 \text{ in at the base of the semicircle}$$

$$D/2 = (0.40 \text{ in})/2 = 0.20 \text{ in}$$

Because $r_{i2} > D/2$, we use the equation,

$$ASF_2 = r_i \pi - D - \pi \sqrt{r_{i2}^2 - D^2/4} + 2 \sqrt{r_{i2}^2 - D^2/4} [\sin^{-1}(D/2r_{i2})]$$

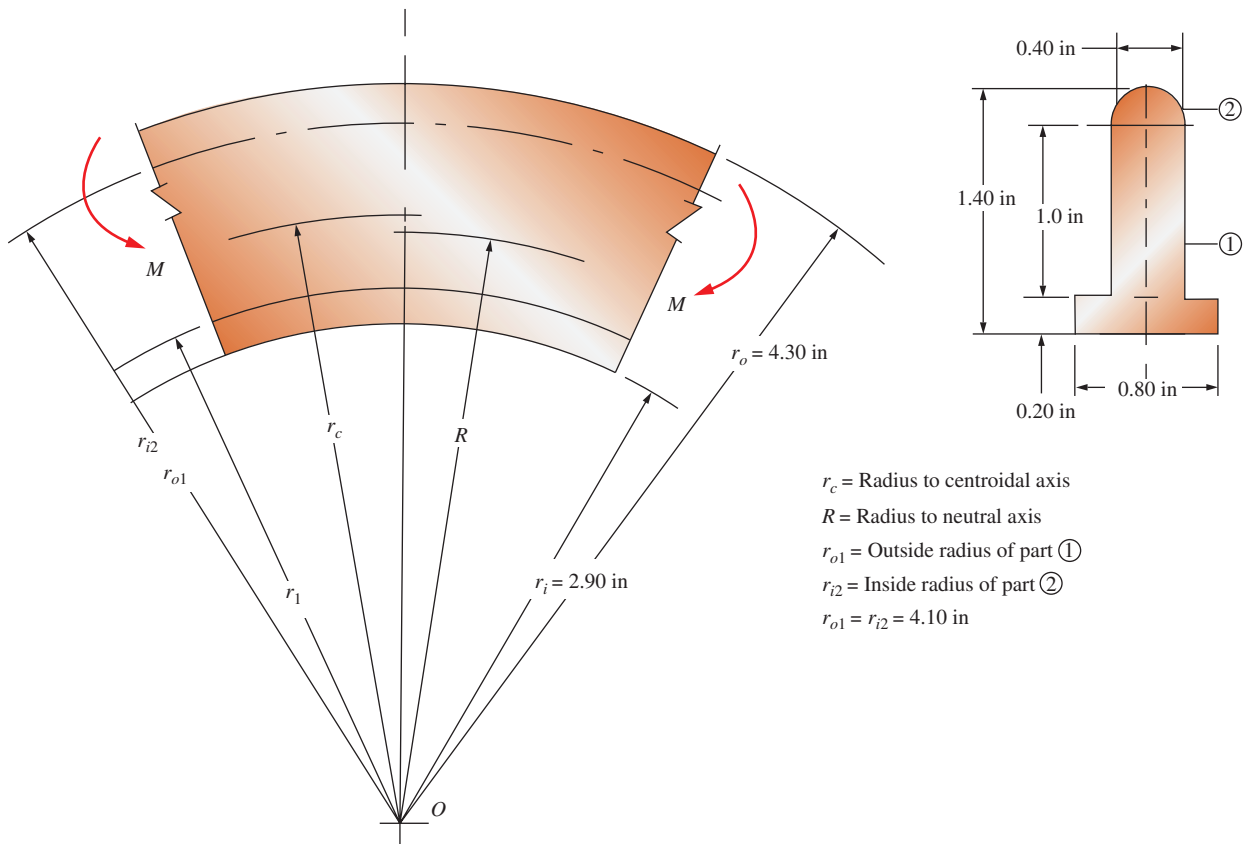


FIGURE 3-28 Curved beam with a composite shape carrying a negative bending moment for Example Problem 3-18

[Note: Argument of inverse sine is in radians]

$$ASF_2 = (4.10)\pi - 0.40 - \pi\sqrt{(4.1)^2 - (0.40)^2}/4 + 2\sqrt{(4.1)^2 - (0.40)^2}/4 [\sin^{-1}(0.40)/(2)(4.1)] = 0.015016 \text{ in}$$

Compute the radius, R , from the center of curvature to the neutral axis from:

$$R = A/\Sigma(ASF_i) = 0.62283 \text{ in}^2/(0.16519 + 0.015016) \text{ in}$$

$$R = 3.4563 \text{ in}$$

Now compute:

$$r_c - R = 3.5049 \text{ in} - 3.4563 \text{ in} = 0.0486 \text{ in} = e$$

$$R - r_o = 3.4563 \text{ in} - 4.30 \text{ in} = -0.8437 \text{ in}$$

$$R - r_i = 3.4563 \text{ in} - 2.90 \text{ in} = 0.5563 \text{ in}$$

This locates the neutral axis from the inner surface.

Now compute the stress at outer surface, using Equation (3-28):

$$\sigma_o = \frac{M(R - r_o)}{Ar_o(r_c - R)} = \frac{(-640 \text{ lb} \cdot \text{in})(-0.8437 \text{ in})}{(0.62283 \text{ in}^2)(4.30 \text{ in})(0.0486 \text{ in})}$$

$$\sigma_o = 4149 \text{ lb/in}^2 = 4149 \text{ psi}$$

This is the maximum tensile stress in the bar.

Stress at inner surface, using Equation (3-29):

$$\sigma_i = \frac{M(R - r_i)}{Ar_i(r_c - R)} = \frac{(-640 \text{ lb} \cdot \text{in})(0.5563 \text{ in})}{(0.62283 \text{ in}^2)(2.90 \text{ in})(0.0486 \text{ in})}$$

$$\sigma_i = -4056 \text{ lb/in}^2 = -4056 \text{ psi}$$

This is the maximum compressive stress in the bar.

Comment This problem demonstrated the process for analyzing a composite section. Note that the maximum tensile and compressive stresses are very nearly equal for this design, a desirable condition for homogeneous, isotropic materials.

Curved Beams with Combined Bending Moment and Normal Load. Each of the example curved beams considered thus far carried only a moment that tended to either increase or decrease the radius of curvature for the beam. Only the bending stress was computed. However, many curved beams carry a combination of a moment and a normal load acting perpendicular to the cross section of the beam. Examples are a crane hook and a C-clamp.

The complete stress analysis requires that the normal stress, $\sigma = P/A$, be added to the bending stress computed from Equations (3-28) and (3-29). This process is discussed in Section 3-21, “Superposition Principle.” Apply Equation (3-30) as illustrated in Example Problem 3-19.

Alternate Approaches to Analyzing Stresses in Curved Beams. An alternate approach to analyzing curved beams is reported in Reference 5. Using the concept of stress concentration factors, data are reported for an equivalent K_t value for curved beams having five different shapes: Circular, elliptical, hollow circular, rectangular, and I-shapes for one specific set of proportions. The controlling parameter is r_c/c , where r_c is the radius to the centroidal axis of the shape and c is one-half of the total height of each section. Then K_t is used in the same manner that is discussed in Section 3-22. That is,

$$K_t = \frac{\sigma_{\max}}{\sigma_{\text{nom}}} \quad \text{or} \quad \sigma_{\max} = K_t \sigma_{\text{nom}} = K_t (Mc/I)$$

Note that this is *not* strictly a stress concentration in the sense that there is no geometric discontinuity or abrupt change in shape or size of the cross section. However, the approach gives a reasonable approximation of the maximum stress in the curved beam.

Reference 7 presents a similar approach to the analysis of curved beams but data are presented in tabular form. Again, the primary parameter is r_c/c which is used to compute the eccentricity, e , the radial distance from the centroidal axis to the true neutral axis. Then two factors

are determined from the table: k_o for the stress at the outside surface and k_i for the stress at the inside surface. Then,

$$\begin{aligned}\sigma_{o-\max} &= k_o \sigma_{\text{nom}} = k_o (Mc/I) \quad \text{and} \\ \sigma_{i-\max} &= k_i \sigma_{\text{nom}} = k_i (Mc/I)\end{aligned}$$

3-21 SUPERPOSITION PRINCIPLE

When the same cross section of a load-carrying member is subjected to both a direct tensile or compressive stress and a stress due to bending, the resulting normal stress can be computed by the method of superposition. The formula is

$$\sigma = \pm Mc/I \pm F/A \quad (3-32)$$

where *tensile stresses are positive and compressive stresses are negative.*

An example of a load-carrying member subjected to combined bending and axial tension is shown in Figure 3-29. It shows a beam subjected to a load applied downward and to the right through a bracket below the beam. Resolving the load into horizontal and vertical components shows that its effect can be broken into three parts:

1. The vertical component tends to place the beam in bending with tension on the top and compression on the bottom.
2. The horizontal component, because it acts away from the neutral axis of the beam, causes bending with tension on the bottom and compression on the top.
3. The horizontal component causes direct tensile stress across the entire cross section.

We can proceed with the stress analysis by using the techniques from the previous section to prepare the shearing force and bending moment diagrams and then using Equation (3-30) to combine the effects of the bending stress and the direct tensile stress at any point. The details are shown within Example Problem 3-19.

Example Problem 3-19

The cantilever beam in Figure 3-29 is a steel American Standard beam, S6×12.5. The force F is 10 000 lb, and it acts at an angle of 30° below the horizontal, as shown. Use $a = 24$ in and $e = 6.0$ in. Draw the free-body diagram and the shearing force and bending moment diagrams for the beam. Then compute the maximum tensile and maximum compressive stresses in the beam and show where they occur.

Solution

Objective Determine the maximum tensile and compressive stresses in the beam.

Given The layout from Figure 3-29(a). Force = $F = 10\,000$ lb; angle $\theta = 30^\circ$.
The beam shape: S6×12.5; length = $a = 24$ in.
Section modulus = $S = 7.37$ in³; area = $A = 3.67$ in² (Appendix 15-10).

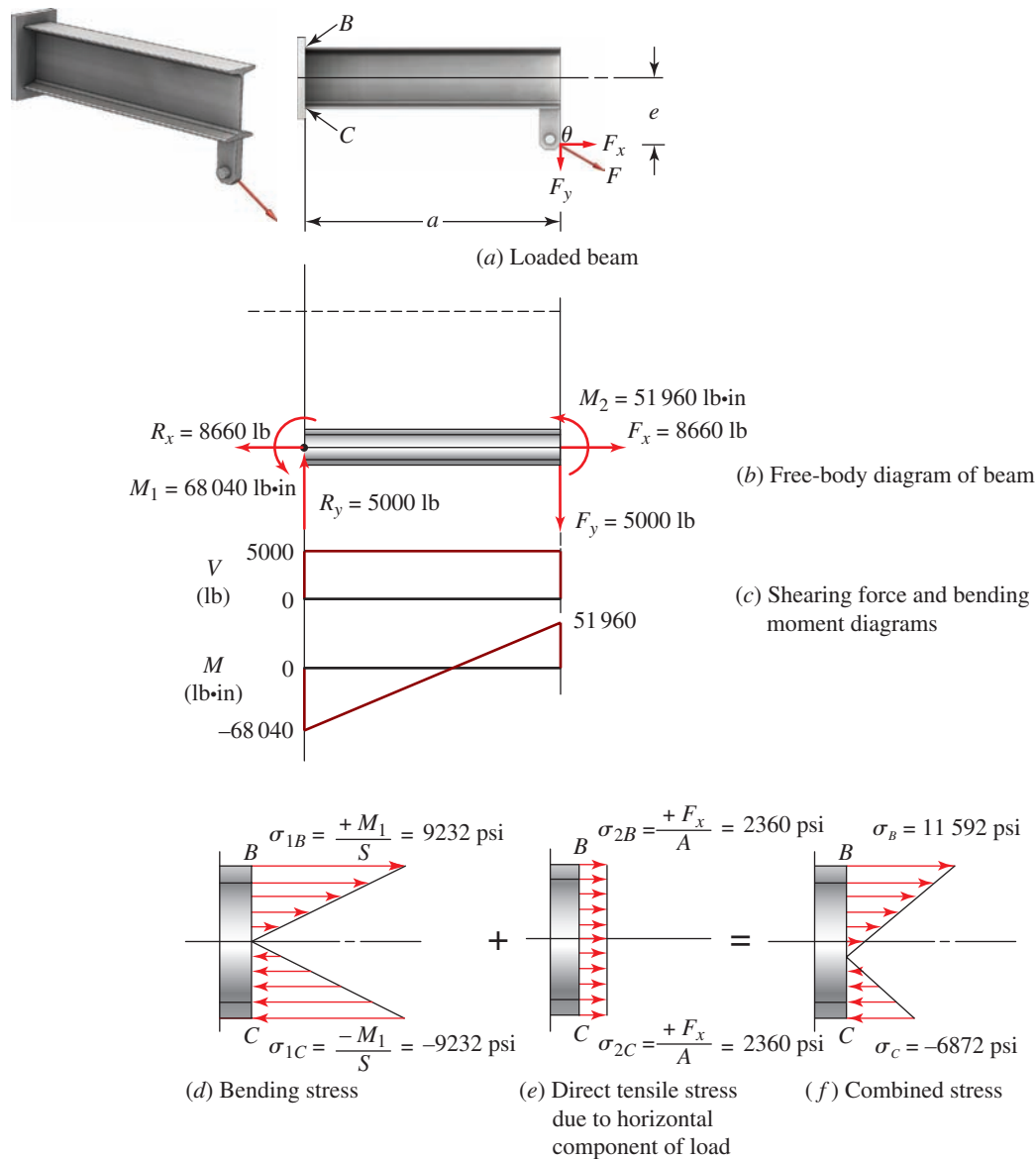


FIGURE 3–29 Beam subjected to combined stresses

Eccentricity of the load = $e = 6.0$ in from the neutral axis of the beam to the line of action of the horizontal component of the applied load.

Analysis The analysis takes the following steps:

1. Resolve the applied force into its vertical and horizontal components.
2. Transfer the horizontal component to an equivalent loading at the neutral axis having a direct tensile force and a moment due to the eccentric placement of the force.
3. Prepare the free-body diagram using the techniques from Section 3–16.
4. Draw the shearing force and bending moment diagrams and determine where the maximum bending moment occurs.
5. Complete the stress analysis at that section, computing both the maximum tensile and maximum compressive stresses.

Results The components of the applied force are:

$$F_x = F \cos(30^\circ) = (10\,000 \text{ lb})[\cos(30^\circ)] = 8660 \text{ lb acting to the right}$$

$$F_y = F \sin(30^\circ) = (10\,000 \text{ lb})[\sin(30^\circ)] = 5000 \text{ lb acting downward}$$

The horizontal force produces a counterclockwise concentrated moment at the right end of the beam with a magnitude of:

$$M_2 = F_x(6.0 \text{ in}) = (8660 \text{ lb})(6.0 \text{ in}) = 51\,960 \text{ lb} \cdot \text{in}$$

The free-body diagram of the beam is shown in Figure 3–29(b).

Figure 3–29(c) shows the shearing force and bending moment diagrams.

The maximum bending moment, 68 040 lb in, occurs at the left end of the beam where it is attached firmly to a column.

The bending moment, taken alone, produces a tensile stress (+) on the top surface at point *B* and a compressive stress (–) on the bottom surface at *C*. The magnitudes of these stresses are:

$$\sigma_1 = \pm M_1/S = \pm (68\,040 \text{ lb in})/(7.37 \text{ in}^3) = \pm 9232 \text{ psi}$$

Figure 3–29(d) shows the stress distribution due only to the bending stress.

Now we compute the tensile stress due to the axial force of 8660 lb.

$$\sigma_2 = F_x/A = (8660 \text{ lb})/(3.67 \text{ in}^2) = 2360 \text{ psi}$$

Figure 3–29(e) shows this stress distribution, uniform across the entire section.

Next, let's compute the combined stress at *B* on the top of the beam.

$$\sigma_B = +\sigma_1 + \sigma_2 = 9232 \text{ psi} + 2360 \text{ psi} = 11\,592 \text{ psi—Tensile}$$

At *C* on the bottom of the beam, the stress is:

$$\sigma_C = -\sigma_1 + \sigma_2 = -9232 \text{ psi} + 2360 \text{ psi} = -6872 \text{ psi—Compressive}$$

Figure 3–29(f) shows the combined stress condition that exists on the cross section of the beam at its left end at the support. It is a superposition of the component stresses shown in Figure 3–29(d) and (e).

3-22 STRESS CONCENTRATIONS

The formulas reviewed earlier for computing simple stresses due to direct tensile and compressive forces, bending moments, and torsional moments are applicable under certain conditions. One condition is that the geometry of the member is uniform throughout the section of interest.

In many typical machine design situations, inherent geometric discontinuities are necessary for the parts to perform their desired functions. For example, as shown in Figure 12–2 shafts carrying gears, chain sprockets, or belt sheaves usually have several diameters that create a series of shoulders that seat the power transmission members and support bearings. Grooves in the shaft allow the installation of retaining rings. Keyseats milled into the shaft enable keys to drive the elements. Similarly, tension members in linkages may be designed with retaining ring grooves, radial holes for pins, screw threads, or reduced sections.

Any of these geometric discontinuities will cause the actual maximum stress in the part to be higher than the simple formulas predict. Defining *stress concentration factors* as the factors by which the actual maximum stress exceeds the nominal stress, σ_{nom} or τ_{nom} , predicted from the simple equations allows the designer to analyze these situations. The symbol for these factors is K_t . In general, the K_t factors are used as follows:

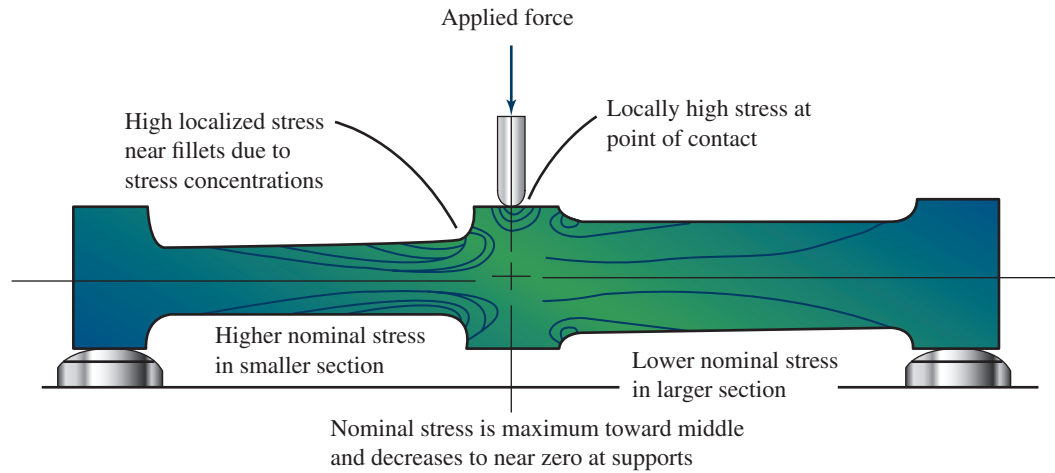
$$\sigma_{\text{max}} = K_t \sigma_{\text{nom}} \quad \text{or} \quad \tau_{\text{max}} = K_t \tau_{\text{nom}} \quad (3-33)$$

depending on the kind of stress produced for the particular loading. The value of K_t depends on the shape of the discontinuity, the specific geometry, and the type of stress.

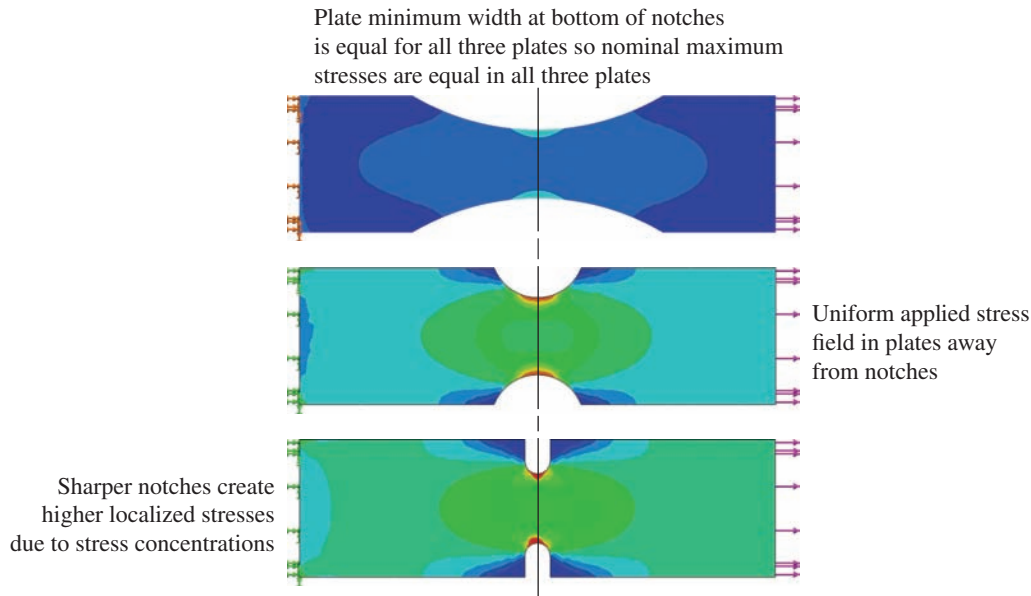
Figure 3–30 shows two examples of how geometric discontinuities affect the stress levels in mechanical components, creating stress concentrations. Figure 3–30 (a) shows a drawing of a model of a beam made from a flat plate using a photoelastic material that reacts to stress levels by producing light and dark areas proportional to variations in stresses when viewed through a polarizing filter. Each dark line, called a *fringe*, indicates a change in stress of a certain amount.

The beam is simply supported at its ends and carries a concentrated load at its middle, where the breadth of the section is made larger. To the right side of the middle, there is a reduction in the breadth with a circular fillet. To the left side of the middle, the breadth of the section is much smaller and the circular fillet is also used. You should visualize that for this simply supported beam, the maximum bending moment occurs at the middle and it decreases linearly to zero at the supports. The predicted maximum bending stress would be in the left portion where the cross section is smaller and toward the right end of that portion where the moment is higher. The fringe lines verify this observation because the number of fringes increases as you look toward the right end of the smaller section. However, note that near where the fillet is located, there are many more, closely spaced fringe lines. This indicates that the local stress around this geometric discontinuity is much higher than would be predicted by simple bending stress calculations. This illustrates the concept of *stress concentrations*. A similar pattern exists in the right portion of the beam, but the overall stress levels are lower because the section size is larger.

Another detail that can be observed from the beam in Figure 3–30(a) is in the area under where the force is applied. The numerous, closely spaced fringes there illustrate the concept of *contact stress* that can be very high



(a) Beam in bending with load at middle applied to enlarged section; reduced sections with circular fillets on either side; left side smaller than right side



(b) Flat plates with uniform tensile stress field applied; reduced sections at middle have equal widths; increasingly sharper notch curvature in three plates

FIGURE 3–30 Sketches of stress distributions near stress concentrations at circular fillets in a beam in bending

locally. Special care should be applied in the design of that part of the beam to withstand the high contact stress. This concept is also pertinent to stresses in gear teeth (Chapters 9 and 10) and rolling contact bearings (Chapter 14).

Figure 3–30(b) shows a different loading and stress concentration phenomenon. Three flat plates are each subjected to identical levels of uniformly distributed tensile stresses in the larger part of each plate toward the ends. Each plate has an equal-width, reduced-breadth section at the middle but with quite different geometry for each. The top plate's width is reduced very gradually and the stress in the material increases essentially only due to the reduction in cross-sectional area. In the plate in the middle, the reduction is made by machining a circular notch with a relatively large radius. The stress at

the notch again increases because of the reduction in cross-sectional area, but there is also an increased stress level near the two notches because of the sudden change in geometry. The larger number of fringe lines is the indication of a *stress concentration* near the notches. The lower plate also contains a notch, but it is much narrower and the radius at its bottom is much sharper than that for the middle plate. Here, there are even more fringe lines, more closely spaced, indicating a much higher local stress near the small notches.

Stress concentrations are generally highly local phenomena, affecting only the region close to the geometric discontinuity. The entire component must be examined to determine where maximum stresses may occur and often two or more areas must be analyzed to ensure that no failure occurs. More is said about this in Chapter 5.

As a designer, you are responsible for applying stress concentration factors correctly and careful judgment is required. Numerous geometries and loading cases have been evaluated analytically, empirically, and by the use of finite element stress analysis methods to produce data from which values for K_t can be found. This section summarizes the basic principles and shows examples of the application of stress concentration factors. General guidelines are listed next.

General Guidelines for the Use of Stress Concentration Factors

1. Stress concentration factors must always be applied for brittle materials because locally high stresses typically lead to fracture.
2. For ductile materials under static tensile loading, it is typical to ignore stress concentrations in small local areas because the ductility permits yielding that redistributes the stress over a larger area and ultimate failure does not occur. However, some designers prefer to ensure that no yielding occurs anywhere in the component.
3. Areas of ductile members under cyclical compressive stress are unlikely to produce failure unless the local maximum stress exceeds the ultimate compressive stress. At lower stresses, a small amount of material near the highest stress area may yield and permit the load to be redistributed over a larger area.
4. Flexing a member in tension tends to cause excessively high local stresses where there are geometric discontinuities, where small internal voids occur, or where surface imperfections occur such as cracks, tool marks, nicks, roughness, corrosion, plating, or scratches. Small cracks appear that will grow to cover larger areas over time, leading to ultimate failure. This type of failure, called *fatigue*, is discussed further in Chapter 5. *Therefore, always apply stress concentration factors in areas of cyclical tensile stress where fatigue failures initiate.*
5. Designing components to avoid the occurrence of factors listed in item 4 is recommended. Maintaining smooth surfaces and homogeneous internal structure of the material and avoiding damage in manufacturing and handling are critical.
6. Stress concentration factors are highest in regions of acutely abrupt changes in geometry. Therefore, good design will call for more gradual changes such as by providing generous fillet radii and blending mating surfaces smoothly.

Reference 5 is arguably the most comprehensive source of data for stress concentration factors. Reference 7 also contains much useful data.

Stress concentration factors are often presented in graphical form to aid designers in visualizing the effect

of specific geometrical decisions on local stresses. Examples of the general forms of such graphs are shown in Figure 3–31. The horizontal axis is typically defined in terms of the ratio of the sizes for primary geometrical features such as diameters, widths, and fillet radii. *It is essential that you understand the basis for the nominal stress for use in Equation (3–33).* For each case in Figure 3–31, the equation for the nominal stress is reported. It is typical for the geometry of the smaller section to be used to compute nominal stress because that will give the largest stress in the region around the discontinuity before accounting for the specific geometry. However, exceptions occur. Additional discussion of the four general types of graphs follows.

- (a) **Stepped flat plate in tension:** The magnitude of the ratio of the larger width, H , to the smaller width, h , is a major factor and it is typically shown as a family of curves like the three in the figure. The horizontal axis is the ratio of r/h , where r is the fillet radius at the step. Note that the larger the ratio, the smaller the value of K_t . Small fillet radii should be avoided if possible because the stress concentration factor typically increases exponentially for very small r/t ratios. *The nominal stress is based on the applied force, F , divided by the minimum cross-sectional area in the smaller section as shown in the figure.*
- (b) **Round bar in bending:** A family of curves is shown for the ratio of the larger diameter, D , to the smaller diameter, d , similar to the discussion for Part (a). The horizontal axis is the ratio of r/d and, again, it is obvious that small fillet radii should be avoided. *The nominal stress is based on the applied bending moment, M , divided by the section modulus of the smaller cross section as shown in the figure.*
- (c) **Flat plate in tension with a central hole:** This case assumes that the applied force is uniformly distributed to all parts of the plate in areas away from the hole. The maximum stress rises in the near vicinity of the hole and it occurs at the top and bottom of the hole. The horizontal axis of the chart is the ratio of d/w , where d is the hole diameter and w is the width of the plate. Note that the plate thickness is called t . *The nominal stress is based on the applied force divided by the net area found at the cross section through the hole.* The figure shows that $A_{\text{net}} = (w - d)t$.
- (d) **Circular bar in torsion with a central hole:** *The stress concentration factor in this case is based on the gross polar section modulus for the full cross section of the bar.* This is one of the few cases where the gross section is used instead of the net section. The primary reason is the complexity of the calculation for the polar section modulus around the hole. The result is that the value of K_t accounts for both the reduction in polar section modulus and the increase in the stress due to the discontinuity around the hole, resulting in rather large values for K_t . The horizontal

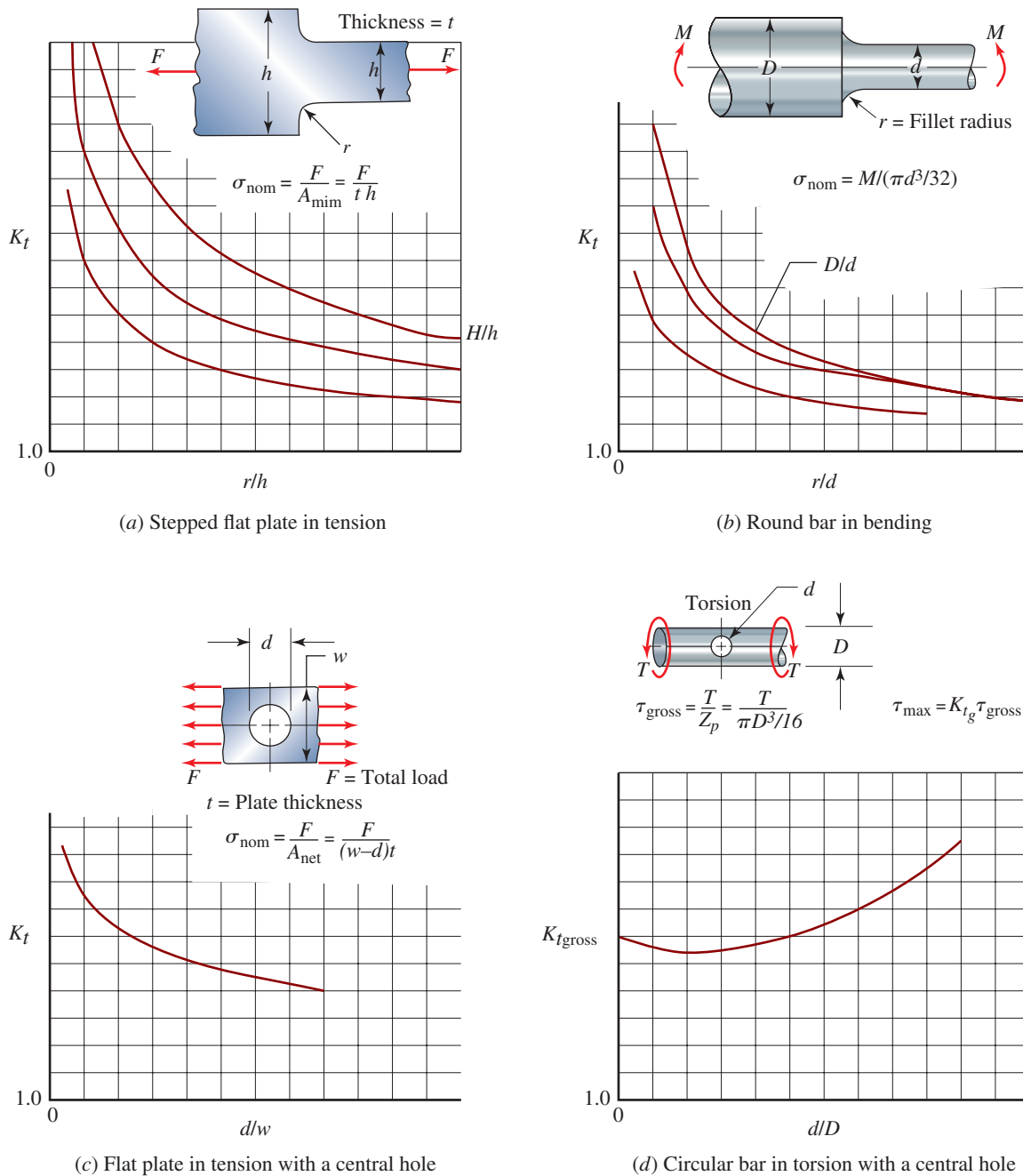


FIGURE 3-31 General Forms of Stress Concentration Factor Curves

axis is the ratio of d/D , where D is the full diameter of the bar and d is the diameter of the hole.

Appendix 18 includes 10 examples of the types of data available for stress concentration factors and should be used for problems in this book. Figures 18-1 through 18-4 are for shapes loaded in direct tension. Figures 18-5 through 18-8 are for bending loads. Figure 18-9 and 18-10 are for torsional loads.

References 5 and 7 are excellent sources of large numbers of cases for stress concentration factors.

Internet site 3 is another useful source for K_t data. Four categories of shapes, listed below, are included on

the site with multiple kinds of stresses, typically axial tension, bending, and torsion.

1. Rectangular bars with several types of notches, holes, slots, and changes in thickness (17 geometry types; 31 cases).
2. Round bars and shafts with steps in diameter, grooves, keyseats, and holes (8 geometry types; 23 cases).
3. Infinite or semi-infinite plates (3 geometry types; 7 cases).
4. Special shapes (3 geometry types; 6 cases).

Example Problem 3-20

Compute the maximum stress in a stepped flat plate subjected to an axial tensile force of 9800 N. The geometry is shown in Figure 3-32.

Solution

Objective Compute the maximum stress in the stepped flat plate in Figure 3-32.

Plate thickness = $t = 6.0$ mm.

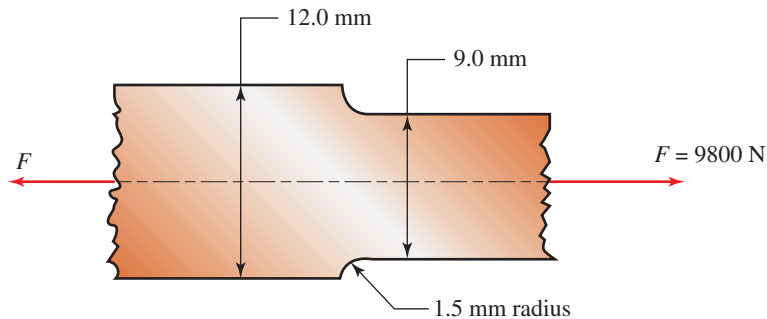


FIGURE 3-32 Stepped flat plate for Example Problem 3-20

Given The layout from Figure 3-32. Force = $F = 9800$ N
Using the notation from Figure 3-31(a): widths $H = 12.0$ mm; $h = 9.0$ mm
Plate thickness: $t = 6.0$ mm
Fillet radius at the step: $r = 1.50$ mm

Analysis The presence of the change in width at the step causes a stress concentration to occur. The geometry is of the type shown in Figure 3-31(a) and Appendix A18-2 that we will use to find the value of K_t for this problem. That value is used in Equation (3-33) to compute the maximum stress.

Results The values of the ratios H/h and r/h are required:

$$H/h = 12.0 \text{ mm}/9.0 \text{ mm} = 1.33$$

$$r/h = 1.5 \text{ mm}/9.0 \text{ mm} = 0.167$$

The value of $K_t = 1.83$ can be read from Appendix A18-2.

The nominal stress is computed for the small section having a cross section of 6.0 mm by 9.0 mm.

$$A_{\text{net}} = h \cdot t = (9.0 \text{ mm})(6.0 \text{ mm}) = 54.0 \text{ mm}^2$$

The nominal stress is:

$$\sigma_{\text{nom}} = F/A_{\text{net}} = (9800 \text{ N})/(54.0 \text{ mm}^2) = 181.5 \text{ N/mm}^2 = 181.5 \text{ MPa}$$

The maximum stress in the area of the fillet at the step is:

$$\sigma_{\text{max}} = K_t \sigma_{\text{nom}} = (1.83)(181.5 \text{ MPa}) = 332 \text{ MPa}$$

Comments The maximum stress of 332 MPa occurs in the fillet area at both the top and bottom of the small section. A short distance to the right of the fillet, the local stress reduces to the nominal value of 181.5 MPa. The stress in the larger section is obviously much smaller because of the larger cross-sectional area of the plate. Note that specifying a smaller fillet radius would cause a much larger maximum stress because the curve for K_t increases sharply as the ratio of r/t decreases. Modestly smaller maximum stress would be produced for larger fillet radii.

Stress Concentration Factors for Lug Joints.

A common machine element is the familiar *clevis joint* or *lug joint*, sketched in Figure 3-33. For this discussion, the bar-shaped plate is called the *lug* that may have either a flat, square-ended design as shown or a rounded end that is sometimes used to provide clearance for rotation of the lug. The generic name for the lug in stress analysis is *flat plate with a central hole with a load applied through a pin in tension*. Figure 3-34

shows approximate data for the stress concentration factors that occur around the hole for typical conditions, described below. The plate is assumed to have a uniform distribution of the load across any cross section well away from the hole. The critical section with regard to axial tensile stress in the plate occurs in the section through the hole where the minimum net cross-sectional area occurs. The stress concentration factor accounts for the geometric discontinuity at this section.

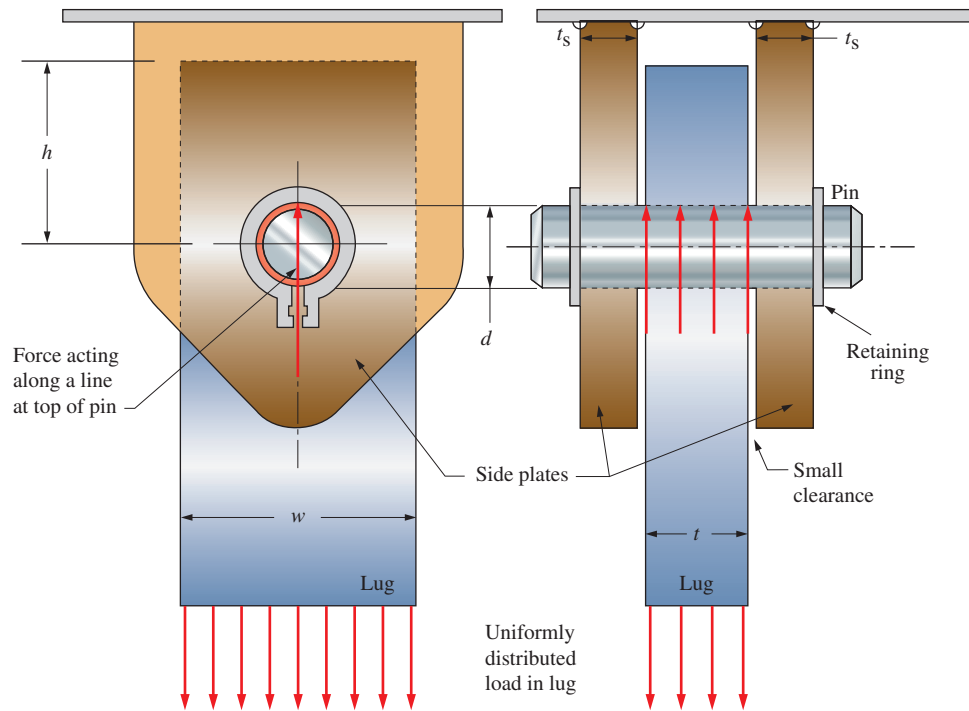


FIGURE 3-33 Clevis joint showing a pin-lug connection including a flat plate with a central hole as used in Figure 3-34.

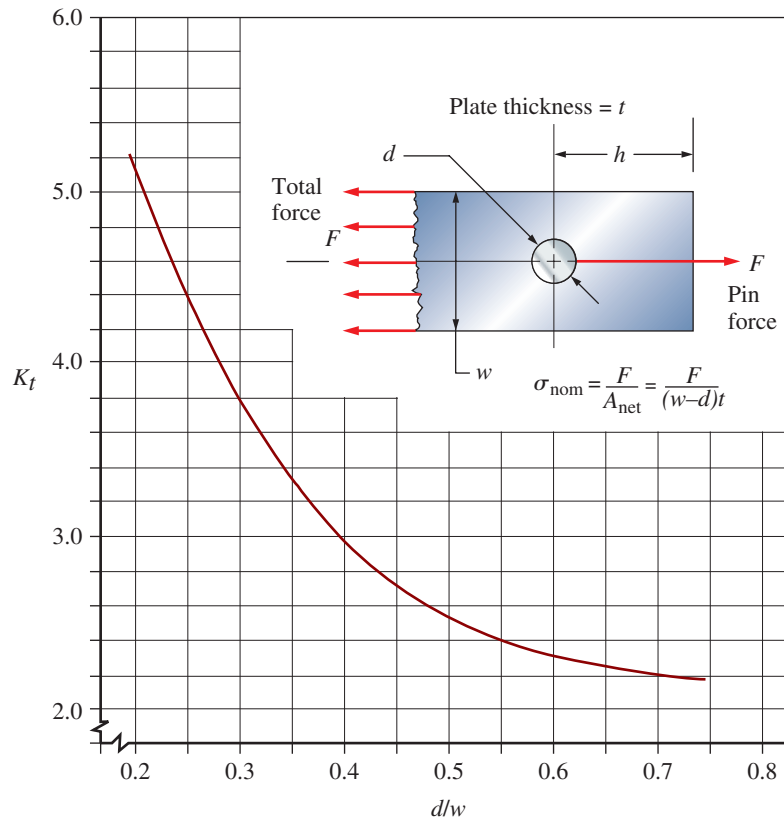


FIGURE 3-34 Stress concentration factor for flat plate with tensile load applied through a pin

Example Problem 3-21

The lug for a clevis joint of the type shown in Figure 3-33 is being designed for an applied force of 8.75 kN. The steel pin nominal diameter has been specified to be $d = 12.0$ mm. Specify the dimensions w , h , and t for the steel lug to be consistent with the parameters listed in this section and a d/w ratio of 0.40. Also specify the nominal clearance between the pin and the hole in the lug. Then determine the expected maximum stress in the lug.

Objective Specify w , h , and t and the nominal clearance.

Given $d = 12.0$ mm; $d/w = 0.40$. $F = 8.75$ kN

Analysis *Lug thickness, t :* Recommended $t/d \leq 0.50$. Using the upper limit gives:
 $t = 0.50 d = (0.50)(12.0 \text{ mm}) = 6.0 \text{ mm}$

Lug width, w : Using $d/w = 0.40$, $w = d/0.40 = 12 \text{ mm}/0.40 = 30.0 \text{ mm}$

End distance, h : Recommended ratio $h/w = 1.0$. Then,

$$h = w = 30.0 \text{ mm}$$

Close-fitting pin: $d_{\text{hole}} \cong d_{\text{pin}}(1.002) = 12.0 \text{ mm}(1.002) = 12.024 \text{ mm}$

Maximum stress: $\sigma_{\text{max}} = K_t \times \sigma_{\text{nom}}$

$$\sigma_{\text{nom}} = F/(w - d)t = [8.75 \text{ kN}/(30.0 - 12.0)(6.0 \text{ mm}^2)](1000 \text{ N/kN})$$

$$\sigma_{\text{nom}} = 81.8 \text{ N/mm}^2 = 81.8 \text{ MPa}$$

To find K_t : At $d/w = 0.40$ in Figure 3-33, $K_t \cong 3.0$. Then

$$\sigma_{\text{max}} = K_t \times \sigma_{\text{nom}} = (3.0)(81.8 \text{ MPa}) = 243 \text{ MPa}$$

Results The design details for the lug of the clevis joint in Figure 3-33 are:

$$d = 12.0 \text{ mm}; w = 30.0 \text{ mm}; t = 6.0 \text{ mm}; h = 30.0 \text{ mm}$$

Nominal clearance between pin and hole: $\delta = 0.02 \text{ mm}$

Maximum stress in lug at hole: $\sigma_{\text{max}} = 243 \text{ MPa}$

In general, the axial tensile stress near the hole in the lug depends on several factors:

1. The width and thickness of the lug and the size and placement of the hole relative to the end of the lug.
2. The clearances between the faces of the lug and the side plates of the clevis.
3. The diameter and length of the pin.
4. The clearance between the pin and the hole.
5. The materials of the pin and the lug.

A typical lug connection should have the following features:

- a square-ended lug.
- close-fitting pin: $d_{\text{hole}} \cong d_{\text{pin}}(1.002)$.
- $t/d \leq 0.5$ [t = thickness of lug parallel to axis of pin; d = nominal hole diameter].
- Both pin and lug have the same modulus of elasticity; $E_{\text{pin}}/E_{\text{lug}} = 1.0$.
- $h/w = 1.0$.
- h = distance from centerline of hole to the top of the lug.
- w = width of the lug across the location of the hole.

Problems in this book will assume the conditions listed above and use the values for K_t from Internet site 3. The curve in Figure 3-34 gives approximate data for K_t vs. the ratio d/w .

Reference 5 includes extensive data that allow consideration of factors different from those assumed above.

Very loose fitting pins should be avoided because:

- The pin makes a virtual line contact on the top of the hole.
- The stress concentration factor for the lug is somewhat higher than shown in Figure 3-34.
- The maximum stress occurs at the line of contact.
- There is the danger that the contact stress between the pin and the hole will cause failure in bearing, a local crushing of the pin or the inside surface of the hole.

Other factors to consider in the design of a clevis or lug joint are:

1. The pin should be as short as practical to minimize the tendency for it to bend, causing nonuniform contact of the pin in the hole.
2. The diameter of the pin must be adequate to resist failure by direct shear and to keep bending deflection to an acceptable level.

3. The distance, h , from the centerline of the hole to the top of the lug should be nominally equal to the width of the plate, w , and as large as practical to minimize the bending stresses in the upper part of the lug and to prevent shearing tear-out of the lug.
4. The bearing stress between the pin surface and the inside of the hole of the lug must be acceptable, typically less than $0.9s_y$.
5. Similarly, the bearing stress between the pin surface and the holes in the side plates must be acceptable.
6. The clearance between the faces of the lug and the side plates of the clevis should be small to minimize bending of the pin.

3-23 NOTCH SENSITIVITY AND STRENGTH REDUCTION FACTOR

The amount by which a load-carrying member is weakened by the presence of a stress concentration (notch), considering both the material and the sharpness of the notch, is defined as

K_f = fatigue strength reduction factor

$$K_f = \frac{\text{endurance limit of a notch-free specimen}}{\text{endurance limit of a notched specimen}}$$

This factor could be determined by actual test. However, it is typically found by combining the stress concentration factor, K_t , defined in Section 3-22, and a material factor called the *notch sensitivity*, q . We define

$$q = (K_f - 1)/(K_t - 1) \quad (3-34)$$

When q is known, K_f can be computed from

$$K_f = 1 + q(K_t - 1) \quad (3-35)$$

Values of q range from 0 to 1.0, and therefore K_f varies from 1.0 to K_t . Under repeated bending loads, very ductile steels typically exhibit values of q from 0.5 to 0.7. High-strength steels with hardness approximately HB 400 ($s_u \cong 200$ ksi or 1400 MPa) have values of q from 0.90 to 0.95. (See Reference 2 for further discussion of values of q .)

Because reliable values of q are difficult to obtain, the problems in this book will typically assume that $q = 1.0$ and $K_f = K_t$, the safest, most conservative value.

REFERENCES

1. Boresi, Arthur P., and Richard J. Schmidt. *Advanced Mechanics of Materials*. 6th ed. New York: John Wiley & Sons, 2003.
2. Javidinejad, Amir. *Essentials of Mechanical Stress Analysis*. Boca Raton, FL: CRC Press, 2014.
3. Huston, Ronald, and Harold Josephs. *Practical Stress Analysis in Engineering Design*. 3rd ed. Boca Raton, FL: CRC Press, Taylor & Francis Group, 2009.
4. Mott, Robert L. *Applied Strength of Materials*. 6th ed. Boca Raton, FL: CRC Press, 2017.
5. Pilkey, Walter D., and Deborah F. Pilkey. *Peterson's Stress Concentration Factors*. 3rd ed. New York: John Wiley & Sons, 2008.
6. Timoshenko, S. *Strength of Materials, Part II—Advanced Theory and Problems*. New York: D. Van Nostrand Co., 1930.
7. Young, Warren C., and Richard G. Budynas. *Roark's Formulas for Stress and Strain*. 8th ed. New York: McGraw-Hill, 2012.

INTERNET SITES RELATED TO STRESS AND DEFORMATION ANALYSIS

1. **BEAM 2D-Stress Analysis 3.1** Software for mechanical, structural, civil, and architectural designers providing detailed analysis of statically indeterminate and determinate beams. From Orand Systems.
2. **MDSolids** Educational software devoted to introductory mechanics of materials. Includes modules on basic stress and strain; beam and strut axial problems; trusses; statically indeterminate axial structures; torsion; determinate beams; section properties; general analysis of axial, torsion, and beam members; column buckling; pressure vessels; and Mohr's circle transformations.
3. **eFatigue.com** An Internet source for consultation and information related to design of machine components subjected to fatigue loading. From the home page, select "Constant Amplitude" then "Stress Concentrations." This site permits online calculation of stress concentration factors for 60 cases covering a variety of geometry types and loading conditions.

PROBLEMS

Direct Tension and Compression

1. A tensile member in a machine structure is subjected to a steady load of 4.50 kN. It has a length of 750 mm and is made from a steel tube having an outside diameter of 18 mm and an inside diameter of 12 mm. Compute the tensile stress in the tube and the axial deformation.
2. Compute the stress in a round bar having a diameter of 10.0 mm and subjected to a direct tensile force of 3500 N.
3. Compute the stress in a rectangular bar having cross-sectional dimensions of 10.0 mm by 30.0 mm when a direct tensile force of 20.0 kN is applied.
4. A link in a packaging machine mechanism has a square cross section 0.40 in on a side. It is subjected to a tensile force of 860 lb. Compute the stress in the link.
5. Two circular rods support the 3800 lb weight of a space heater in a warehouse. Each rod has a diameter of 0.375 in and carries 1/2 of the total load. Compute the stress in the rods.

6. A tensile load of 5.00 kN is applied to a square bar, 12 mm on a side and having a length of 1.65 m. Compute the stress and the axial deformation in the bar if it is made from (a) SAE 1020 hot-rolled steel, (b) SAE 8650 OQT 1000 steel, (c) ductile iron A536(60-40-18), (d) aluminum 6061-T6, (e) titanium Ti-6Al-4V, (f) rigid PVC plastic, and (g) phenolic plastic.
7. An aluminum rod is made in the form of a hollow square tube, 2.25 in outside, with a wall thickness of 0.120 in. Its length is 16.0 in. What axial compressive force would cause the tube to shorten by 0.004 in? Compute the resulting compressive stress in the aluminum.
8. Compute the stress in the middle portion of rod AC in Figure P3-8 if the vertical force on the boom is 2500 lb. The rod is rectangular, 1.50 in by 3.50 in.

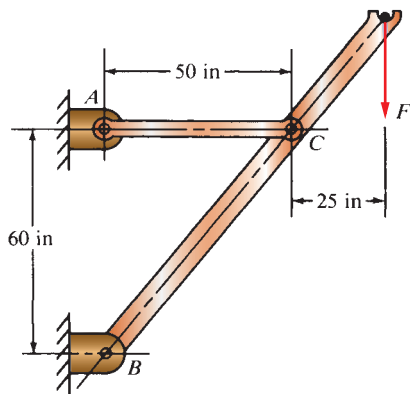


FIGURE P3-8 (Problems 8, 16, 56, and 80)

9. Compute the forces in the two angled rods in Figure P3-9 for an applied force, $F = 1500$ lb, if the angle θ is 45° .

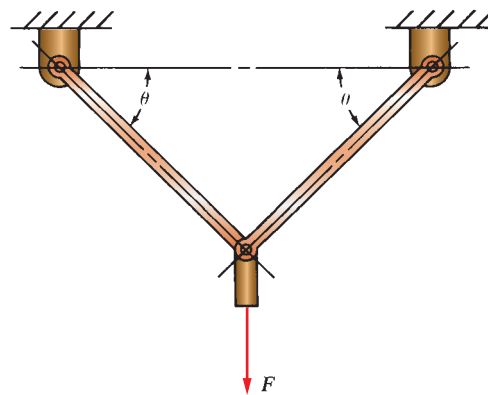
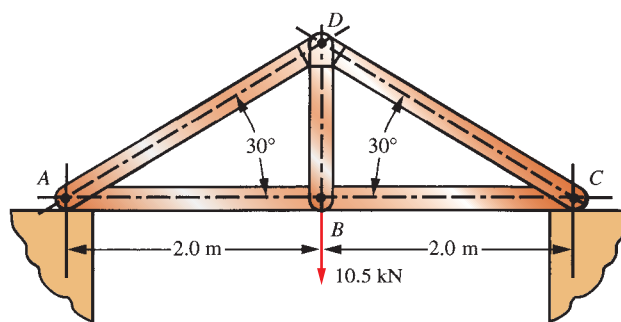
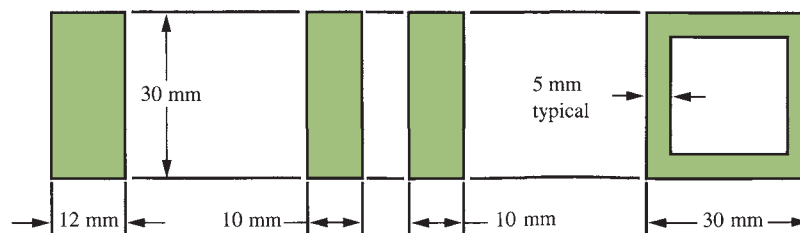


FIGURE P3-9 (Problems 9, 10, 11, 17, and 18)

10. If the rods from Problem 9 are circular, determine their required diameter if the load is static and the allowable stress is 18 000 psi.
11. Repeat Problems 9 and 10 if the angle θ is 15° .
12. Figure P3-12 shows a small truss spanning between solid supports and suspending a 10.5 kN load. The cross sections for the three main types of truss members are shown. Compute the stresses in all of the members of the truss near their midpoints away from the connections. Consider all joints to be pinned.
13. The truss shown in Figure P3-13 spans a total space of 18.0 ft and carries two concentrated loads on its top chord. The members are made from standard steel angle and channel shapes as indicated in the figure. Consider all joints to be pinned. Compute the stresses in all members near their midpoints away from the connections.



(a) Truss with load applied to joint B.

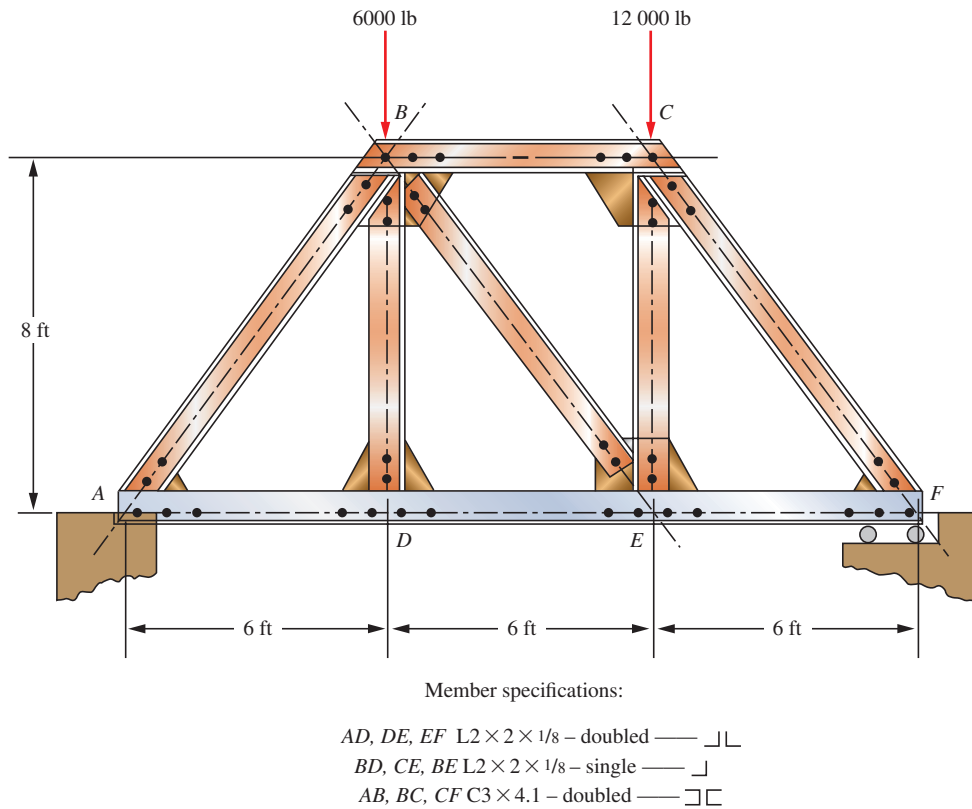


(b) Cross section of members AB, BC

(c) Cross section of member BD

(d) Cross section of members AD, CD

FIGURE P3-12 (Problem 12)



14. Figure P3–14 shows a short leg for a machine that carries a direct compression load. Compute the compressive stress if the cross section has the shape shown and the applied force is $F = 52\,000$ lb.
15. Consider the short compression member shown in Figure P3–15. Compute the compressive stress if the cross section has the shape shown and the applied load is 640 kN.

Direct Shear Stress

16. Refer Figure P3–8. Each of the pins at A , B , and C has a diameter of 0.50 in and is loaded in double shear. Compute the shear stress in each pin.
17. Compute the shear stress in the pins connecting the rods shown in Figure P3–9 when a load of $F = 1500$ lb is carried. The pins have a diameter of 0.75 in. The angle $\theta = 40^\circ$.

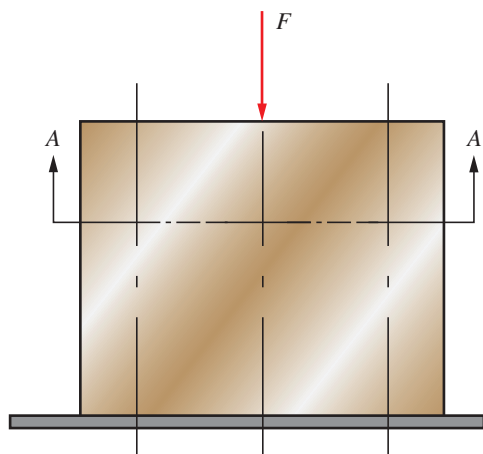
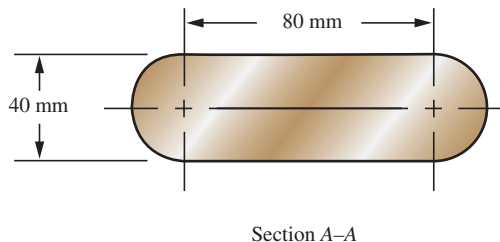


FIGURE P3-15 (Problem 15)



Section A-A

18. Repeat Problem 17, but change the angle to $\theta = 15^\circ$.
19. Refer Figure 3-7. Compute the shear stress in the key if the shaft transmits a torque of $1600 \text{ N}\cdot\text{m}$. The shaft diameter is 60 mm. The key is square with $b = 12 \text{ mm}$, and it has a length of 45 mm.
20. A punch is attempting to cut a slug having the shape shown in Figure P3-20 from a sheet of aluminum having a thickness of 0.060 in. Compute the shearing stress in the aluminum when a force of 52 000 lb is applied by the punch.

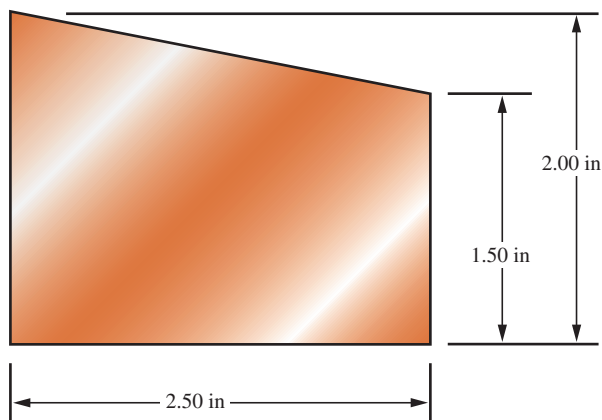


FIGURE P3-20 (Problem 20)

FIGURE P3-21 (Problem 21)

21. Figure P3-21 shows the shape of a slug that is to be cut from a sheet of steel having a thickness of 2.0 mm. If the punch exerts a force of 225 kN, compute the shearing stress in the steel.

Torsion

22. Compute the torsional shear stress in a circular shaft with a diameter of 50 mm that is subjected to a torque of $800 \text{ N}\cdot\text{m}$.
23. If the shaft of Problem 22 is 850 mm long and is made of steel, compute the angle of twist of one end in relation to the other.
24. Compute the torsional shear stress due to a torque of $88.0 \text{ lb}\cdot\text{in}$ in a circular shaft having a 0.40-in diameter.

25. Compute the torsional shear stress in a solid circular shaft having a diameter of 1.25 in that is transmitting 110 hp at a speed of 560 rpm.
26. Compute the torsional shear stress in a hollow shaft with an outside diameter of 40 mm and an inside diameter of 30 mm when transmitting 28 kilowatts (kW) of power at a speed of 45 rad/s.
27. Compute the angle of twist for the hollow shaft of Problem 26 over a length of 400 mm. The shaft is steel.

Noncircular Members in Torsion

28. A square steel bar, 25 mm on a side and 650 mm long, is subjected to a torque of $230 \text{ N}\cdot\text{m}$. Compute the shear stress and the angle of twist for the bar.
29. A 3.00 in-diameter steel bar has a flat milled on one side, as shown in Figure P3-29. If the shaft is 44.0 in long and carries a torque of $10\,600 \text{ lb}\cdot\text{in}$, compute the stress and the angle of twist.
30. A commercial steel supplier lists rectangular steel tubing having outside dimensions of 4.00 by 2.00 in and a wall thickness of 0.109 in. Compute the maximum torque that can be applied to such a tube if the shear stress is to be limited to 6000 psi. For this torque, compute the angle of twist of the tube over a length of 6.5 ft.

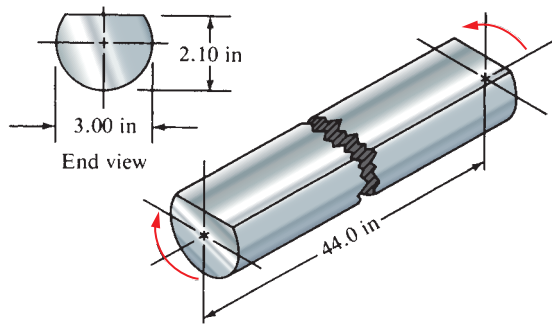


FIGURE P3-29 (Problem 29)

Beams

31. A beam is simply supported and carries the load shown in Figure P3-31. Specify suitable dimensions for the beam if it is steel and the stress is limited to 18 000 psi, for the following shapes:
- Square
 - Rectangle with height three times the width
 - Rectangle with height one-third the width
 - Solid circular section
 - American Standard beam section
 - American Standard channel with the legs down
 - Standard steel pipe

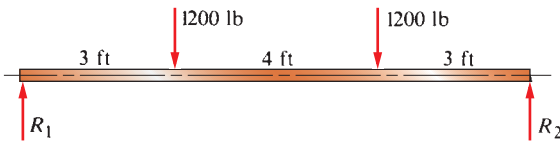


FIGURE P3-31 (Problems 31, 32, and 33)

32. For each beam of Problem 31, compute its weight if the steel weighs 0.283 lb/in^3 .
33. For each beam of Problem 31, compute the maximum deflection and the deflection at the loads.
34. For the beam loading of Figure P3-34, draw the complete shearing force and bending moment diagrams, and determine the bending moments at points A, B, and C.
35. For the beam loading of Figure P3-34, design the beam choosing a commercially available shape in standard SI units from Appendix 15 with the smallest cross-sectional area that will limit the bending stress to 100 MPa.

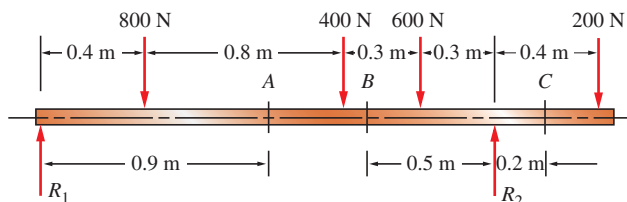


FIGURE P3-34 (Problems 34 and 35)

36. Figure P3-36 shows a beam made from 4 in schedule 40 steel pipe. Compute the deflection at points A and B for two cases: (a) the simple cantilever and (b) the supported cantilever.

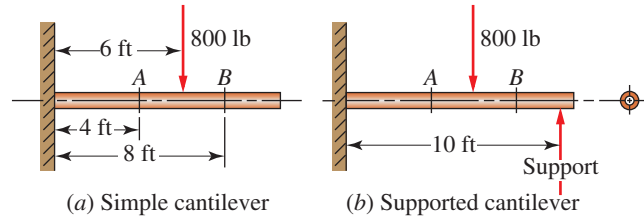


FIGURE P3-36 (Problem 36)

37. Select an aluminum I-beam shape to carry the load shown in Figure P3-37 with a maximum stress of 12 000 psi. Then compute the deflection at each load.

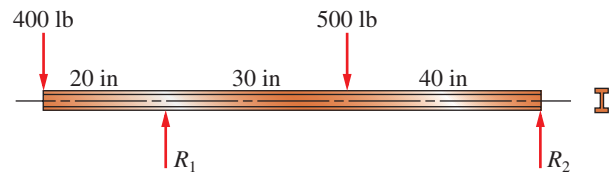


FIGURE P3-37 (Problem 37)

38. Figure P3-38 represents a wood joist for a platform, carrying a uniformly distributed load of 120 lb/ft and two concentrated loads applied by some machinery. Compute the maximum stress due to bending in the joist and the maximum vertical shear stress.

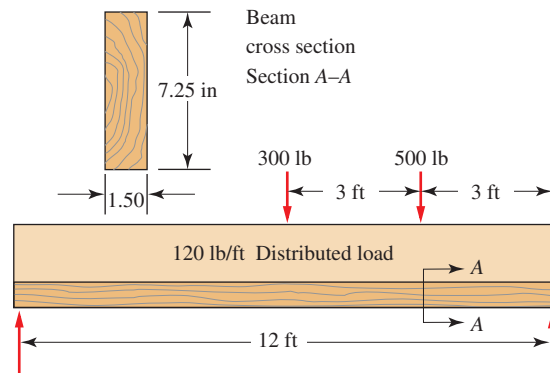


FIGURE P3-38 (Problem 38)

Beams with Concentrated Bending Moments

For Problems 39 through 50, draw the free-body diagram of only the horizontal beam portion of the given figures. Then draw the complete shear and bending moment diagrams. Where used, the symbol X indicates a simple support capable of exerting a reaction force in any direction but having no moment resistance. For beams having unbalanced axial loads, you may specify which support offers the reaction.

39. Use Figure P3-39.

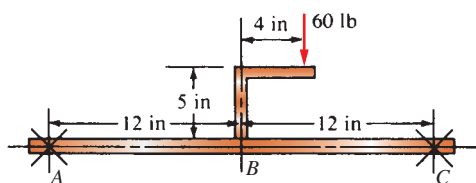


FIGURE P3-39 (Problems 39 and 57)

40. Use Figure P3-40.

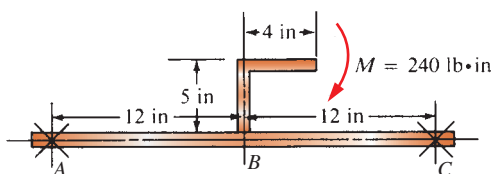


FIGURE P3-40 (Problem 40)

41. Use Figure P3-41.

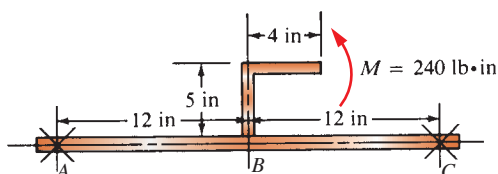


FIGURE P3-41 (Problem 41)

42. Use Figure P3-42.

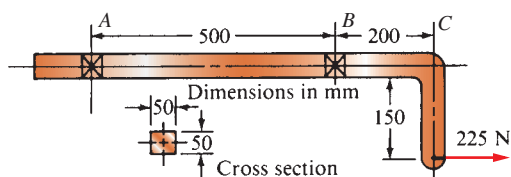


FIGURE P3-42 (Problems 42 and 58)

43. Use Figure P3-43.

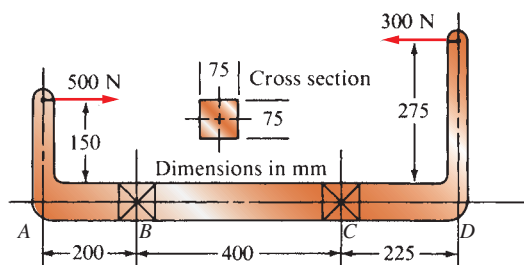


FIGURE P3-43 (Problems 43 and 59)

44. Use Figure P3-44.

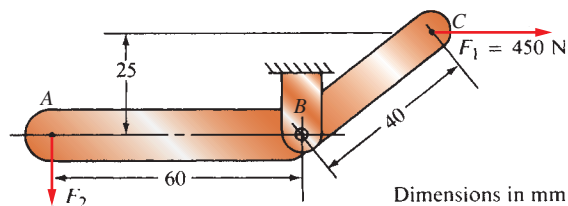


FIGURE P3-44 (Problem 44)

45. Use Figure P3-45.

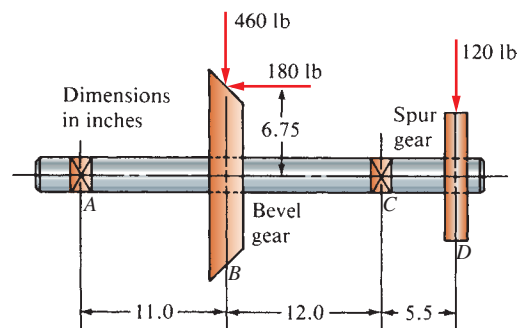


FIGURE P3-45 (Problem 45)

46. Use Figure P3-46.

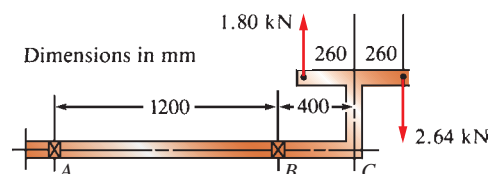


FIGURE P3-46 (Problem 46)

47. Use Figure P3-47.

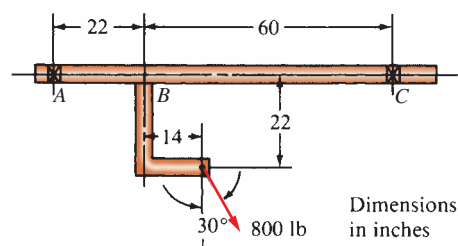


FIGURE P3-47 (Problems 47 and 60)

For Problems 48–50, draw the free-body diagram of the main shaft portion, labeled A, B, and C. Include any unbalanced torque on the shaft that tends to rotate it about the z -axis. In each case, the reaction to the unbalanced torque is taken at the right end of the shaft labeled C. Then draw the complete shearing force and bending moment diagrams for loading in the y - z plane. Also prepare a graph of the torque in the shaft as a function of position along the shaft from A to C.

48. Use Figure P3–48.

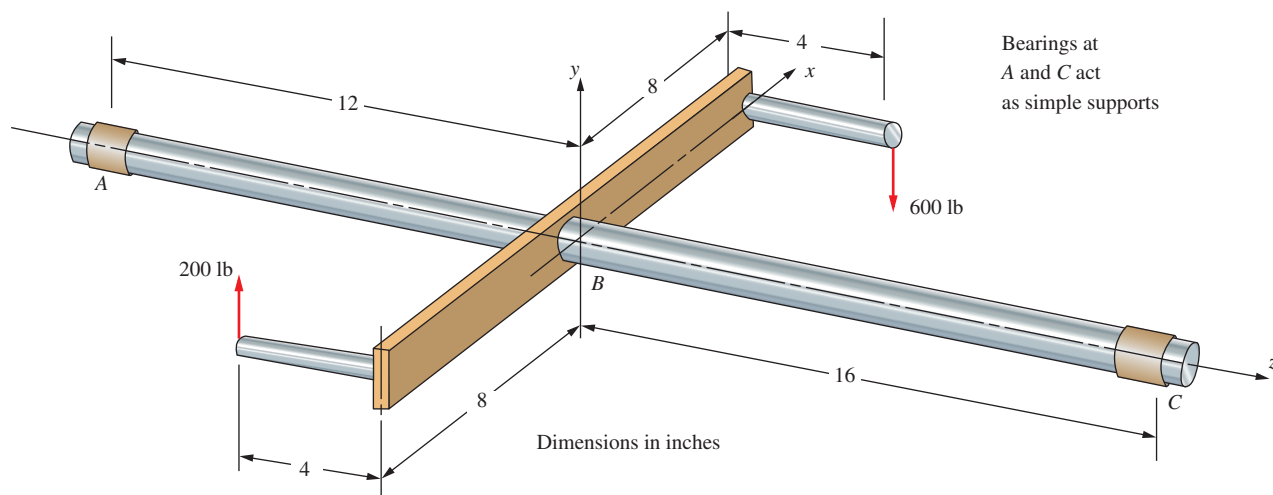


FIGURE P3–48 (Problem 48)

49. Use Figure P3–49.

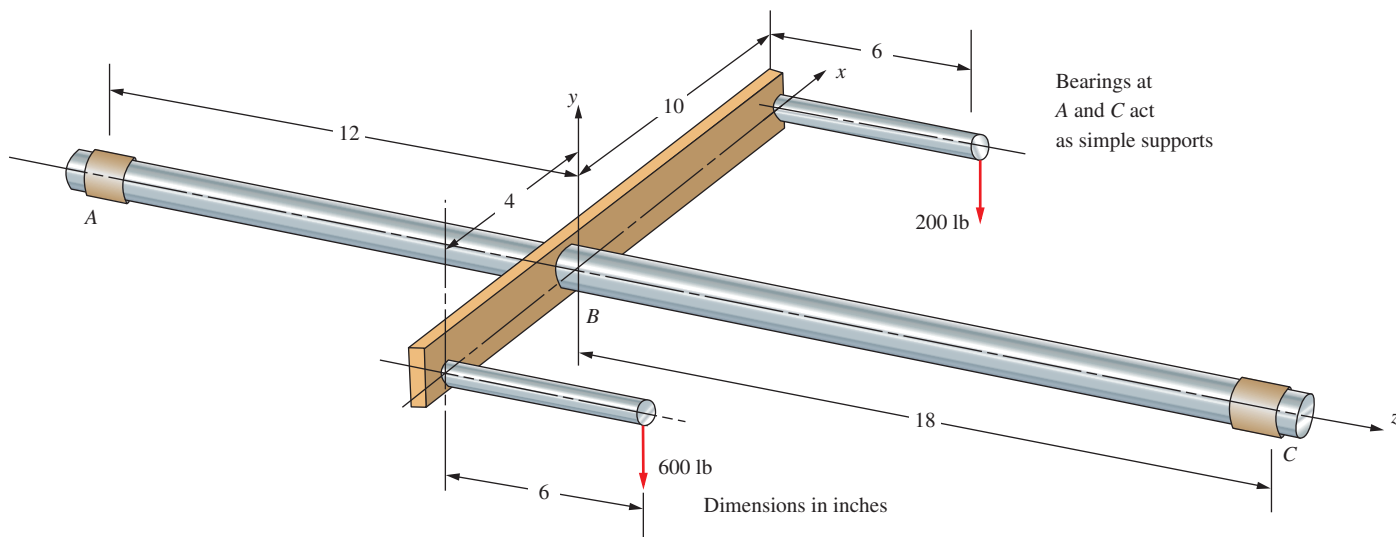


FIGURE P3–49 (Problem 49)

50. Use Figure P3–50.

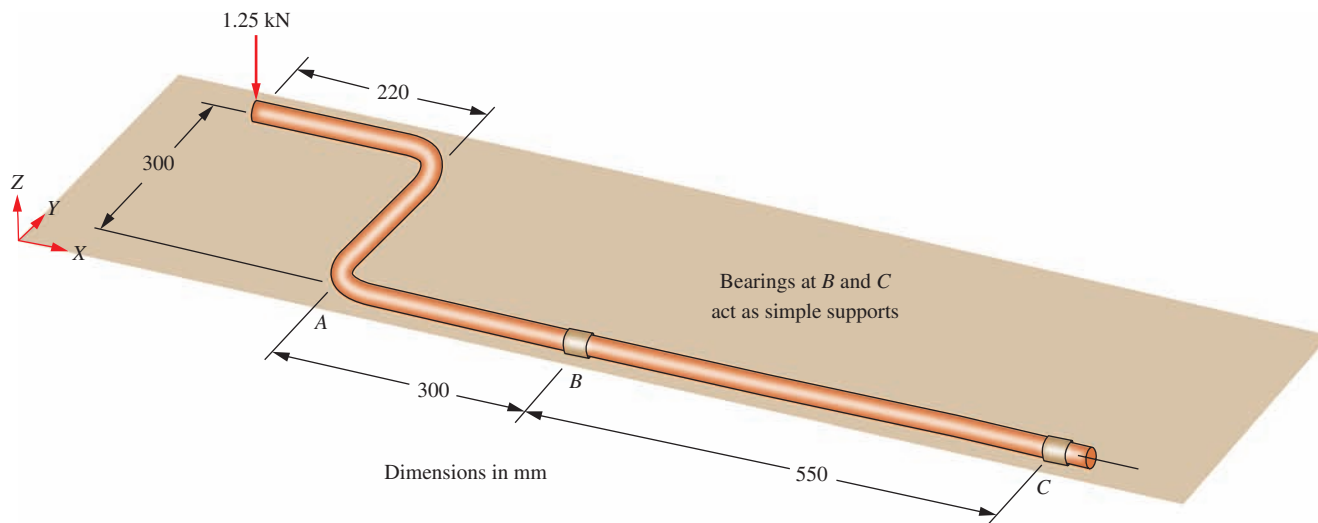


FIGURE P3–50 (Problem 50)

Combined Normal Stresses

51. Compute the maximum tensile stress in the bracket shown in Figure P3–51.

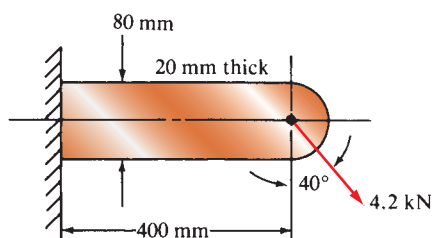


FIGURE P3–51 (Problem 51)

52. Compute the maximum tensile and compressive stresses in the horizontal beam shown in Figure P3–52.

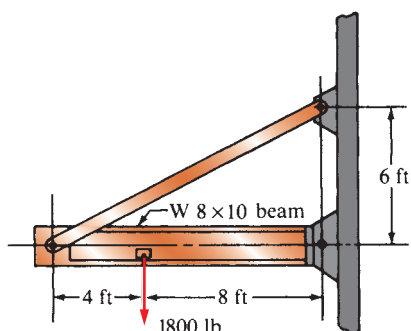


FIGURE P3–52 (Problem 52)

53. For the lever shown in Figure P3–53(a), compute the stress at section A near the fixed end. Then redesign the lever to the tapered form shown in Figure P3–5 (b) by adjusting only the height of the cross section at

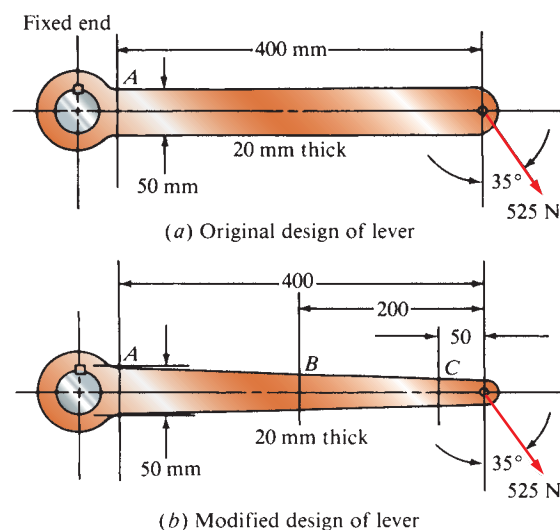


FIGURE P3–53 (Problem 53)

sections B and C so that they have no greater stress than section A.

54. Compute the maximum tensile stress at sections A and B on the crane boom shown in Figure P3–54.
55. Refer to Figure 3–22. Compute the maximum tensile stress in the print head just to the right of the right guide. The head has a rectangular cross section, 5.0 mm high in the plane of the paper and 2.4 mm thick.
56. Refer to Figure P3–8. Compute the maximum tensile and compressive stresses in the member B–C if the load F is 1800 lb. The cross section of B–C is a HSS 6×4×1/4 rectangular tube.
57. Refer to P3–39. The vertical member is to be made from steel with a maximum allowable stress of 12 000 psi. Specify the required size of a standard

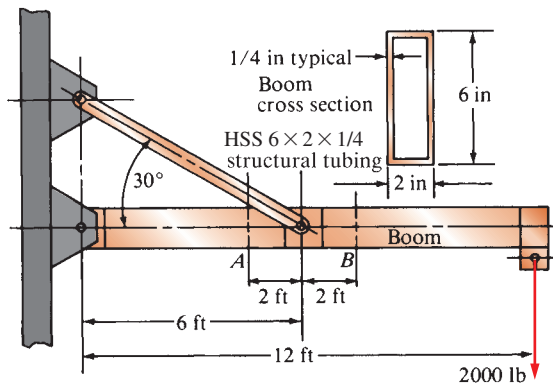


FIGURE P3-54 (Problem 54)

square cross section if sizes are available in increments of 1/16 in.

58. Refer to P3-42. Compute the maximum stress in the horizontal portion of the bar, and tell where it occurs on the cross section. The left support resists the axial force.
59. Refer to P3-43. Compute the maximum stress in the horizontal portion of the bar, and indicate where it occurs on the cross section. The right support resists the unbalanced axial force.
60. Refer to P3-47. Specify a suitable diameter for a solid circular bar to be used for the top horizontal member, which is supported in the bearings. The left bearing resists the axial load. The allowable normal stress is 25 000 psi.

Stress Concentrations

61. Figure P3-61 shows a valve stem from an engine subjected to an axial tensile load applied by the valve spring. For a force of 1.25 kN, compute the maximum stress at the fillet under the shoulder.
62. The conveyor fixture shown in Figure P3-62 carries three heavy assemblies (1200 lb each). Compute the maximum stress in the fixture, considering stress concentrations at the fillets and assuming that the load acts axially.
63. For the flat plate in tension in Figure P3-63, compute the stress at each hole, assuming that the holes are sufficiently far apart that their effects do not interact.

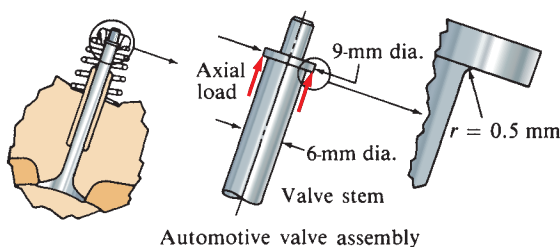


FIGURE P3-61 (Problem 61)

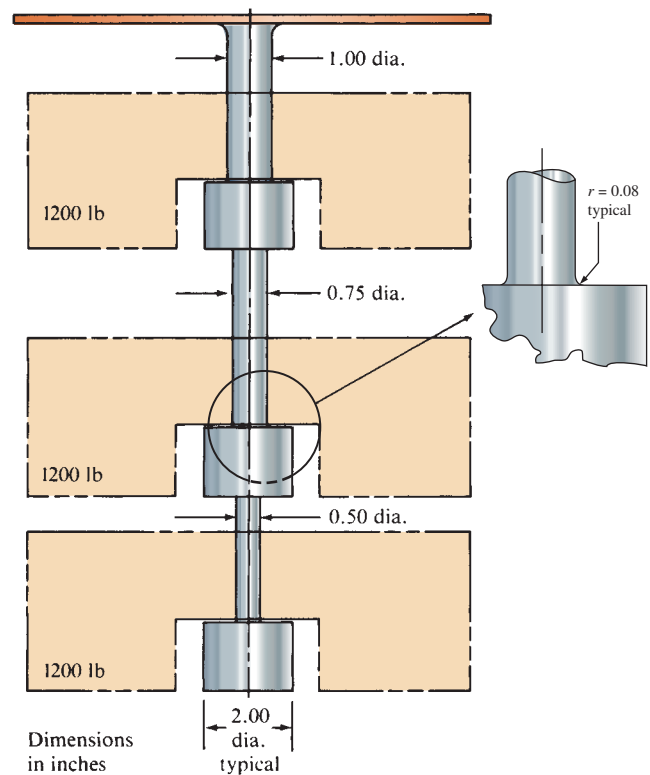


FIGURE P3-62 (Problem 62)

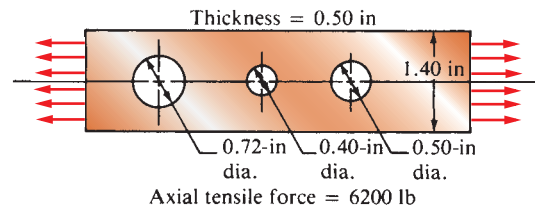


FIGURE P3-63 (Problem 63)

For Problems 64 through 68, compute the maximum stress in the member, considering stress concentrations.

64. Use Figure P3-64.

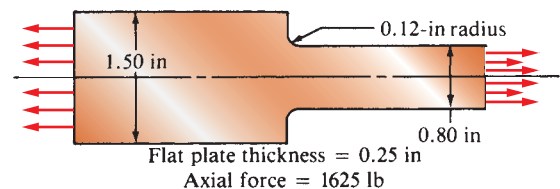


FIGURE P3-64 (Problem 64)

65. Use Figure P3-65.

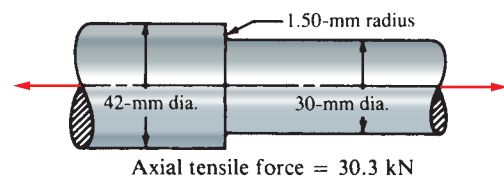


FIGURE P3-65 (Problem 65)

66. Use Figure P3-66.

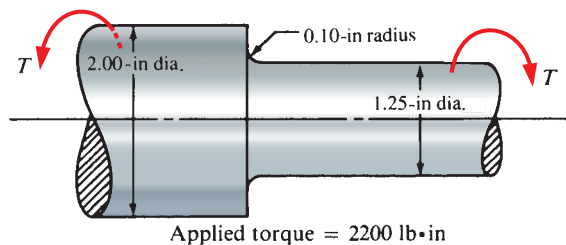


FIGURE P3-66 (Problem 66)

67. Use Figure P3-67.

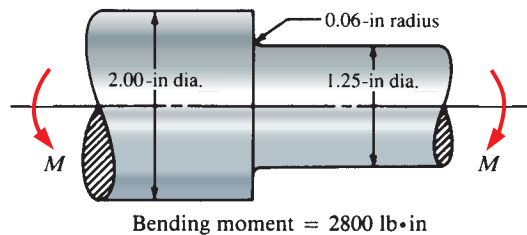


FIGURE P3-67 (Problem 67)

68. Use Figure P3-68.

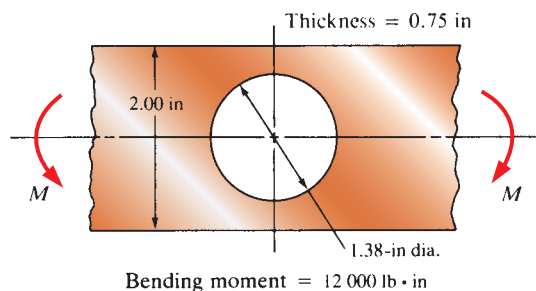


FIGURE P3-68 (Problem 68)

Problems of a General Nature

69. Figure P3-69 shows a horizontal beam supported by a vertical tension link. The cross sections of both the beam and the link are 20 mm square. All connections use 8.00 mm-diameter cylindrical pins in double shear. Compute the tensile stress in member A-B, the

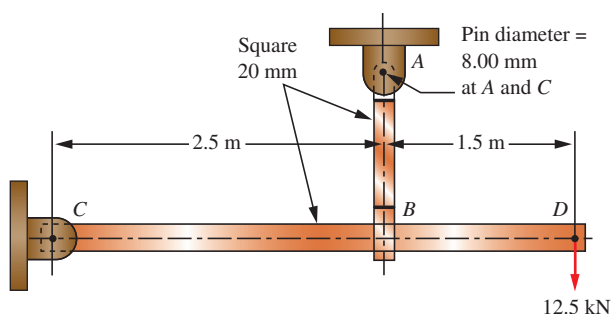


FIGURE P3-69 (Problem 69)

stress due to bending in C-D, and the shearing stress in the pins A and C.

70. Figure P3-70 shows a tapered flat bar that has a uniform thickness of 20 mm. The depth tapers from $h_1 = 40$ mm near the load to $h_2 = 20$ mm at each support. Compute the stress due to bending in the bar at points spaced 40 mm apart from the support to the load. Let the load $P = 5.0$ kN.

71. For the flat bar shown in Figure P3-70, compute the stress in the middle of the bar if a hole of 25 mm diameter is drilled directly under the load on the horizontal centerline. The load is $P = 5.0$ kN. See data in Problem 70.

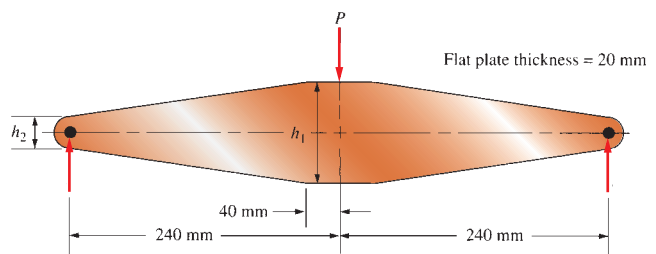


FIGURE P3-70 Tapered flat bar for Problems 70 and 71

72. The beam shown in Figure P3-72 is a stepped, flat bar having a constant thickness of 1.20 in. It carries a single concentrated load at C of 1500 lb. Compare the stresses at the following locations:

- In the vicinity of the load
- At the section through the smaller hole to the right of section C
- At the section through the larger hole to the right of section C

73. Figure P3-73 shows a stepped, flat bar having a constant thickness of 8.0 mm. It carries three concentrated loads as shown. Let $P = 200$ N, $L_1 = 180$ mm, $L_2 = 80$ mm, and $L_3 = 40$ mm. Compute the maximum stress due to bending, and state where it occurs. The bar is braced against lateral bending and twisting. Note that the dimensions in the figure are not drawn to scale.

74. Figure P3-74 shows a bracket carrying opposing forces of $F = 2500$ N. Compute the stress in the upper horizontal part through one of the holes as at B. Use $d = 15.0$ mm for the diameter of the holes.

75. Repeat Problem 74, but use a hole diameter of $d = 12.0$ mm.

76. Figure P3-76 shows a lever made from a rectangular bar of steel. Compute the stress due to bending at the fulcrum (20 in from the pivot) and at the section through the bottom hole. The diameter of each hole is 1.25 in.

77. For the lever in P3-76, determine the maximum stress if the attachment point is moved to each of the other two holes.

78. Figure P3-78 shows a shaft that is loaded only in bending. Bearings are located at points B and D to allow the shaft to rotate. Pulleys at A, C, and E carry cables that support loads from below while allowing

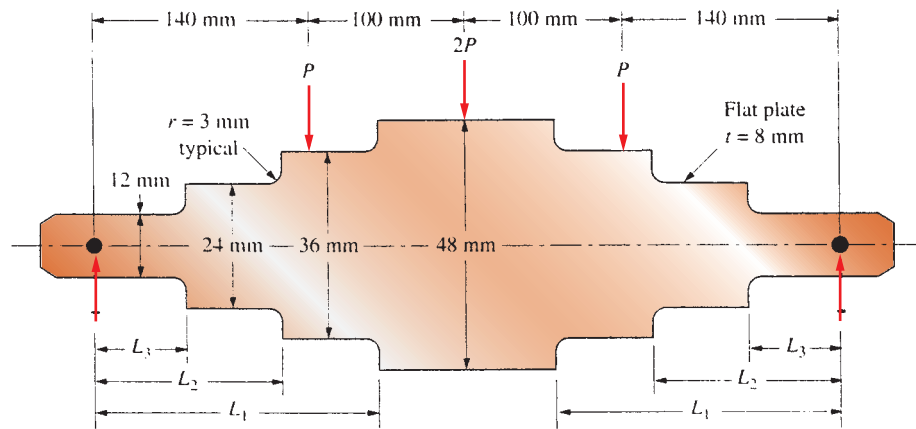
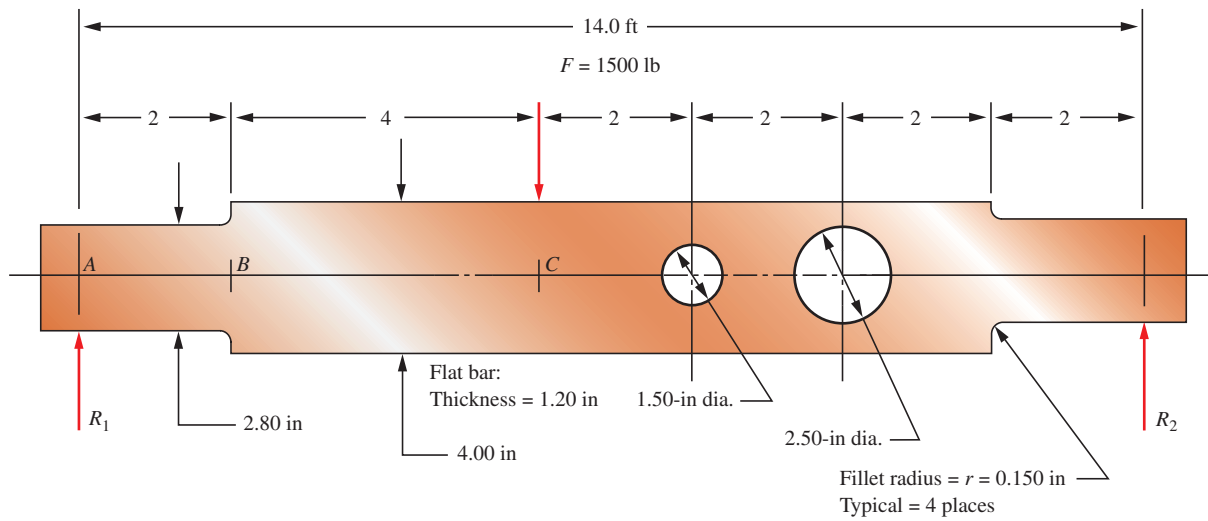


FIGURE P3-73 Stepped flat bar for Problem 73

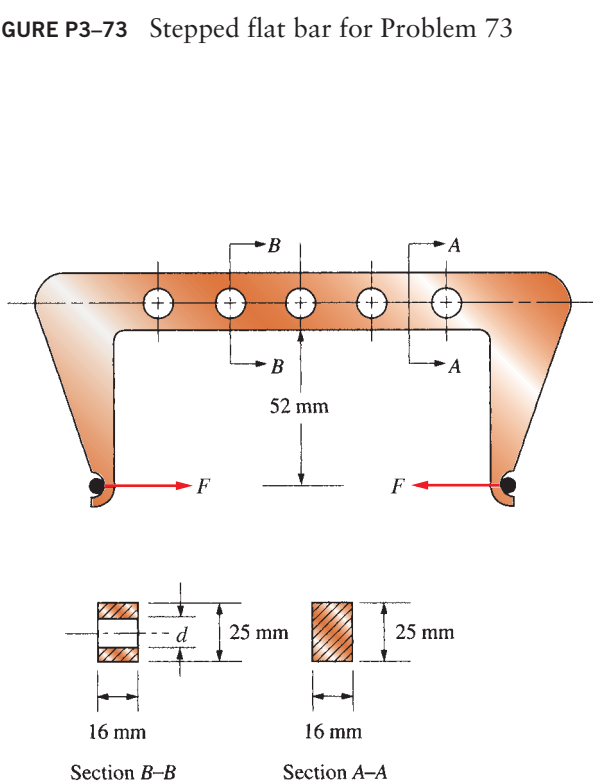


FIGURE P3-74 Bracket for Problems 74 and 75

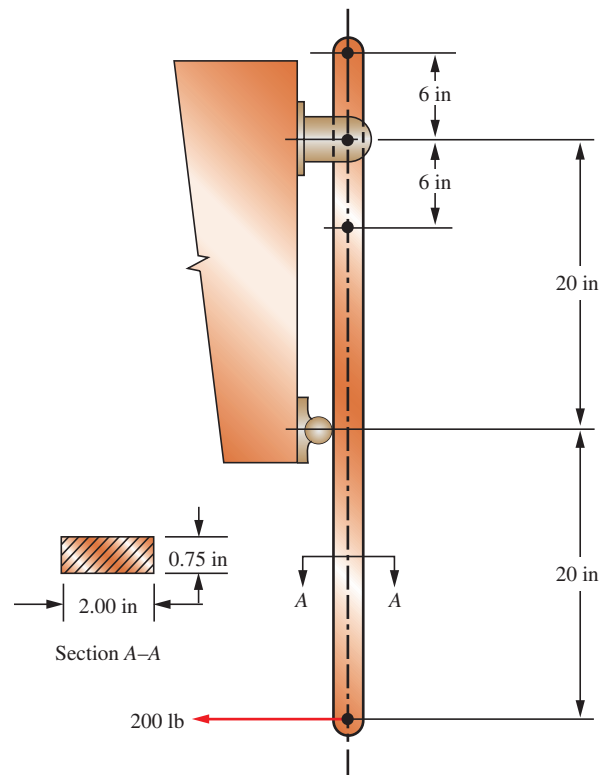


FIGURE P3-76 Lever for Problems 76 and 77

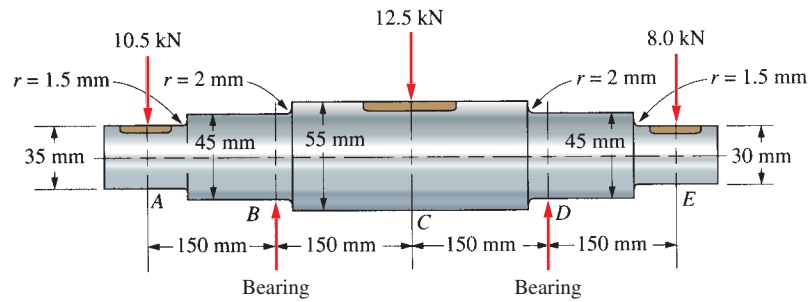


FIGURE P3-78 Data for Problem 78

the shaft to rotate. Compute the maximum stress due to bending in the shaft considering stress concentrations.

Lug Joints

79. For the vertical beam support shown in Figure P3-69, design the clevis joint at point A according to the recommendations described in Section 3-21 on stress concentrations. Use the given pin diameter, $d = 8.0$ mm, and lug width, $w = 20.0$ mm, using the terminology shown in Figure 3-28. The primary design decisions are the thickness of the lug, t , and the materials for the lug and the pin. Work toward a design factor of $N = 5$ based on ultimate strength for both tension in the lug and shearing of the pin.
80. Repeat Problem 3-79 for Joint A in Figure P3-8, using the data from Problem 3-8. All dimensions for the pin and the lug of the clevis are to be specified.

Curved Beams

81. A hanger is made from ASTM A36 structural steel bar with a square cross section, 10 mm on a side, as shown in Figure P3-81. The radius of curvature is 150 mm to the inside surface of the bar. Determine the load F that would cause yielding of the steel.

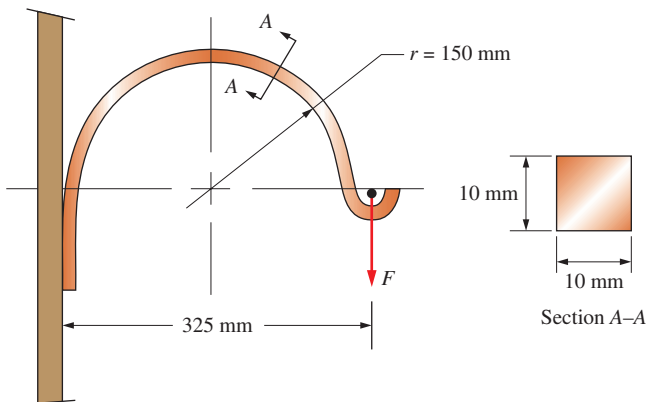


FIGURE P3-81 Hanger for Problem 3-81

82. A coping saw frame shown in Figure P3-82 is made from SAE 1020 CD steel. A screw thread in the handle draws the blade of the saw into a tension of 120 N. Determine the resulting design factor based on yield strength in the area of the corner radii of the frame.

83. Repeat Problem 3-82 for the hacksaw frame shown in Figure P3-83 when the tensile force in the blade is 480 N.

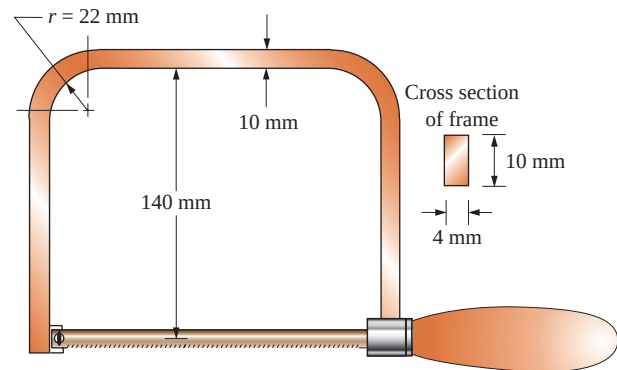


FIGURE P3-82 Coping saw frame for Problem 3-82

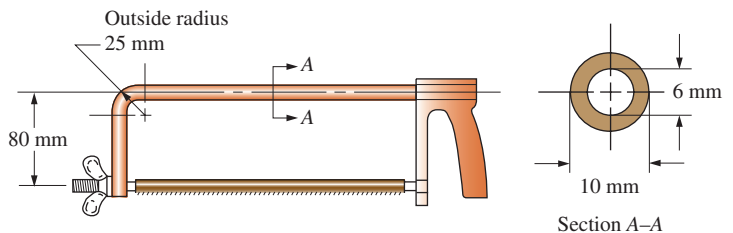


FIGURE P3-83 Hacksaw frame for Problem 3-83

84. Figure P3-84 shows a hand garden tool used to break up soil. Compute the force applied to the end of one prong that would cause yielding in the curved area. The tool is made from cast aluminum, alloy 356.0-T6.
85. Figure P3-85 shows a basketball backboard and goal attached to a steel pipe that is firmly cemented into the ground. The force, $F = 230$ lb, represents a husky player hanging from the back of the rim. Compute the design factor based on yield strength for the pipe if it is made from ASTM A53 Grade B structural steel.
86. The C-clamp in Figure P3-86 is made of cast zinc, ZA12. Determine the force that the clamp can exert for a design factor of 3 based on ultimate strength in either tension or compression.

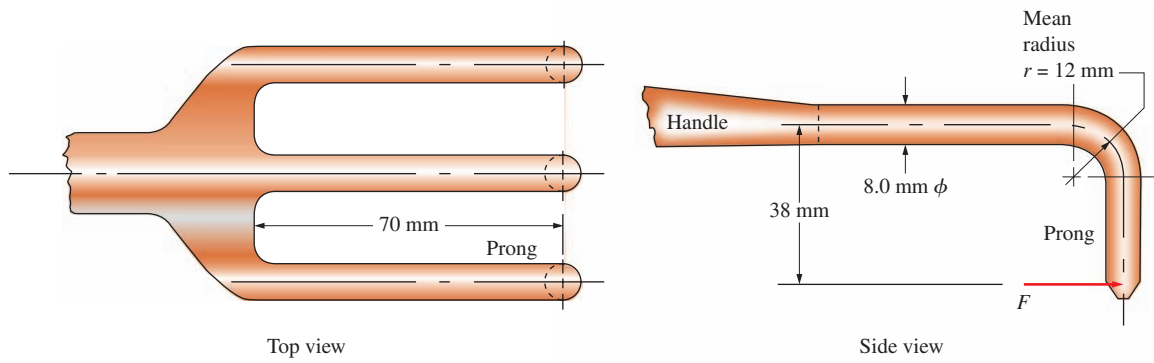


FIGURE P3-84 Garden tool for Problem 3-84

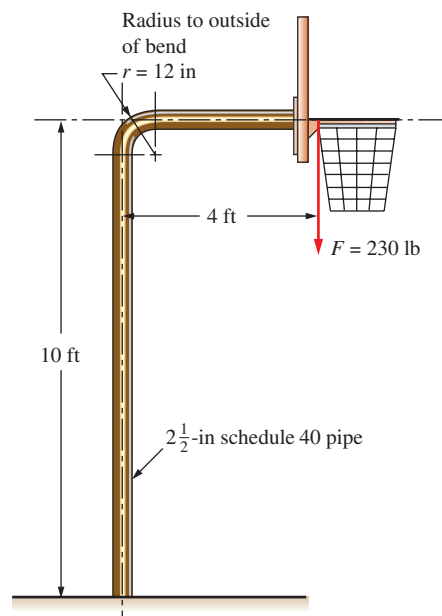


FIGURE P3-85 Basketball backboard for Problem 3-85

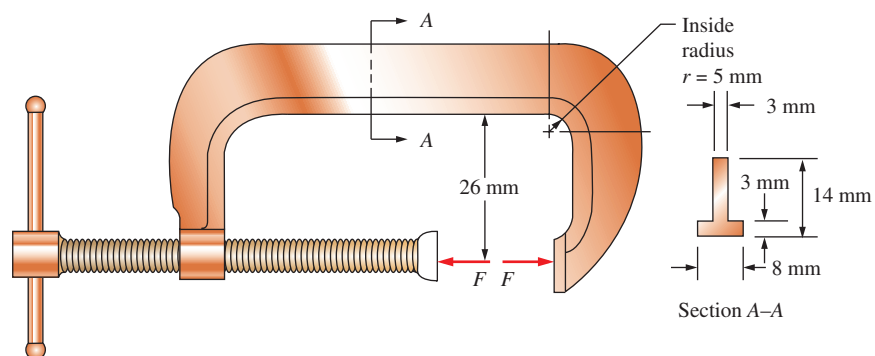


FIGURE P3-86 C-clamp for problem 3-86

COMBINED STRESSES AND STRESS TRANSFORMATION

The Big Picture

You Are the Designer

- 4-1 Objectives of This Chapter
- 4-2 General Case of Combined Stress
- 4-3 Stress Transformation
- 4-4 Mohr's Circle and Tresca and von Mises Stresses
- 4-5 Mohr's Circle Practice Problems
- 4-6 Mohr's Circle for Special Stress Conditions
- 4-7 Analysis of Complex Loading Conditions

THE BIG PICTURE

Combined Stresses and Stress Transformation

Discussion Map

- You must build your ability to analyze more complex parts and loading patterns.

Discover

Find products around you that have complex geometries or loading patterns.

Discuss these products with your colleagues.

This chapter helps you analyze complex objects to determine maximum stresses. We will use *Mohr's circle*, a graphical tool for stress transformation, as an aid in understanding how stresses can be transformed to obtain principal stresses.

In Chapter 3, you reviewed the basic principles of stress and deformation analysis, practiced the application of those principles to machine design problems, and solved some problems by superposition when two or more types of loads caused normal, either tensile or compressive, stresses.

But what happens when the loading pattern is more complex?

Many practical machine components experience combinations of normal and shear stresses. Sometimes the pattern of loading or the geometry of the component causes the analysis to be very difficult to solve directly using the methods of basic stress analysis.

Look around you and identify products, parts of structures, or machine components that have a more complex loading or geometry. Perhaps some of those identified in **The Big Picture** for Chapter 3 have this characteristic.

Discuss how the selected items are loaded, where the maximum stresses are likely to occur, and how the loads and the geometry are related. Did the designer

tailor the shape of the object to be able to carry the applied loads in an efficient manner? How are the shape and the size of critical parts of the item related to the expected stresses?

When we move on to **Chapter 5: Design for Different Types of Loading**, we will need tools to determine the magnitude and the direction of maximum shear stresses or maximum principal (normal) stresses.

Completing this chapter will help you develop a clear understanding of the distribution of stress in a load-carrying member, and it will help you determine the maximum stresses, either normal or shear, so that you can complete a reliable design or analysis.

Some of the techniques of combining stresses require the application of fairly involved equations. A graphical tool, *Mohr's circle*, can be used as an aid in completing the analysis. Applied properly, the method is precise and should aid you in understanding how the stresses can be represented in a complex load-carrying member. It should also help you correctly use the commercially available stress analysis software.

YOU ARE THE DESIGNER

Your company is designing a special machine to test a high-strength fabric under prolonged exposure to a static load to determine whether it continues to deform a greater amount with time. The tests will be run at a variety of temperatures requiring a controlled environment around the test specimen. Figure 4-1 shows the general layout of one proposed design. Two rigid supports are available

at the rear of the machine with a 24-in gap between them. The line of action of the load on the test fabric is centered on this gap and 15.0 in out from the middle of the supports. You are asked to design a bracket to hold the upper end of the load frame.

Assume that one of your design concepts uses the arrangement shown in Figure 4-2. Two circular bars are bent 90°. One end of

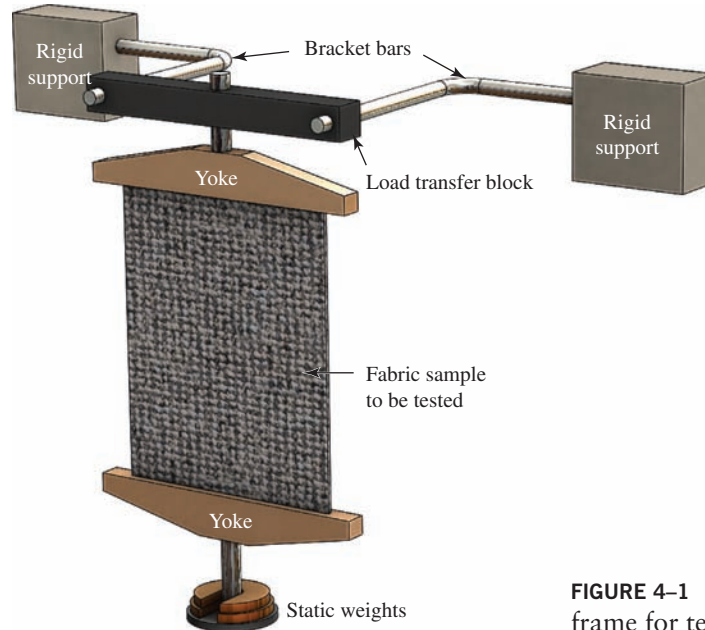


FIGURE 4-1 General concept for a load frame for testing fabric strength

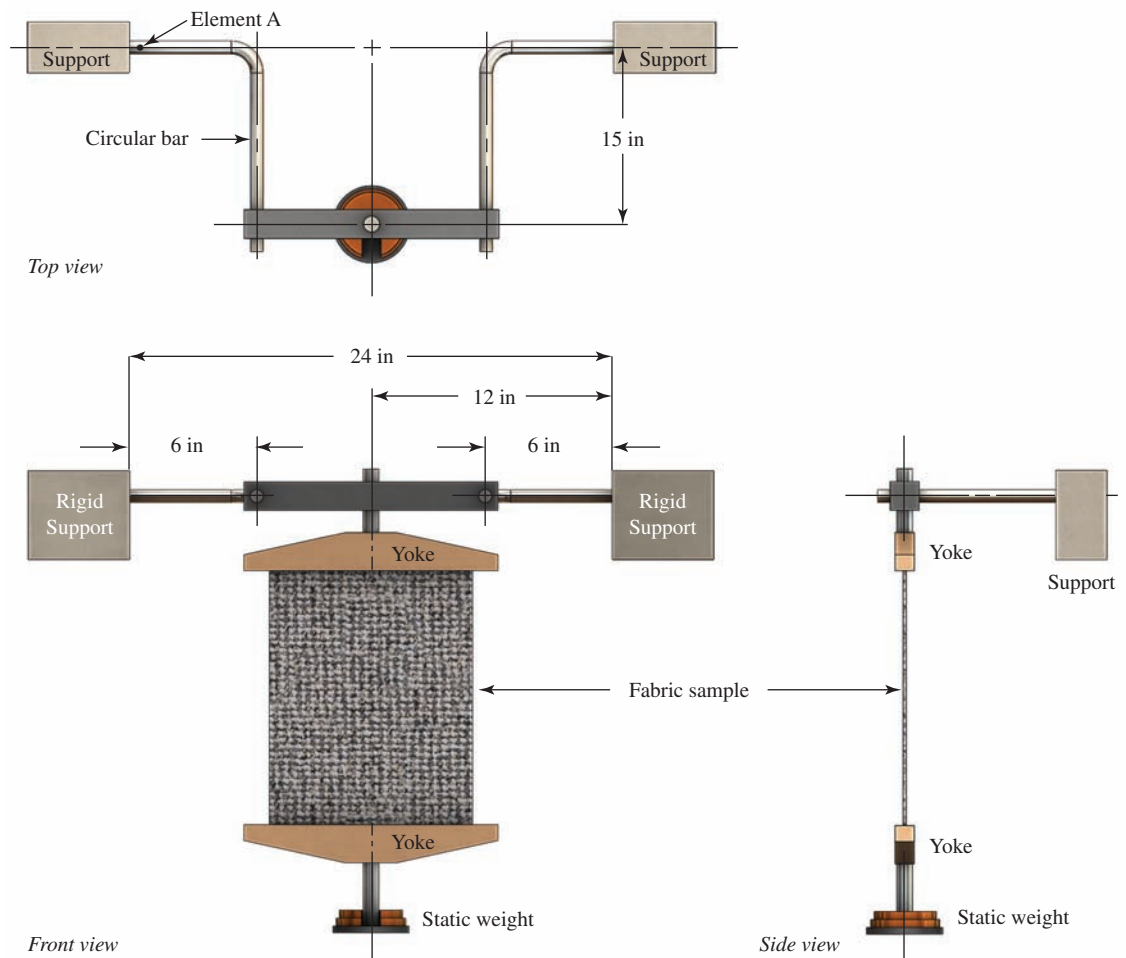


FIGURE 4-2
Proposed
bracket design

each bar is securely welded to the vertical support surface. A flat load transfer block is attached across the outboard end of each bar so that the load is shared evenly by the two bars.

One of your design problems is to determine the maximum stress that exists in the bent bars to ensure that they are safe. What kinds of stress are developed in the bars? Where are the stresses likely to be the greatest? How can the magnitude of the stresses be computed? Note that the part of the bar near its attachment to the support has a combination of stresses exerted on it.

Consider the element on the top surface of the bar, labeled element *A* in Figure 4–2. The moment caused by the force acting at an

extension of 6.0 in from the support places element *A* in tension due to the bending action. The torque caused by the force acting 15.0 in out from the axis of the bar at its point of support creates a torsional shear stress on element *A*. Both of these stresses act in the *x*–*y* plane, subjecting element *A* to a combined normal and shear stress. How do you analyze such a stress condition? How do the tensile and shear stresses act together? What are the maximum normal stress and the maximum shear stress on element *A*, and where do they occur?

You would need such answers to complete the design of the bars. The material in this chapter will enable you to complete the necessary analyses. ■

4-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Illustrate a variety of combined stresses on stress elements.
2. Analyze a load-carrying member subjected to combined stress to determine the principal stresses, the maximum normal stress and the maximum shear stress on any given element.
3. Determine the coordinate system in which the principal stresses are aligned.
4. Determine the state of stress on an element in any specified coordinate system.
5. Draw the complete Mohr's circle as an aid in completing the analyses for the maximum stresses.

4-2 GENERAL CASE OF COMBINED STRESS

To visualize the general case of combined stress, it is helpful to consider a small element of the load-carrying member on which combined normal and shear stresses act. Figure 4–3(a) shows the general case of a three-dimensional stress element.

The most general representation of stress at a point is three dimensional (triaxial), visualized as a cube-like structure shown in Figure 4–3(a) on an *x*-, *y*-, and *z*-axis system. The following characteristics describe the system of stresses that can act on the cubic element.

1. Any of the six faces of the cube can be subjected to a normal stress, either tension or compression. The normal stress vectors act in pairs on parallel opposite faces to either pull or push on the sides of the cube.
2. Similarly, two shearing stresses can act on any face, perpendicular to each other.
 - a. Note that each shearing stress on a given face has an equal counterpart on the parallel opposite face to create the shearing action.
 - b. The pairs of shearing stresses acting on perpendicular faces are numerically equal but act in opposite directions in order to maintain equilibrium.

Therefore, the set of possible stresses acting on the element is

Normal stresses:	σ_x	σ_y	σ_z
Shearing stresses:	τ_{xy} and τ_{yx}	τ_{xz} and τ_{zx}	τ_{yz} and τ_{zy}

- c. Since many combined-stress cases in the real world are in a two-dimensional stress state (plane stress), as shown in Figures 4–3(b) and (c), we will consider a two-dimensional stress condition in this discussion. The *x*- and *y*-axes are aligned with corresponding axes on the member being analyzed. Only shearing stresses on the visible faces are shown in part (b) of the figure.
3. The normal stresses, σ_x and σ_y , could be due to a direct tensile force or due to bending. All normal stresses are shown as tensile, positive vectors in

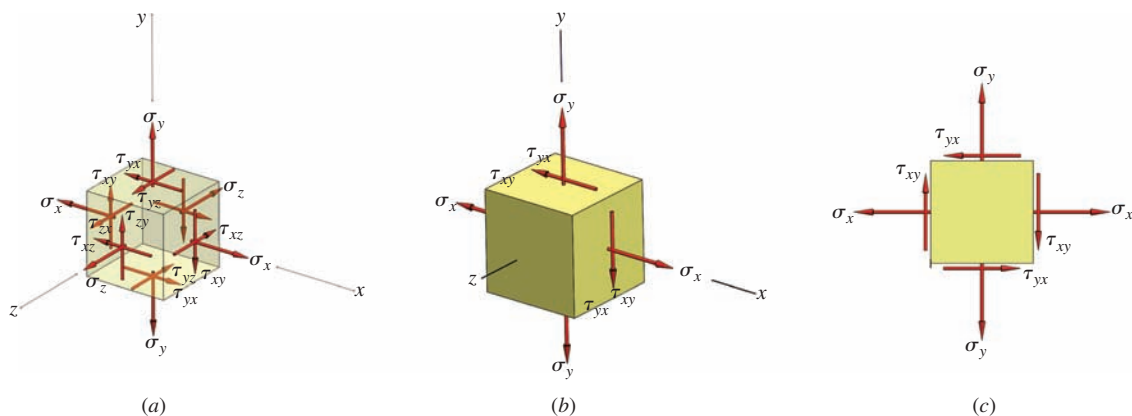


FIGURE 4-3 General three-dimensional (a) and two-dimensional stress states (b) and (c)

Figure 4–3. If the normal stresses were compressive (negative), the vectors would be pointing in the opposite sense, into the stress element.

The shear stress could be due to direct shear, torsional shear, or vertical shear stress. The double-subscript notation helps to orient the direction of shear stresses. For example, τ_{xy} indicates the shear stress acting on the element face that is perpendicular to the x -axis, and the direction of the shear stress is parallel to the y -axis.

Sign Convention

A positive shear stress is one that tends to rotate the stress element clockwise. In Figure 4–3, τ_{xy} is positive and τ_{yx} is negative. Their magnitudes must be equal to maintain the element in equilibrium. It is necessary to determine the magnitudes and the signs of each of these stresses in order to show them properly on the stress element. Example Problem 4–1, which follows the definition of principal stresses, illustrates the process.

With the stress element defined, the objectives of the remaining analysis are to use stress transformation techniques to determine the principal stresses, the maximum and minimum normal stresses, the maximum shear stress, and the planes on which these stresses occur.

4-3 STRESS TRANSFORMATION

In mechanical design, we need to evaluate if the applied stresses would exceed the capability of the selected material. The capability of the material (e.g., yield strength) is typically obtained from one-dimensional uniaxial tensile test. If the stress state at the location of interest is one dimensional, the comparison is straightforward. In reality, a stress element is always in three-dimensional, and it can have six stress components. We will need a methodology to reduce the complexity of the stress state to do the comparison. For static loading, for example, we will reduce these six stress components to one single value and compare it to the yield strength of the material. To achieve this objective, the first step is to perform stress transformation to find the principal stresses. The second step is to find the “effective stress,” a single value, from the three principal stresses for the comparison. In this section, we show the following:

- The process of reducing six stress components (σ_x , σ_y , σ_z , τ_{xy} , τ_{xz} , and τ_{yz}) shown in part (a) of Figure 4–3, to three principal stresses (σ_1 , σ_2 , and σ_3) in a three-dimensional stress state is introduced.
- Similarly, for the two-dimensional plane-stress condition, in part (b) of the figure, the three stress components (σ_x , σ_y , and τ_{xy}), are reduced to three principle stresses. Note that in this case, one of the principal stresses is zero.
- It is important to note that the stress states before and after the transformation are “equivalent.”

Considering a small two-dimensional stress element, the applied normal and shear stresses, σ_x , σ_y , and τ_{xy} , in a given (x – y) coordinate system can be represented as a different set of normal and shear stresses, $\sigma_{x'}$, $\sigma_{y'}$, and $\tau_{x'y'}$, in a different (x – y) coordinate system. The stress components in the new coordinate system can be obtained from the stress components in the original coordinate system through rotating the coordinate frame. The governing formulas for such *plane-stress transformation* follow. (See Reference 1 for the derivations.) Using ϕ as the angle of rotation of the coordinate system,

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\phi + \tau_{xy} \sin 2\phi \quad (4-1)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\phi - \tau_{xy} \sin 2\phi \quad (4-2)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\phi + \tau_{xy} \cos 2\phi \quad (4-3)$$

Maximum and Minimum Principal Stresses

The maximum and the minimum normal stresses can be obtained from differentiating Equations (4–1) and (4–2) with respect to ϕ and setting the results to zero. Using ϕ_σ for the rotation angle required to obtain the maximum and minimum normal stresses, we get:

$$\tan 2\phi_\sigma = \frac{\tau_{xy}}{\left(\frac{\sigma_x - \sigma_y}{2}\right)} \quad (4-4)$$

Solving for ϕ_σ and substituting the result into Equations (4–1) and (4–2) leads to the following:

Maximum normal stress

$$\sigma_1 = \frac{1}{2}(\sigma_x + \sigma_y) + \sqrt{\left[\frac{1}{2}(\sigma_x - \sigma_y)\right]^2 + \tau_{xy}^2} \quad (4-5)$$

Minimum normal stress

$$\sigma_2 = \frac{1}{2}(\sigma_x + \sigma_y) - \sqrt{\left[\frac{1}{2}(\sigma_x - \sigma_y)\right]^2 + \tau_{xy}^2} \quad (4-6)$$

These two stresses are also called the maximum and the minimum principal stresses, such that

$$\sigma_1 > \sigma_2.$$

Particularly in experimental stress analysis, it is important to know the orientation of the principal stresses. The angle of inclination of the planes on which the principal stresses act, called the principal planes, can be found from Equation (4–4).

Angle for Principal Stress Element

$$\phi_\sigma = \frac{1}{2} \arctan \left[\frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2} \right] \quad (4-7)$$

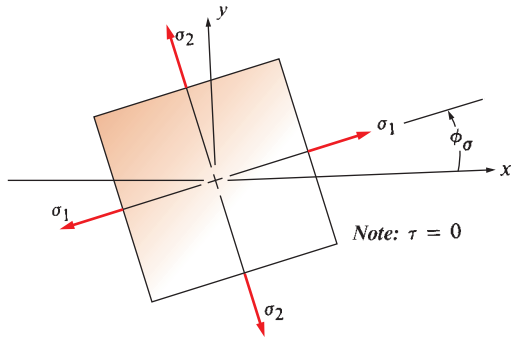


FIGURE 4-4 Principal stress element

The angle ϕ_σ is measured from the positive x -axis of the original stress element to the maximum principal stress σ_1 . Then the minimum principal stress, σ_2 , is on the plane 90° from σ_1 . When the stress element is oriented as discussed so that the principal stresses are acting on it, the shear stress is zero ($\tau_{x'y'} = \tau_{12} = 0$). The resulting stress element is shown in Figure 4-4.

Maximum Shear Stress

When the stress element is rotated to a different orientation, the maximum shear stress can occur. This can be derived by differentiating Equation (4-3) and setting the result equal to zero to obtain

$$\tan 2\phi_\tau = \frac{\left(\frac{\sigma_x - \sigma_y}{2}\right)}{\tau_{xy}} \quad (4-8)$$

where ϕ_τ is the rotation angle to obtain the maximum shear stress. Substituting ϕ_τ into Equation (4-3) results in the following:

Maximum shear stress

$$\tau_{\max} = \sqrt{\left[\frac{1}{2}(\sigma_x - \sigma_y)\right]^2 + \tau_{xy}^2} \quad (4-9)$$

The angle of inclination of the element on which the maximum shear stress occurs is computed from Equation (4-8):

Angle for maximum shear stress element

$$\phi_\tau = \frac{1}{2} \arctan \left[\frac{-(\sigma_x - \sigma_y)/2}{\tau_{xy}} \right] \quad (4-10)$$

The angle between the principal stress element and the maximum shear stress element is always 45° . On the maximum shear stress element, there will be normal stresses of equal magnitude acting perpendicular to the planes on which the maximum shear stresses are acting. These normal stresses have the following value:

Average normal stress

$$\sigma_{\text{avg}} = (\sigma_x + \sigma_y)/2 \quad (4-11)$$

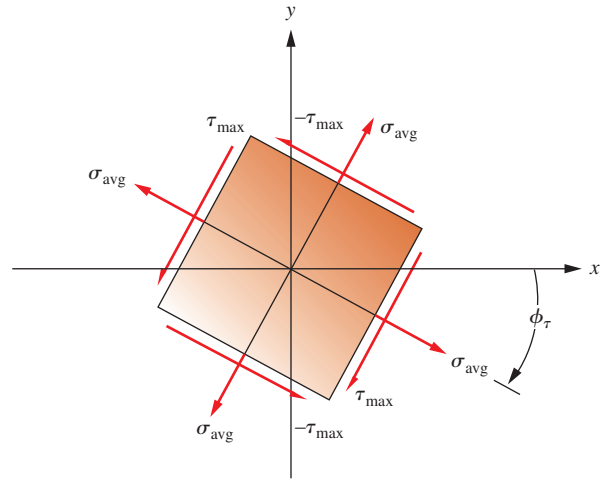


FIGURE 4-5 Maximum shear stress element

Note that this is the average of the two applied normal stresses. The resulting maximum shear stress element is shown in Figure 4-5. Note, as stated above, that the angle between the principal stress element and the maximum shear stress element is always 45° .

Three-Dimensional Stress Transformation

As discussed earlier, the most general state of a stress element has three normal and three shear stress components: σ_x , σ_y , σ_z , τ_{xy} , τ_{yz} , and τ_{zx} . The three-dimensional stress element can be rotated in three dimensions to a certain configuration such that there are three principal stresses, σ_1 , σ_2 , and σ_3 (and no shear stress) acting on the stress element. The derivation of the three-dimensional stress transformation is not presented here. The three principal stresses, however, can be determined by finding the roots of the following cubic equation:

$$\sigma^3 - C_2\sigma^2 + C_1\sigma - C_0 = 0 \quad (4-12)$$

where

$$C_2 = \sigma_x + \sigma_y + \sigma_z$$

$$C_1 = \sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{zx}^2$$

$$C_0 = \sigma_x\sigma_y\sigma_z + 2\tau_{xy}\tau_{yz}\tau_{zx} - \sigma_x\tau_{yz}^2 - \sigma_y\tau_{zx}^2 - \sigma_z\tau_{xy}^2$$

For design decision making (to be presented in Chapter 5), the principal stresses are typically ranked such that

$$\sigma_1 > \sigma_2 > \sigma_3 \quad (4-13)$$

Summary and General Procedure for Analyzing Combined Stresses

The following list gives a summary of the techniques presented in this section; it also outlines the general procedure for applying the techniques to a given stress analysis problem.

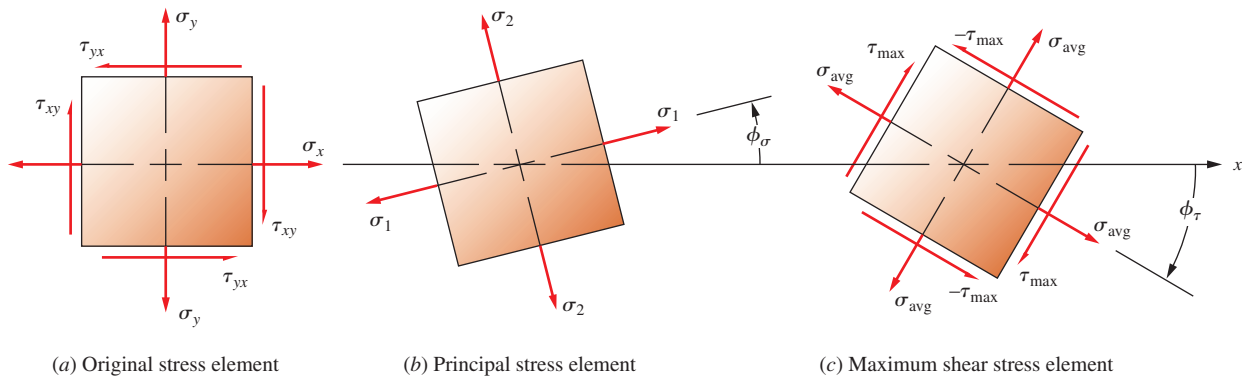


FIGURE 4-6 Relationships among original stress element, principal stress element, and maximum shear stress element for a given loading

GENERAL PROCEDURE FOR COMPUTING PRINCIPAL STRESSES AND MAXIMUM SHEAR STRESSES ▼

1. Decide for which point you want to compute the stresses.
2. Clearly specify the coordinate system for the object, the free-body diagram, and the magnitude and direction of forces.
3. Compute the stresses on the selected point due to the applied forces, and show the stresses acting on a stress element at the desired point with careful attention to directions. Figure 4-3 is a model for how to show these stresses.
4. Compute the principal stresses on the point and the directions in which they act. Use Equations (4-5) to (4-7).
5. Draw the stress element on which the principal stresses act, and show its orientation relative to the original x -axis.

It is recommended that the principal stress element be drawn beside the original stress element to illustrate the relationship between them.

6. Compute the maximum shear stress on the element and the orientation of the plane on which it acts. Also, compute the normal stress that acts on the maximum shear stress element. Use Equations (4-9) to (4-11).
7. Draw the stress element on which the maximum shear stress acts, and show its orientation to the original x -axis. It is recommended that the maximum shear stress element be drawn beside the maximum principal stress element to illustrate the relationship between them.
8. The resulting set of three stress elements will appear as shown in Figure 4-6.

The following example problem illustrates the use of this procedure.

Example Problem 4-1

The shaft shown in Figure 4-7 is supported by two bearings and carries two V-belt sheaves. The tensions in the belts exert horizontal forces on the shaft, tending to bend it in the x - z plane. Sheave B exerts a clockwise torque on the shaft when viewed toward the origin of the coordinate system along the x -axis. Sheave C exerts an equal but opposite torque on the shaft. For the loading condition shown, determine the principal stresses and the maximum shear stress on element K on the front surface of the shaft (on the positive z -side) just to the right of sheave B . Follow the general procedure for analyzing combined stresses given in this section.

Solution

Objective Compute the principal stresses and the maximum shear stresses on element K .

Given Shaft and loading pattern shown in Figure 4-7. The forces at locations B and C are identified as 550 lb and 275 lb, respectively. These forces are determined by the tight side and slack side of the belt forces discussed in Section 12-3.

Analysis Use the general procedure for analyzing combined stresses.

Results Element K is subjected to bending that produces a tensile stress acting in the x -direction. Also, there is a torsional shear stress acting at K . Figure 4-8 shows the shearing force and bending moment diagrams for the shaft and indicates that the bending moment at K is 1540 lb · in. The bending stress is therefore

$$\sigma_x = M/S$$

$$S = \pi D^3/32 = [\pi(1.25 \text{ in})^3]/32 = 0.192 \text{ in}^3$$

$$\sigma_x = (1540 \text{ lb} \cdot \text{in})/(0.192 \text{ in}^3) = 8030 \text{ psi}$$

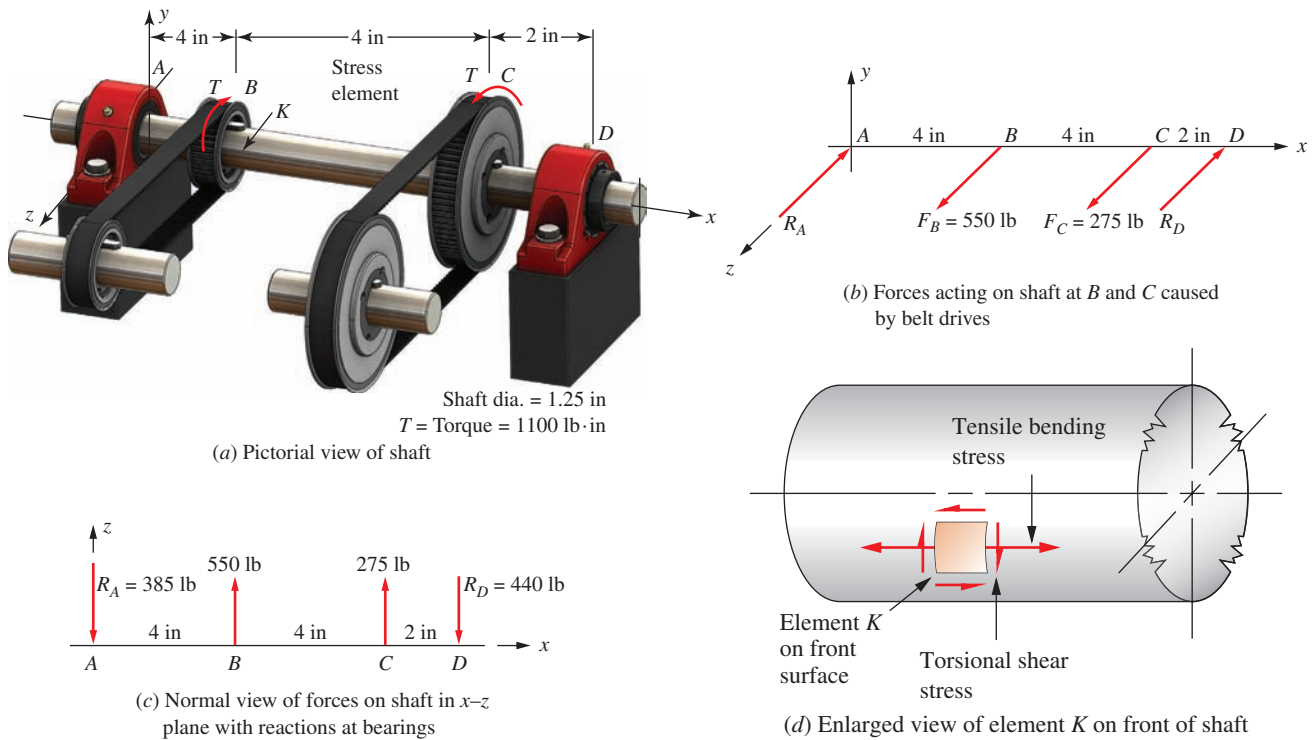


FIGURE 4-7 Shaft supported by two bearings and carrying two V-belt sheaves

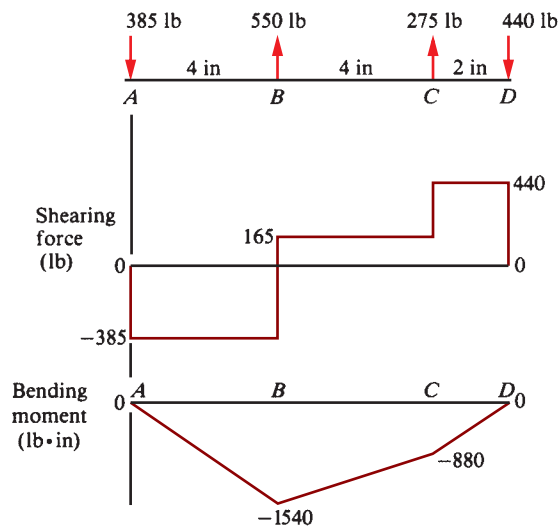


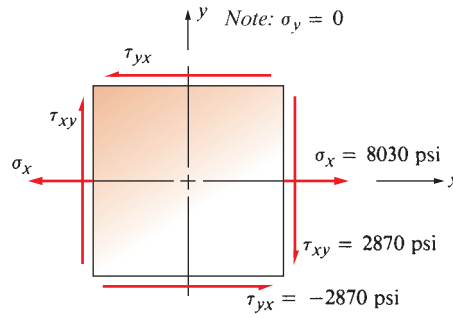
FIGURE 4-8 Shearing force and bending moment diagrams for the shaft

The torsional shear stress acts on element K in a way that causes a downward shear stress on the right side of the element and an upward shear stress on the left side. This action results in a tendency to rotate the element in a *clockwise* direction, which is the *positive* direction for shear stresses according to the standard convention. Also, the notation for shear stresses uses double subscripts. For example, τ_{xy} indicates the shear stress acting on the face of an element that is perpendicular to the x -axis and parallel to the y -axis. Thus, for element K ,

$$\tau_{xy} = T/Z_p$$

$$Z_p = \pi D^3/16 = \pi (1.25 \text{ in})^3/16 = 0.383 \text{ in}^3$$

$$\tau_{xy} = (1100 \text{ lb}\cdot\text{in})/(0.383 \text{ in}^3) = 2870 \text{ psi}$$

FIGURE 4-9 Stresses on element K

The values of the normal stress, σ_x , and the shear stress, τ_{xy} , are shown on the stress element K in Figure 4-9. Note that the stress in the y -direction is zero for this loading. Also, the value of the shear stress, τ_{yx} , must be equal to τ_{xy} , and it must act as shown in order for the element to be in equilibrium.

We can now compute the principal stresses on the element, using previously developed equations. Noting that σ_y is zero, the maximum principal stress is

$$\begin{aligned}\sigma_1 &= \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ \sigma_1 &= [(8030 - 0 \text{ psi})/2] + \sqrt{[(8030 \text{ psi} - 0 \text{ psi})/2]^2 + (2870 \text{ psi})^2} \\ \sigma_1 &= 4015 \text{ psi} + 4935 \text{ psi} = 8950 \text{ psi}\end{aligned}\quad (4-5)$$

The minimum principal stress is

$$\begin{aligned}\sigma_2 &= \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ \sigma_2 &= [(8030 \text{ psi} - 0 \text{ psi})/2] - \sqrt{[(8030 \text{ psi} - 0 \text{ psi})/2]^2 + (2870 \text{ psi})^2} \\ \sigma_2 &= 4015 \text{ psi} - 4935 \text{ psi} = -920 \text{ psi (compression)}\end{aligned}\quad (4-6)$$

The direction in which the maximum principal stress acts is

$$\begin{aligned}\phi_\sigma &= \frac{1}{2} \arctan [2\tau_{xy}/(\sigma_x - \sigma_y)] \\ \phi_\sigma &= \frac{1}{2} \arctan [(2)(2870 \text{ psi})/(8030 \text{ psi} - 0 \text{ psi})] = +17.8^\circ\end{aligned}\quad (4-7)$$

The positive sign calls for a *clockwise* rotation of the element.

The principal stresses can be shown on a stress element as illustrated in Figure 4-10(b). Note that the element is shown in relation to the original element to emphasize the direction of the principal stresses in relation to the original x -axis. The positive sign for ϕ_σ indicates that the principal stress element is rotated *clockwise* from its original position.

Now the maximum shear stress element can be defined using Equations (4-9) through (4-11):

$$\begin{aligned}\tau_{\max} &= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ \tau_{\max} &= \sqrt{(8030 \text{ psi} - 0)/2)^2 + (2870 \text{ psi})^2} \\ \tau_{\max} &= \pm 4935 \text{ psi}\end{aligned}\quad (4-9)$$

The two pairs of shear stresses $+\tau_{\max}$ and $-\tau_{\max}$ are equal in magnitude but opposite in direction.

The orientation of the element on which the maximum shear stress acts is found from Equation (4-10):

$$\begin{aligned}\phi_\tau &= \frac{1}{2} \arctan [-(\sigma_x - \sigma_y)/2\tau_{xy}] \\ \phi_\tau &= \frac{1}{2} \arctan [-(8030 \text{ psi} - 0)/(2)(2870 \text{ psi})] = -27.2^\circ\end{aligned}\quad (4-10)$$

The negative sign calls for a *counterclockwise* rotation of the element.

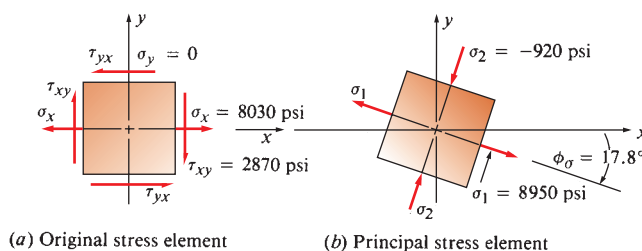


FIGURE 4-10 Principal stress element

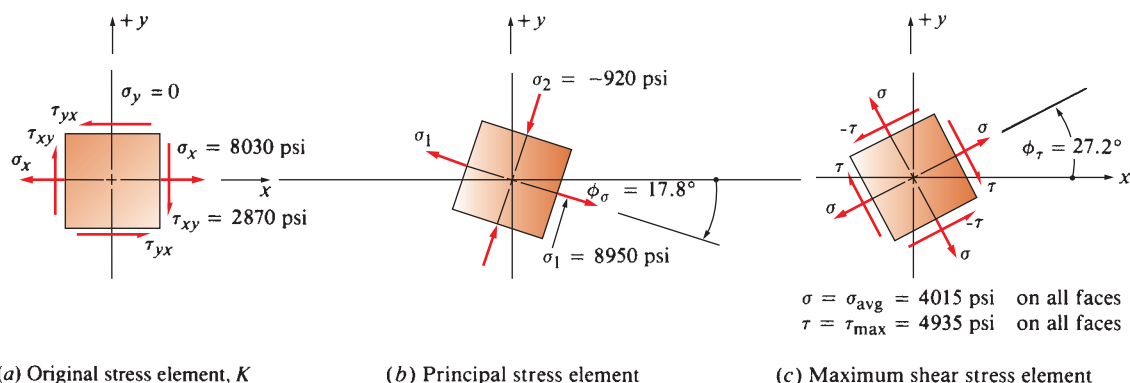


FIGURE 4-11 Relation of maximum shear stress element to the original stress element and the principal stress element

There are equal normal stresses acting on the faces of this stress element, which have the value of

$$\sigma_{\text{avg}} = (\sigma_x + \sigma_y)/2 \quad (4-11)$$

$$\sigma_{\text{avg}} = (8030 \text{ psi} - 0)/2 = 4015 \text{ psi}$$

Comments Figure 4-11(c) shows the stress element on which the maximum shear stress acts in relation to the original stress element. Note that the angle between this element and the principal stress element is 45° . It is typical to show all three stress elements side by side as shown in Figure 4-9.

Examine the results of Example Problem 4-1. The maximum principal stress, $\sigma_1 = 8950$ psi, is 11 percent greater than the value of $\sigma_x = 8030$ psi computed for the bending stress in the shaft acting in the x -direction. The maximum shear stress, $\tau_{\text{max}} = 4935$ psi, is 72% greater than the computed applied torsional shear stress of $\tau_{xy} = 2870$ psi. You will see in Chapter 5 that either the maximum normal stress or the maximum shear stress is often required for accurate failure prediction and for safe design decisions. The angles of the final stress elements also predict the alignment of the most damaging stresses that can be an aid in experimental stress analysis and the analysis of actual failed components.

4-4 MOHR'S CIRCLE

The process of computing the principal stresses and the maximum shear stress shown in Example Problem 4-1 may seem somewhat abstract. These same results can be obtained using a method called *Mohr's circle*, which

is discussed next. This method uses a combination of a graphical aid and simple calculations. With practice, the use of Mohr's circle should provide you with a more intuitive feel for the variations in stress that exist at a point in relation to the angle of orientation of the stress element. In addition, it provides a streamlined approach to determining the stress condition on any plane of interest. Because of the many terms and signs involved, and the many calculations required in the computation of the principal stresses and the maximum shear stress, there is a rather high probability of error. Using the graphic aid Mohr's circle helps to minimize errors and gives a better "feel" for the stress condition at the point of interest.

After Mohr's circle is constructed, it can be used for the following:

1. Finding the maximum and minimum principal stresses and the directions in which they act.
2. Finding the maximum shear stresses and the orientation of the planes on which they act.

3. Finding the value of the normal stresses that act on the planes where the maximum shear stresses act.
4. Finding the values of the normal and shear stresses that act on an element with any orientation.

The data needed to construct Mohr's circle are, of course, the same as those needed to compute the preceding values, because the graphical approach is an exact analogy to the computations.

If the normal and shear stresses that act on any two mutually perpendicular planes of an element are known, the circle can be constructed and any of items 1 through 4 can be found.

Mohr's circle is actually a plot of the combinations of normal and shearing stresses that exist on a stress element for all possible angles of orientation of the element. This method is particularly valuable in experimental stress analysis work because the results obtained from many types of standard strain gage instrumentation techniques give the necessary inputs for the creation of Mohr's circle. (See Reference 1.) When the principal stresses and the maximum shear stress are known, the complete design and analysis can be done, using the various theories of failure discussed in Chapter 5.

The following *Procedure for Constructing Mohr's Circle* first focuses on given stresses in a 2D plane only. Finding the principal stresses 1 and 2 are the primary goal along with the maximum shear stress and pertinent angles.

After this procedure is developed, an additional section called *Mohr's Circles for Three-Dimensional Stresses* describes how the more general 3D stress system can be analyzed.

PROCEDURE FOR CONSTRUCTING MOHR'S CIRCLE FOR A 2D STRESS SYSTEM ▼

1. Perform the stress analysis to determine the magnitudes and directions of the normal and shear stresses acting at the point of interest.
2. Draw the stress element at the point of interest as shown in Figure 4-12(a). Normal stresses on any two mutually perpendicular planes are drawn with tensile stresses positive—projecting outward from the element. Compressive stresses are negative—directed inward on the face. Note that the *resultants* of all normal stresses acting in the chosen directions are plotted. Shear stresses are considered to be positive if they tend to rotate the element in a *clockwise* (cw) direction, and negative otherwise. Note that on the stress element illustrated, σ_x is positive, σ_y is negative, τ_{xy} is positive, and τ_{yx} is negative. This assignment is arbitrary for the purpose of illustration. In general, any combination of positive and negative values could exist.
3. Refer to Figure 4-12(b). Set up a rectangular coordinate system in which the positive horizontal axis represents positive (tensile) normal stresses, and the positive vertical axis represents positive (clockwise) shear stresses. Thus, the plane created will be referred to as the σ - τ plane.

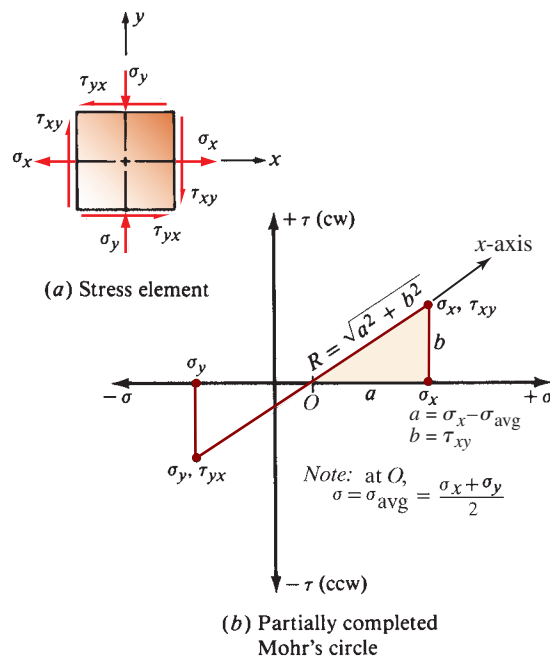


FIGURE 4-12 General form of a partially completed 2D Mohr's circle, Steps 1-7

4. Plot points on the σ - τ plane corresponding to the stresses acting on the faces of the stress element. If the element is drawn in the x - y plane, the two points to be plotted are σ_x , τ_{xy} and σ_y , τ_{yx} . These are two points on Mohr's circle.
5. Draw the line connecting the two points. The resulting line crosses the σ -axis at the center of Mohr's circle at the average of the two applied normal stresses, where

$$\sigma_{\text{avg}} = (\sigma_x + \sigma_y)/2$$

The center of Mohr's circle is called O in Figure 4-12.

6. Now it is important to note that the line from the center of Mohr's circle, at O , and through the first plotted point, (σ_x, τ_{xy}) , represents the x -axis from the original stress element. **Draw and label an extension of this line at this time.** The direction of this line, of course, corresponds to a known direction on the actual component being analyzed. Angles of rotation for the principle stress element and the maximum shear stress element, found from Steps 11-14 later in this process, will be measured from this x -axis.
7. Note in Figure 4-12(b) that a right triangle has been formed, having the sides a , b , and R , where

$$R = \sqrt{a^2 + b^2}$$

By inspection, we can see that

$$\begin{aligned} a &= \sigma_x - \sigma_{\text{avg}} \\ b &= \tau_{xy} \end{aligned}$$

Also compute the angle α within the triangle between lines a and R . You can see that

$$\alpha = \tan^{-1}(b/a) \quad \text{or} \quad \alpha = \sin^{-1}(b/R)$$

we can now proceed with the construction of the circle.

8. Draw the complete circle with the center at O and a radius of R , as shown in Figure 4-13(a).

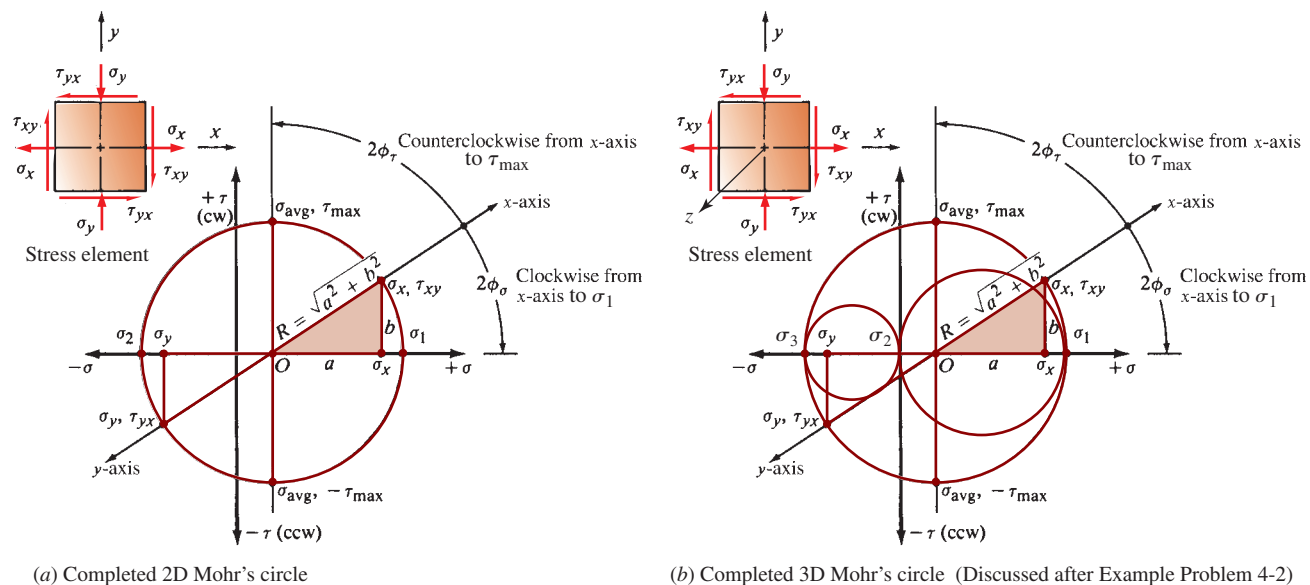


FIGURE 4-13 General form for a completed 2D Mohr's circle, Steps 8–14; Complete 3D Mohr's circle (discussed later)

9. Find σ_1 , the maximum principal stress, at the point where the circle crosses the σ -axis at the right. Note that $\sigma_1 = \sigma_{avg} + R$. Because the Mohr's circle represents a plot of all possible combinations of normal and shear stresses on the element for any angle of orientation, it stands to reason that σ_1 lies at the right end of the horizontal diameter.
10. Find σ_2 , the minimum principal stress, at the left end of the horizontal diameter. Note that $\sigma_2 = \sigma_{avg} - R$.
11. Find τ_{max} , the maximum shear stress, at the top end of the vertical diameter of the circle. By observation, $\tau_{max} = R$, the radius of the circle. Note also that the coordinates for the point at the top of the circle are $(\sigma_{avg}, \tau_{max})$.

At this point in the process, it is important to realize that angles on Mohr's circle are actually double the true angles.

The following steps define the general method for finding the angles of orientation of the principal stress element and the maximum shear stress element in relation to the original x -axis, located in Step 6 and shown in Figure 4-13(a). Now we can observe that the line from the center of the circle to the second

plotted point, (σ_y, τ_{yx}) , represents the y -axis from the original stress element. Of course, the x -axis and the y -axis are truly 90° apart, whereas they are 180° apart on Mohr's circle, illustrating the double-angle phenomenon.

12. Find the angle called $2\phi_\sigma$, always measured **from the x -axis to the σ -axis**, and **note the direction—either clockwise or counterclockwise**. In the **current problem**, we can observe that $2\phi_\sigma = \alpha$, the angle found in Step 7, and the rotation is clockwise. **However, in other problems, the angle must be found from the geometry of the circle.** (This is illustrated in example practice problems that follow.) Now compute $\phi_\sigma = 2\phi_\sigma/2$.
13. Draw the *principal stress element* as shown in Figure 4-14(b). In Part (a) of the figure, we have reproduced the *original stress element* to indicate the direction of the x -axis and we draw a new element in relation to that axis at an angle of ϕ_σ . The new element must be rotated in the same direction, clockwise or counterclockwise, as observed in Step 12.
 - a. On the face of the element found from the rotation, draw the vector σ_1 .

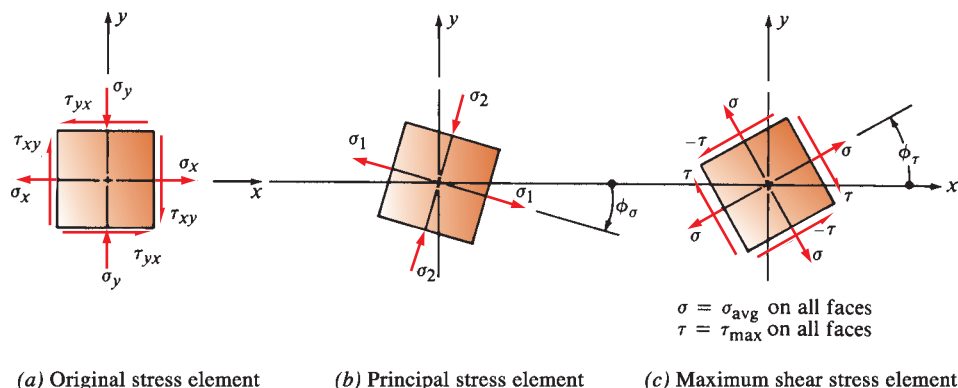


FIGURE 4-14 Display of results from 2D Mohr's circle

- b. Then show that the vector σ_1 is acting on the opposite parallel face in the opposite sense to indicate the tension or compression stress.
 - c. Draw two σ_2 vectors on the other perpendicular faces in the proper sense.
 - d. Note that **on the principal stress element, the shear stress is always zero**. This is evident from the Mohr's circle where the coordinates of the principal stresses are always $(\sigma_1, 0)$ and $(\sigma_2, 0)$, indicating zero shear stress.
14. Find the angle called $2\varphi_r$, always measured **from the x -axis to the $\tau_{\max}$ -axis**, and **note the direction—either clockwise or counterclockwise**. In the **current problem**, we can observe that $2\varphi_r = 90^\circ - \alpha$, the angle found in Step 7, and the rotation is counterclockwise. **However, in other problems, the angle must be found from the geometry of the circle**. Now compute $\varphi_r = 2\varphi_r/2$.
15. Draw the **maximum shear stress element** as shown in Figure 4–14(c). Here, we draw a new element in relation to the x -axis at an angle of φ_r . The new element must be rotated in the direction of either clockwise or counterclockwise, as observed in Step 14.
 - a. On the face of the element found from the rotation, draw the vector $\tau_{\max}$.
 - b. Then show that the vector $\tau_{\max}$ is acting on the opposite parallel face in the opposite sense to indicate the shearing action.
 - c. Draw two $-\tau_{\max}$ vectors on the other perpendicular faces. The combination of all four shearing vectors must indicate equilibrium of the element.
 - d. Note that **on the maximum shear stress element, the average normal stress acts on all four faces**. This is evident from the Mohr's circle where the coordinates of the maximum shear stress is always $(\sigma_{\text{avg}}, \tau_{\max})$ and for the minimum shear stress it is always $(\sigma_{\text{avg}}, -\tau_{\max})$.
16. The final step in the process of preparing the 2D Mohr's circle is to summarize the primary results, typically: σ_1 , σ_2 , $\tau_{\max}$, σ_{avg} , φ_σ , and φ_τ .

We will now illustrate the construction of a 2D Mohr's circle by using the same data as in Example Problem 4–1, in which the principal stresses and the maximum shear stress were computed directly from the equations.

Example Problem 4–2

The shaft shown in Figure 4–7 is supported by two bearings and carries two V-belt sheaves. The tensions in the belts exert horizontal forces on the shaft, tending to bend it in the x - z plane. Sheave B exerts a clockwise torque on the shaft when viewed toward the origin of the coordinate system along the x -axis. Sheave C exerts an equal but opposite torque on the shaft. For the loading condition shown, determine the principal stresses and the maximum shear stress on element K on the front surface of the shaft (on the positive z -side) just to the right of sheave B . Use the procedure for constructing the 2D Mohr's circle in this section.

Solution

Objective Determine the principal stresses and the maximum shear stresses on element K .

Given Shaft and loading pattern shown in Figure 4–7.

Analysis Use the *Procedure for Constructing a 2D Mohr's Circle*. Some intermediate results will be taken from the solution to Example Problem 4–1 and from Figures 4–7 to 4–9.

Results **Steps 1 and 2.** The stress analysis for the given loading was completed in Example Problem 4–1. Figure 4–15 is identical to Figure 4–9 and represents the results of Step 2 of the 2D Mohr's circle procedure, the original stress element.

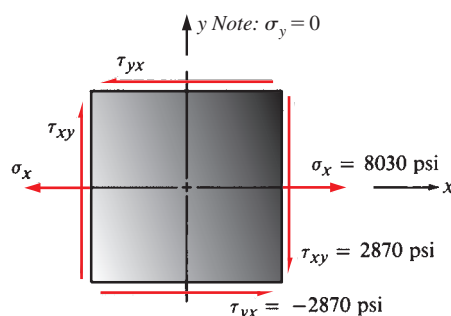


FIGURE 4–15 Stresses on element K

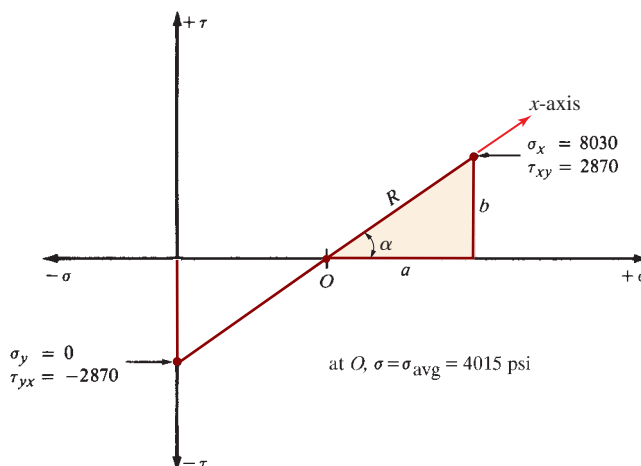


FIGURE 4-16 Partially completed 2D Mohr's circle

Steps 3–6. Figure 4-16 shows the results. The first point plotted was

$$\sigma_x = 8030 \text{ psi}, \tau_{xy} = 2870 \text{ psi}$$

The second point was plotted at

$$\sigma_y = 0 \text{ psi}, \tau_{yx} = -2870 \text{ psi}$$

Then a line was drawn between them, crossing the σ -axis at O . The value of the stress at O is

$$\sigma_{\text{avg}} = (\sigma_x + \sigma_y)/2 = (8030 \text{ psi} + 0 \text{ psi})/2 = 4015 \text{ psi}$$

Extend the line through the point (σ_x, τ_{xy}) and label it, x -axis.

Step 7. We compute the values for a , b , and R from

$$a = (\sigma_x - \sigma_{\text{avg}}) = (8030 \text{ psi} - 4015 \text{ psi}) = 4015 \text{ psi}$$

$$b = \tau_{xy} = 2870 \text{ psi}$$

$$R = \sqrt{a^2 + b^2} = \sqrt{(4015 \text{ psi})^2 + (2870 \text{ psi})^2} = 4935 \text{ psi}$$

Step 8. Figure 4-17(a) shows the completed 2D Mohr's circle. The circle has its center at O and the radius R . Note that the circle passes through the two points originally plotted. It must do so because the circle represents all possible states of stress on the element K .

Step 9. The maximum principal stress is at the right side of the circle.

$$\sigma_1 = \sigma_{\text{avg}} + R$$

$$\sigma_1 = 4015 + 4935 = 8950 \text{ psi}$$

Step 10. The minimum principal stress is at the left side of the circle.

$$\sigma_2 = \sigma_{\text{avg}} - R$$

$$\sigma_2 = 4015 - 4935 = -920 \text{ psi}$$

Step 11. At the top of the circle,

$$\sigma = \sigma_{\text{avg}} = 4015 \text{ psi}$$

$$\tau = \tau_{\text{max}} = R = 4935 \text{ psi}$$

The value of the normal stress on the element that carries the maximum shear stress is the same as the coordinate of O , the center of the circle.

Step 12. Compute the angles α , $2\phi_\sigma$, and then ϕ_σ . Use the circle as a guide.

$$\alpha = 2\phi_\sigma = \arctan(b/a) = \arctan(2870 \text{ psi}/(4015 \text{ psi})) = 35.6^\circ$$

$$\phi_\sigma = 35.6^\circ/2 = 17.8^\circ$$

Note that ϕ_σ must be measured *clockwise* from the original x -axis to the direction of the line of action of σ_1 for this set of data. The principal stress element will be rotated in the same direction as part of step 13.

Step 16. Summary of results:

- Maximum principal stress = $\sigma_1 = 8950$ psi—Tension
- Minimum principal stress = $\sigma_2 = -920$ psi—Compression
- Maximum shear stress = $\tau_{\max} = 4935$ psi
- Angle of rotation of principal stress element = $\varphi_\sigma = 17.8^\circ$ clockwise below the x-axis
- Angle of rotation of the maximum shear stress element = $\sigma_\tau = 27.2^\circ$ counterclockwise above the x-axis
- Figure 4–18 shows the resulting stress elements. These are identical to those shown in Figure 4–11 from the hand calculations performed in Example Problem 4–1.

Mohr's Circles for Three-Dimensional Stresses

Obviously, real load-carrying members are three-dimensional objects, and all stress elements should be three-dimensional elements. In Example Problem 4–1, Figure 4–13(a), the Mohr's circle represents the stress transformation of a two-dimensional, plane-stress element. The maximum and minimum principal stresses are identified as σ_1 and σ_2 . It is clear that the z direction is one of the principal directions, as the stress components involving the z direction are all zero, $\sigma_z = \tau_{yz} = \tau_{zx} = 0$. Rearranging/ranking the three principle stresses, $\sigma_1 > \sigma_2 > \sigma_3$, the Mohr's circles for the three-dimensional stress element in plane stress state is shown in Figure 4–13(b), where $\sigma_2 = \sigma_z = 0$. Note that the additional two circles are drawn from the stress states in the 1-2 plane and the 2-3 plane.

The largest circle is the same as that developed for the 2D example with the maximum stress labeled as σ_1 as before at the right end. The smallest stress is at the left end of the largest circle and it is now labeled as σ_3 . The second circle is drawn with σ_1 as before at the right end and $\sigma_2 = 0$ at the left end. Finally, the third circle is drawn with $\sigma_2 = 0$ at the right end and σ_3 at the left end.

In Example Problem 4–2, the element at the location of interest is subjected to plane stress. In the full

three-dimensional model of the stress element, the principal stresses are again rearranged/reordered:

$$\begin{aligned}\sigma_1 &= 8950 \text{ psi} \\ \sigma_2 &= 0 \text{ psi} \\ \sigma_3 &= -920 \text{ psi}\end{aligned}$$

The corresponding Mohr's circles for three-dimensional stress state is shown in Figure 4–17(b).

In three-dimensional stress transformation, it is possible that all of the roots in Equation (4–12) are non-zero. With three non-zero principal stresses, the resulting Mohr's circles are shown in Figure 4–19. Note that the three principal stresses are again ordered such that $\sigma_1 > \sigma_2 > \sigma_3$.

Tresca Stress

Based on the three principal stresses obtained from stress transformation, a special form of stress, called the Tresca stress (after Henri Tresca), can be calculated for design decision making. The Tresca stress is defined as

$$\sigma' = (\sigma_1 - \sigma_3)/2 \quad (4-14)$$

Observing from the Mohr's circles (three-dimensional), the Tresca stress is the maximum shear stress. In design, a failure criterion can be established to evaluate if yielding will occur. For a ductile material under static loading, a stress element is considered to have failed when $\sigma' > s_{sy}$, where s_{sy} is the shear yield strength of the material. The failure theory involving the Tresca stress is therefore called the Maximum Shear Stress Theory (MSST).

von Mises Stress

A more accurate assessment of the stress state for design with ductile material under static loading is the von Mises stress. It is defined as

$$\sigma_e = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}} \quad (4-15)$$

The failure criterion based on the von Mises stress postulates that material will yield when $\sigma_e > s_y$, where s_y is the yield strength of the material. The application of the Tresca and the von Mises stresses in design will be further discussed in Chapter 5.

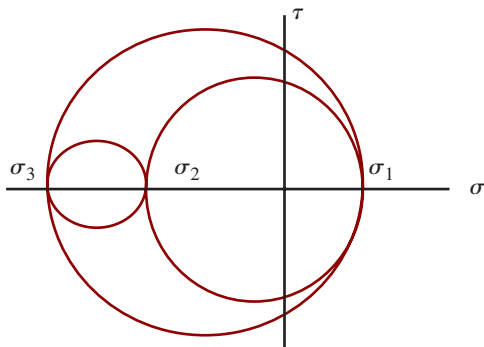


FIGURE 4–19 General example of Mohr's circles for three-dimensional stress state for which all three principal stresses are non-zero

TABLE 4-1 Practice Problems for Mohr's Circle

Example Problem	x-axis	σ_x	σ_y	τ_{xy}	Fig. No.
4-3	1st quadrant	+10.0 ksi	+4.0 ksi	+5.0 ksi	4-20
4-4	2nd quadrant	-80 MPa	+20 MPa	+50 MPa	4-21
4-5	3rd quadrant	-80 MPa	+20 MPa	-50 MPa	4-22
4-6	4th quadrant	+10.0 ksi	-2.0 ksi	-4.0 ksi	4-23

4-5 MOHR'S CIRCLE PRACTICE PROBLEMS

To a person seeing the construction of Mohr's circle for the first time, it may seem long and involved. But with practice under a variety of combinations of normal and shear stresses, you should be able to execute the 16 steps quickly and accurately.

Table 4-1 gives four sets of data (Example Problems 4-3 through 4-6) for normal and shear stresses in the x - y plane. You are advised to complete the Mohr's circle for each before looking at the solutions in Figures 4-20

through 4-23. From the circle, determine the two principal stresses, the maximum shear stress, and the planes on which these stresses act. Then draw the given stress element, the principal stress element, and the maximum shear stress element, all oriented properly with respect to the x - and y -directions. Note that each problem results in the x -axis being in a different quadrant.

After completing the solution for two-dimensional data, values are given for "3D stresses" considering that the z -axis stress is zero. The new values are the three principal stresses, σ_1 , σ_2 , and σ_3 as used in Sections 4-3 and 4-4.

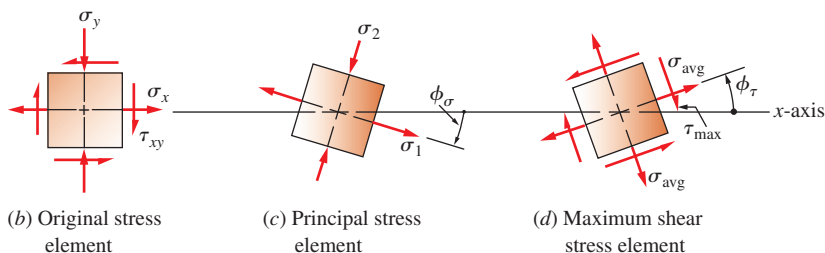
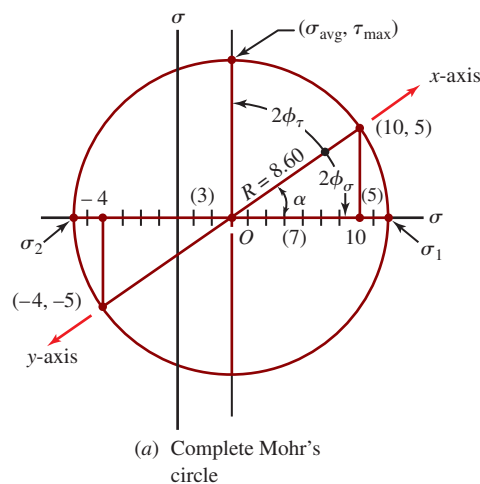
Example Problem 4-3

Given:

$$\begin{aligned}\sigma_x &= +10.0 \text{ ksi} \\ \sigma_y &= -4.0 \text{ ksi} \\ \tau_{xy} &= +5.0 \text{ ksi (cw)}\end{aligned}$$

Results:

$$\begin{aligned}\sigma_1 &= +11.60 \text{ ksi} \\ \sigma_2 &= -5.60 \text{ ksi} \\ \phi_\sigma &= 17.8^\circ \text{ cw} \\ \tau_{\max} &= 8.60 \text{ ksi} \\ \phi_\tau &= 27.2^\circ \text{ ccw} \\ \sigma_{\text{avg}} &= +3.0 \text{ ksi} \\ x\text{-axis in 1st quadrant}\end{aligned}$$



3D stresses: $\sigma_1 = +11.60 \text{ ksi}$, $\sigma_2 = 0 \text{ ksi}$, $\sigma_3 = -5.60 \text{ ksi}$

FIGURE 4-20 Solution for Example Problem 4-3, x -axis in 1st quadrant

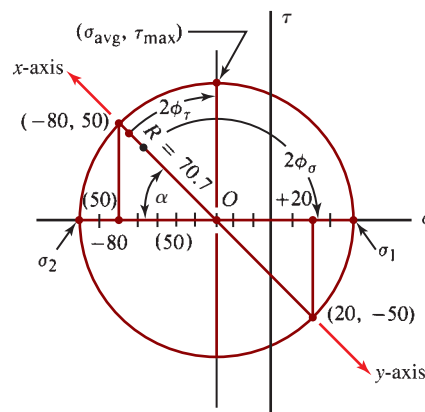
Example Problem 4-4

Given:

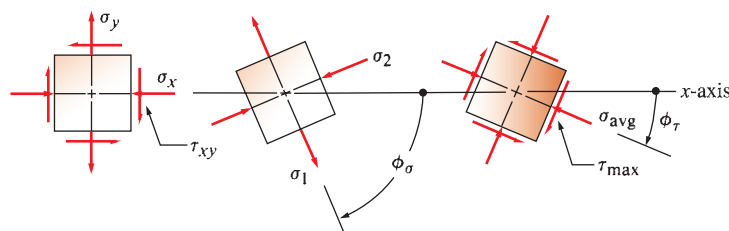
$$\begin{aligned}\sigma_x &= -80 \text{ MPa} \\ \sigma_y &= +20 \text{ MPa} \\ \tau_{xy} &= +50 \text{ MPa}\end{aligned}$$

Results:

$$\begin{aligned}\sigma_1 &= +40.7 \text{ MPa} \\ \sigma_2 &= -100.7 \text{ MPa} \\ \phi_\sigma &= 67.5^\circ \text{ cw} \\ \tau_{\max} &= 70.7 \text{ MPa} \\ \phi_\tau &= 22.5^\circ \text{ cw} \\ \sigma_{\text{avg}} &= -30 \text{ MPa} \\ \text{x-axis in 2nd quadrant}\end{aligned}$$



(a) Complete Mohr's circle



(b) Original stress element (c) Principal stress element (d) Maximum shear stress element

3D stresses: $\sigma_1 = +40.7 \text{ MPa}$, $\sigma_2 = 0 \text{ MPa}$, $\sigma_3 = -100.7 \text{ MPa}$

FIGURE 4-21 Solution for Example Problem 4-4 x-axis in 2nd quadrant

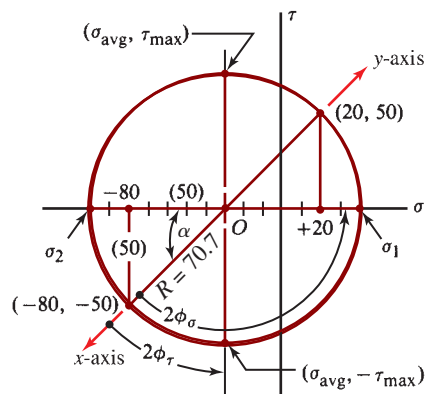
Example Problem 4-5

Given:

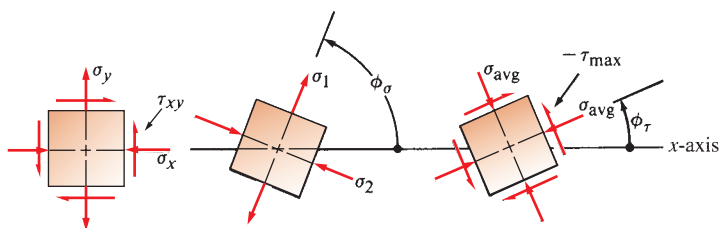
$$\begin{aligned}\sigma_x &= -80 \text{ MPa} \\ \sigma_y &= +20 \text{ MPa} \\ \tau_{xy} &= -50 \text{ MPa}\end{aligned}$$

Results:

$$\begin{aligned}\sigma_1 &= +40.7 \text{ MPa} \\ \sigma_2 &= -100.7 \text{ MPa} \\ \phi_\sigma &= 67.5^\circ \text{ ccw} \\ \tau_{\max} &= 70.7 \text{ MPa} \\ \phi_\tau &= 22.5^\circ \text{ ccw to } -\tau_{\max} \\ \sigma_{\text{avg}} &= -30 \text{ MPa} \\ \text{x-axis in 3rd quadrant}\end{aligned}$$



(a) Complete Mohr's circle



(b) Original stress element (c) Principal stress element (d) Maximum shear stress element

3D stresses: $\sigma_1 = +40.7 \text{ MPa}$, $\sigma_2 = 0 \text{ MPa}$, $\sigma_3 = -100.7 \text{ MPa}$

FIGURE 4-22 Solution for Example Problem 4-5, x-axis in 3rd quadrant

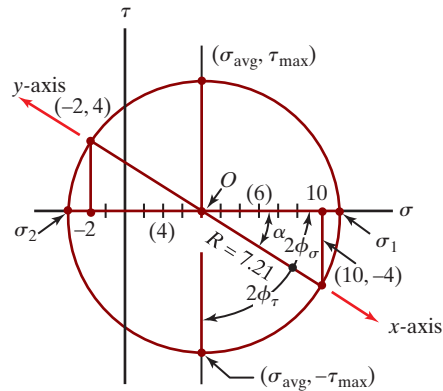
**Example Problem
4-6**

Given:

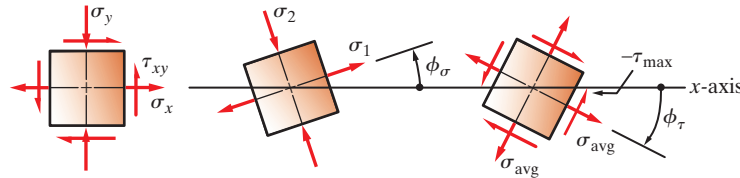
$$\begin{aligned}\sigma_x &= +10.0 \text{ ksi} \\ \sigma_y &= -2.0 \text{ ksi} \\ \tau_{xy} &= -4.0 \text{ ksi (ccw)}\end{aligned}$$

Results:

$$\begin{aligned}\sigma_1 &= +11.21 \text{ ksi} \\ \sigma_2 &= -3.21 \text{ ksi} \\ \phi_\sigma &= 16.8^\circ \text{ ccw} \\ \tau_{\max} &= 7.21 \text{ ksi} \\ \phi_\tau &= 28.2^\circ \text{ cw to } -\tau_{\max} \\ \sigma_{\text{avg}} &= +4.0 \text{ ksi} \\ x\text{-axis in 4th quadrant}\end{aligned}$$



(a) Complete Mohr's circle



(b) Original stress element

(c) Principal stress element

(d) Maximum shear stress element

3D stresses: $\sigma_1 = +11.21 \text{ ksi}$, $\sigma_2 = 0 \text{ ksi}$, $\sigma_3 = -3.21 \text{ ksi}$ **FIGURE 4-23** Solution for Example Problem 4-6, x -axis in 4th quadrant**4-6 MOHR'S CIRCLE FOR
SPECIAL STRESS
CONDITIONS**

Mohr's circle is used here to demonstrate the relationship among the applied stresses, the principal stresses, and the maximum shear stress for the following special cases:

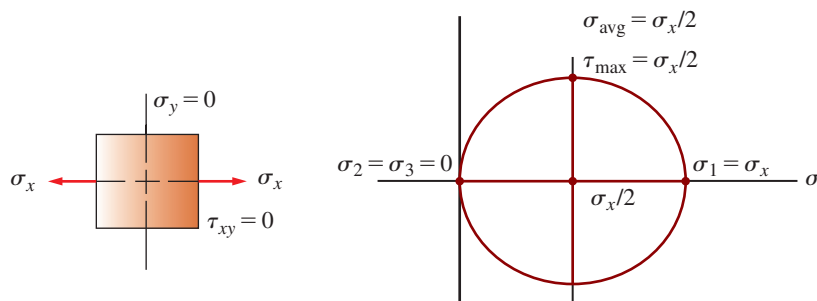
- Uniaxial tension
- Uniaxial compression
- Biaxial tension
- Biaxial tension and compression
- Pure shear
- Combined tension and shear

These are important, frequently encountered stress conditions, and they will be used in later chapters to

illustrate failure theories and design methods. These failure theories are based on the values of the principal stresses and the maximum shear stress.

Uniaxial Tension

The stress condition produced in all parts of a standard tensile test specimen is pure uniaxial tension. Figure 4-24 shows the stress element and the corresponding Mohr's circle. Note that the maximum principal stress, σ_1 , is equal to the applied stress, σ_x ; the minimum principal stress, σ_3 , is zero; and the maximum shear stress, $\tau_{\max}$, is equal to $\sigma_x/2$. Note that in three-dimensional stress state, the Mohr's circle drawn from the 1-2 plane coincides with that from the 1-3 plane, and the Mohr's circle drawn from the 2-3 plane is a point.

**FIGURE 4-24** Mohr's circle for pure uniaxial tension

Uniaxial Compression

Figure 4–25 shows pure uniaxial compression as it would be produced by a standard compression test. Mohr's circle shows that $\sigma_1 = \sigma_2 = 0$ and $\sigma_3 = \sigma_x$ (a negative value); and the magnitude of the maximum shear stress is $\tau_{\max} = \sigma_x/2$. Note that in a three-dimensional stress state, the Mohr's circle drawn from the 1-2 plane is a point, and the Mohr's circle drawn from the 1-3 plane coincides with that from the 2-3 plane.

Biaxial Tension

The stress condition produced in a spherical thin-walled pressure vessel or a filled balloon is biaxial tension. Figure 4–26 shows the stress element, where $\sigma_x = \sigma_y$,

and the corresponding Mohr's circle. Note that the Mohr's circle drawn from the original two-dimensional stress element in the 1-2 plane is a point. Considering the third principal direction, two more circles are added. The Mohr's circle drawn from the 1-3 plane coincides with that from the 2-3 plane. The magnitude of the maximum shear stress is $\tau_{\max} = \sigma_x/2$.

Biaxial Tension and Compression

Figure 4–27 shows that the element is subjected to biaxial tension and compression where σ_x is positive (tension) and σ_y is negative (compression) with the same magnitude. The resulting Mohr's circles show that $\sigma_1 = \sigma_x$, $\sigma_2 = 0$, and $\sigma_3 = \sigma_y$. The magnitude of the maximum shear stress is $\tau_{\max} = \sigma_x$.

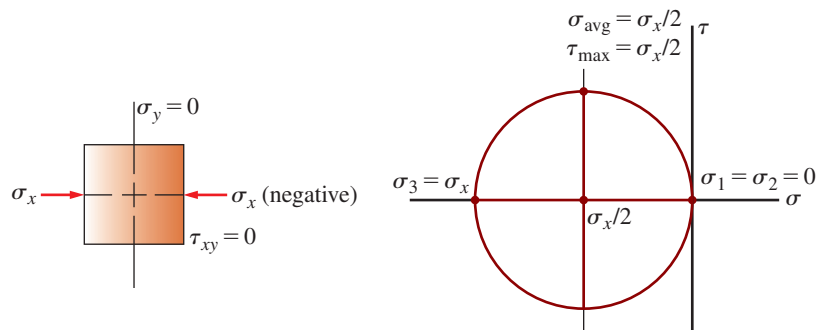


FIGURE 4–25 Mohr's circle for pure uniaxial compression

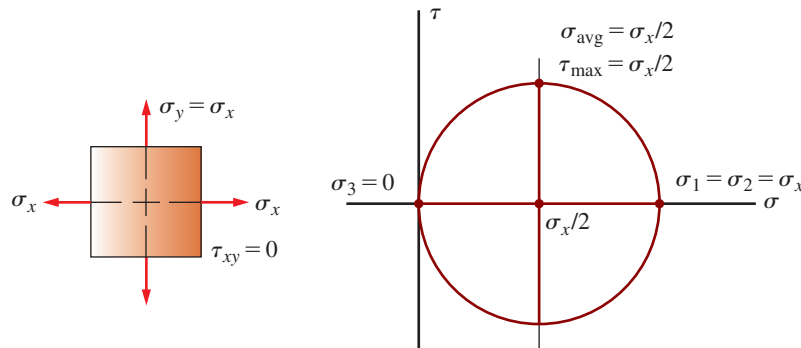


FIGURE 4–26 Mohr's circle for biaxial tension

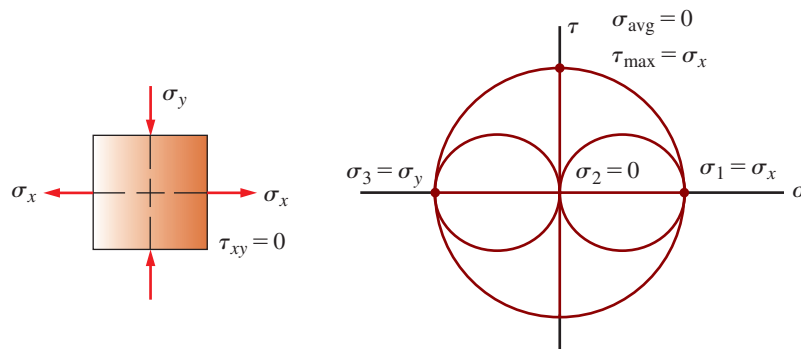


FIGURE 4–27 Mohr's circle for biaxial tension and compression

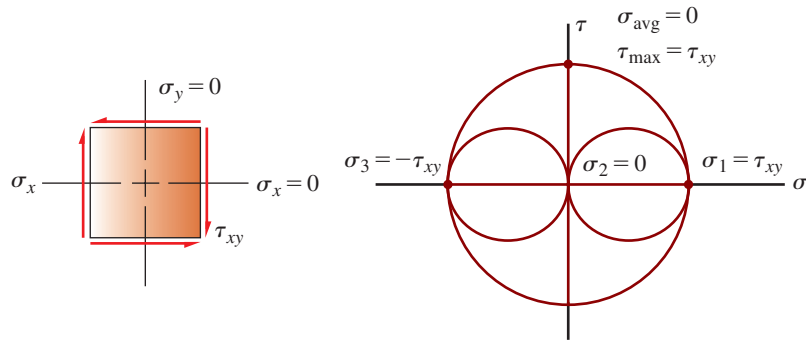


FIGURE 4-28 Mohr's circle for pure shear

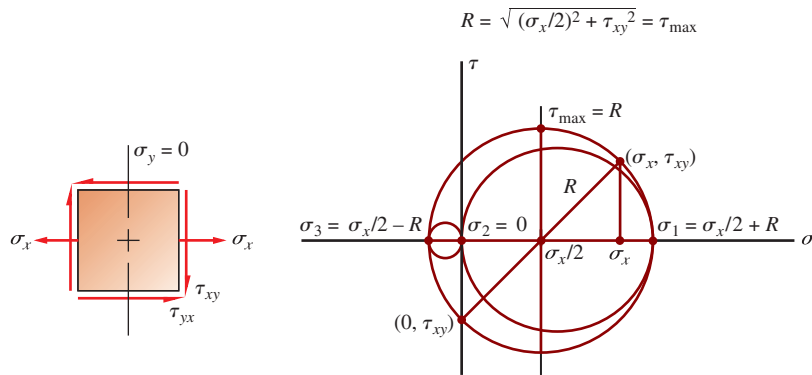


FIGURE 4-29 Mohr's circles for uniaxial tension and torsional shear

Pure Shear

Figure 4-28 shows the Mohr's circles for the case of pure shear that can result from a torsional load. The plane stress circle has its center at the origin of the σ - τ axes and that the radius of the circle is equal to the value of the applied shear stress, τ_{xy} . Considering the third principal direction, we have $\sigma_1 = \tau_{xy}$, $\sigma_2 = 0$, and $\sigma_3 = -\tau_{xy}$. The maximum shear $\tau_{\max} = \tau_{xy}$. From Figures 4-27 and 4-28, it can be observed that if the magnitude of tensile stress σ_x and compressive stress σ_y in Figure 4-27 is the same as that of the shear stress τ_{xy} in Figure 4-28, the Mohr's circles in the two figures are identical. The two stress elements are identified to be in the *equivalent stress state*. That is, the two different stress states have the same effect on the material.

Combined Tension and Shear

This is an important special case because it describes the stress condition in a rotating shaft carrying bending loads while simultaneously transmitting torque. This is the type of stress condition on which the procedure for designing shafts, presented in Chapter 12, is based. If the applied stresses are called σ_x and τ_{xy} , the Mohr's circle in Figure 4-29 shows that

$$\tau_{\max} = R = \text{radius of circle} = \sqrt{(\sigma_x/2)^2 + \tau_{xy}^2} \quad (4-16)$$

$$\sigma_1 = \sigma_x/2 + R = \sigma_x/2 + \sqrt{(\sigma_x/2)^2 + \tau_{xy}^2} \quad (4-17)$$

$$\sigma_2 = \sigma_x/2 - R = \sigma_x/2 - \sqrt{(\sigma_x/2)^2 + \tau_{xy}^2} \quad (4-18)$$

A convenient and useful concept called *equivalent torque* can be developed from Equation (4-16) for the special case of a body subjected to only bending and torsion.

An example is shown in Figure 4-30, where a circular bar is loaded at one end by a downward force and a torsional moment. The force causes bending in the bar with the maximum moment at the point where the bar is attached to the support. The moment causes a tensile stress on the top of the bar in the x -direction at the point called A, where the magnitude of the stress is

$$\sigma_x = M/S \quad (4-19)$$

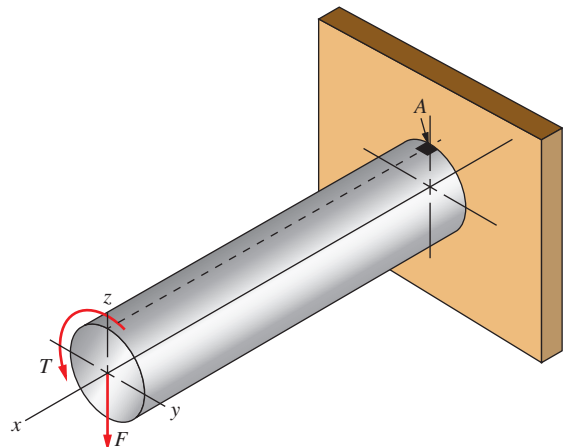


FIGURE 4-30 Circular bar in bending and torsion

where S = section modulus of the round bar.

Now the torsional moment causes torsional shear stress in the x - y plane at point A having a magnitude of

$$\tau_{xy} = T/Z_p \quad (4-20)$$

where Z_p = polar section modulus of the bar.

Point A then is subjected to a tensile stress combined with shear, the special case shown in the Mohr's circle of Figure 4-29. The maximum shear stress can be computed from Equation (4-16). If we substitute Equations (4-19) and (4-20) in to Equation (4-16), we get

$$\tau_{\max} = \sqrt{(M/2S)^2 + (T/Z_p)^2} \quad (4-21)$$

Note from Appendix 1 that $Z_p = 2S$. Equation (4-21) can then be written as

$$\tau_{\max} = \frac{\sqrt{M^2 + T^2}}{Z_p} \quad (4-22)$$

It is convenient to define the quantity in the numerator of this equation to be the *equivalent torque*, T_e . Then the equation becomes

$$\tau_{\max} = T_e/Z_p \quad (4-23)$$

Stresses in a Cylinder with Internal Pressure

Another important special case of stress state is the set of stresses developed in a closed-end cylinder with internal pressure. In strength of materials, you learned that the outer surfaces of the walls of such cylinders are subjected to tensile stresses in two directions: (1) tangential to its circumference and (2) axially, parallel to the axis of the cylinder. The stress perpendicular to the wall at the outer surface is zero.

Figure 4-31 shows the stress condition on an element of the surface of the cylinder. The tangential stress, also called *hoop stress*, is aligned with the x -direction and is labeled σ_x . The axially directed stress, also called *longitudinal stress*, acts in line with the y -direction and is labeled σ_y .

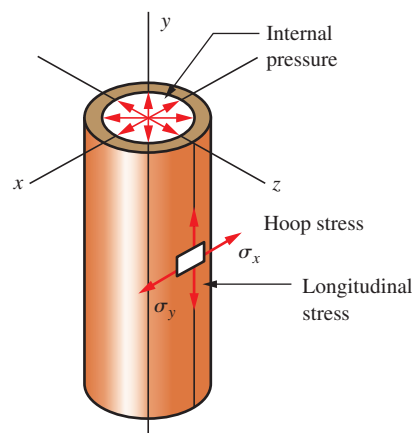


FIGURE 4-31 Thin-walled cylinder subjected to pressure with its ends closed

In strength of materials, you learned that if the wall of the cylinder is relatively thin, the maximum hoop stress is

$$\sigma_x = pD/2t$$

where p = internal pressure in the cylinder
 D = mean diameter of the cylinder
 t = thickness of the cylinder wall

Also, the longitudinal stress is

$$\sigma_y = pD/4t$$

Both stresses are tensile, and the hoop stress is twice as large as the longitudinal stress.

The analysis would be similar for any kind of thin-walled cylindrical vessel carrying an internal pressure. Examples are storage tanks for compressed gases, pipes carrying moving fluids under pressure, and the familiar beverage can that releases internal pressure when the top is popped open.

Let's use the following example to demonstrate the use of Mohr's circles for stress analysis.

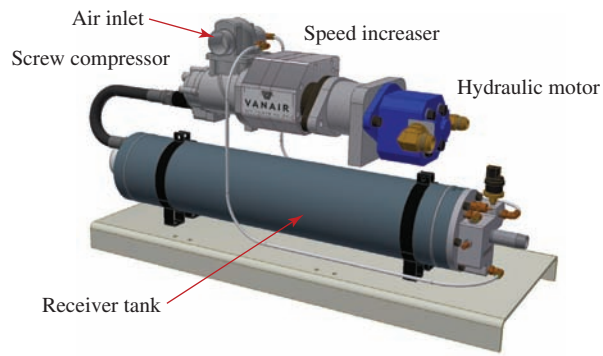
Example Problem 4-7

As shown in Figure 4-32, a hydraulic motor drives a speed increaser gearbox which in turn drives the air screw compressor at 8000 rpm. The air screw compressor draws in outside air, compresses it, and discharges it into the air receiver tank. The receiver tank supplies service air to power air tools. A cut section of a receiver tank is shown below. The maximum design pressure of the system is set at 175 psi. The air receiver tank is made from aluminum tubing and has a 6 in outside diameter with a wall thickness of 1/8 in.

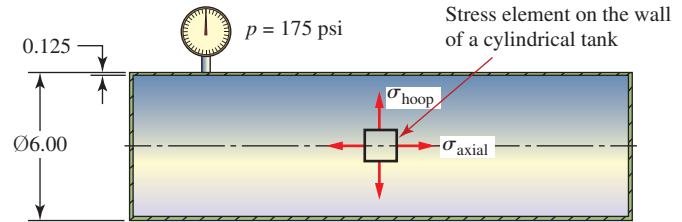
- Can the receiver tank be considered a thin-walled pressure vessel?
- Calculate the longitudinal and circumferential stresses on the receiver tank.
- Use Mohr's circle to calculate the principal stresses and the maximum shear stress.

Solution

Given: The outside diameter of the tank, $D = 6.0$ in and the wall thickness, $t = 0.125$ in. The internal pressure in the tank is 175 psi. [Note: Mean diameter $D_m = D_o - t = 5.875$ in]



(a) Air compressor system



(b) Cut section through the compressed air receiver tank

FIGURE 4-32 Screw-type air compressor system with a receiver tank
(Courtesy of Vanair, Inc., Michigan City, IN)

Results First check to determine if the pressure vessel is to be considered a thin-walled or thick-walled cylinder:

$$\frac{D_m}{t} = \frac{5.875}{0.125} = 47 > 20$$

Therefore, the vessel is considered to be a thin-walled cylinder.

The stress element obtained from the cylindrical wall has the following stress components:

$$\text{Longitudinal/axial stress: } \frac{pD_m}{4t} = \frac{175 \times 5.875}{4 \times 0.125} = 2056 \text{ psi}$$

$$\text{Circumferential/hoop stress: } \sigma_{\text{hoop}} = \frac{pD_m}{2t} = \frac{175 \times 5.875}{2 \times 0.125} = 4113 \text{ psi}$$

$$\text{Radial stress: } \sigma_{\text{radial}} = 0$$

Since there is no shear stress, the stress components are already in principal directions. Ordering the three principal stresses, we have

$$\sigma_1 = \sigma_{\text{hoop}} = 4113 \text{ psi}$$

$$\sigma_2 = \sigma_{\text{axial}} = 2056 \text{ psi}$$

$$\sigma_3 = \sigma_{\text{radial}} = 0$$

The corresponding three-dimensional stress element and Mohr's circles are shown in Figure 4-33. Note that the maximum shear stress is based on the consideration of the largest circle where

$$\tau_{\text{max}} = 2056 \text{ psi}$$

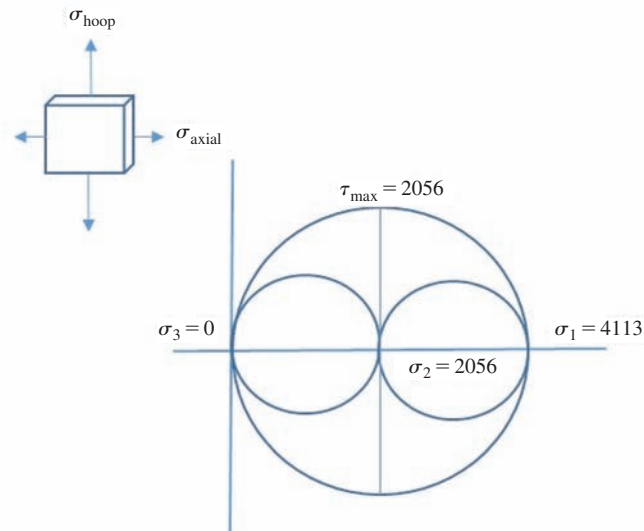


FIGURE 4-33 Stress element on the surface of the cylindrical tank and the resulting Mohr's circles